Proposal title: Safe Food and Feed through an Integrated ToolBox for Mycotoxin Management

MyToolBox Partners

Nr./ Org.type	Partner organisation name	Country	Main role in project
01/(CO) RTD	University of Natural Resources and Life Sciences, Vienna (BOKU)	Austria	Coordination (WP6); Safe use options of contaminated batches; Analytical chemistry
02/RTD	Stichting Dienst Landbouwkundig Onderzoek RIKILT (DLO-RIKILT)	Netherlands	Scientific Management (Deputy-Coordination), Leader WP3 - MyToolBox e-platform; HACCP
03/AG	International Association for Cereal Science and Technology (ICC)	International Organisation	Leader WP5 - Exploitation, dissemination and communication
04/RTD	Harper Adams University (HAU)	United Kingdom	Leader WP1 - Pre-harvest reduction
05/SME	FoodLife International Ltd (FLI)	Turkey	Leader WP4 – Practical implementation
06/IND	BARILLA S.p.A. (BARILLA)	Italy	End-user food industry; Leader WP2 – Post-harvest reduction
07/RTD	Cranfield University (CRANFIELD)	United Kingdom	Real time monitoring of stored grains/peanuts
08/SME	IRIS (IRIS)	Spain	Robust sensor platform for silo monitoring; Pilot scale testing; ICT based e-platform.
09/RTD	University of Novi Sad (UNS)	Serbia	Reduction of aflatoxins in maize (biocontrol, drought resistance)
10/SME	HORTA (HORTA)	Italy	Forecasting; Resistance of maize hybrids
11/RTD	Bundesinstitut f. Risikobewertung (BfR)	Germany	Legislative implications; Risk assessment authority
12/RTD	Suleyman Demiral University (SDU)	Turkey	Real time screening/sorting of dried fruit
13/RTD	Wageningen University (WU)	Netherlands	Socio-economic and stakeholder analysis
14/RTD	Bioforsk (BIOFORSK)	Norway	Efficacy testing of biological control agents to reduce *Fusarium* inoculum
15/RTD	Institute of Agro-Products Processing Science and Technology, Chinese Academy of Agricultural Sciences (CAAS-IAPPST)	China	Cooperation in production/formulation of biocontrol agents; Partner in setting up the EU-China-Mycotoxin Forum
16/IND	Biomin (BIOMIN)	Austria	Use of mycotoxin degrading enzymes for optimal biogas and bioethanol production; detoxifiers as feed additives
17/RTD	Chinese Academy of State Administration of Grain (ASAG)	China	DON - Forecasting; Monitoring of stored grains
18/IND	Südzucker (Südzucker)	Germany	End user – Food and crop energy
19/SME	Axeb Biopesticides (AXEB)	Spain	Bio-pesticide technology
20/IND	Agro LV (Agro LV)	Ukraine	End user - Grain industry (Pre-Harvest)
21/IND	TARIS (TARIS)	Turkey	End user – Dried fruit industry (Post-harvest)
22/SME	D.O.O. Agrocentrum Becej (Agrocentrum)	Serbia	End user – Feed industry (Maize based)
23/RTD	Feed Research Institute, Chinese Academy of Agricultural Sciences (CAAS-FRI)	China	Standard settings for authorisation of mycotoxin-detoxifying feed additives; feeding trials

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1. Excellence

1.1 Objectives

Introduction

The occurrence of fungal and subsequently, **mycotoxin contamination** in various crops is of major concern since it has significant implications for food and feed safety, food security and international trade. The EU produces yearly about 133 million tons (MT) **wheat** (~M€ 29038), 68 MT **maize** (~M€ 13571) and 8 MT **oats** (~M€ 1543)[1,2]. Results using latest state-of-the-art multi-analyte methods show that almost 100% of the crops is contaminated with one or more mycotoxins[3]. With annual losses due to mycotoxins estimated at 5-10%, this equates to €1.2-2.4 billion (Bn) in lost income for wheat alone, a reduction in these losses of only 1% would save €12-24 Mio. Moreover with ''one in eight persons worldwide suffering from chronic undernourishment''[4]; there is clearly a moral obligation to curb these significant food and feed losses.

Grain and foods based on these grain (e.g. pasta, bread, bakery products) account for the largest contribution to mycotoxin exposure in all age classes of the EU population, in particular due to the mycotoxins produced by ***Fusarium* spp.** (deoxynivalenol [DON], T-2/HT-2 toxins, zearalenone [ZEN], ochratoxin A [OTA] and fumonisins). Despite huge research investments prevention and control of these toxic secondary metabolites remains difficult and the agriculture and food industries continue to be vulnerable to problems of contamination: In February–March 2013 Romania, Serbia and Croatia, reported aflatoxin M_1 contamination of milk. **Severe droughts in Serbia in 2012 resulted in 70% of the maize crop being contaminated with aflatoxins[5]**. Use of this maize to feed dairy cattle led to the high levels of aflatoxin M_1 in milk, up to twice the EU legal limit. The milk scandal was fuelled when the permitted level of aflatoxin M_1 in milk in Serbia was temporarily raised to 0.5 µg/kg milk, 10 times the EU legal limit. In recent years, a **dramatic increase in DON contamination in Nordic countries** has been observed. The Norwegian Scientific Committee for Food Safety concluded in a study published in 2013 that "*The estimated mean and high (95-percentile) exposures to DON ...were in the range of, or exceeded the TDI* ***by almost 2 times in 1-year-old infants and 2-year-old children*** *and* ***up to 3.5 times*** *in years with high DON levels which has been especially true for* ***oat based products[6].*** On the other hand the catastrophic floods and the rainy summer in 2014 resulted in low levels of aflatoxin B_1 with high levels of DON. Obviously, extreme weather conditions as a result of **climate change** is increasingly affecting the mycotoxin map in Europe and also world-wide. In 2013 the EU's RASFF[7] (Rapid Alert System for Food and Feed) showed that of the total rejections at the EU borders, 18.7% were due to mycotoxin contamination exceeding the EU legislative limits[8]. The RASFF notification system also reveals that among the susceptible fruit imported into the EU, the commodity with the **highest risk for aflatoxin contamination was dried figs**. In addition, over the last five years RASFF reported 292 alerts relating to aflatoxin levels exceeding legal limits in **peanuts imported from China**.

Exposure of livestock to mycotoxins may lead to substantial economic losses (Wu & Munkvold 2008)[9]. While mycotoxicoses in agricultural animals are regularly reported (Jovanović *et al*. 2015)[10], it is difficult to get an accurate estimate of the burden of market loss associated with animal health and productivity, and loss of animal feed. However, several publications from different parts of the world provide some insight into the potential magnitude of the problem. In the USA, Vardon *et al*. (2003)[11] estimated **a loss of 225 million US$ per annum for feed maize due to contamination with aflatoxins, and an additional loss of livestock of 4 million US$ per annum**. Wu and Munkvold (2008) estimated the losses due to decreased weight gain in pigs resulting from fumonisin contamination of **DDGS** in the USA at **147 million US$** annually (20% DDGS in the diets and market penetration is 100%). A study

[1] EUROSTAT, Selling prices of crop products (absolute prices) - annual price (from 2000 onwards). http://appsso.eurostat.ec.europa.eu/nui/show.do?dataset=apri_ap_crpouta&lang=en.
[2] EUROSTAT, Crops products - annual data. http://appsso.eurostat.ec.europa.eu/nui/show.do?dataset=apro_cpp_crop&lang=en.
[3] Berthiller F. *et al*. (2014) Developments in mycotoxin analysis: an update for 2012-2013. World Mycotoxin J 7: 3-33.
[4] Schmale III D.G. and Munkvold G.P. 2009. Mycotoxins in Crops: A Threat to Human and Domestic Animal Health. https://www.apsnet.org/edcenter/intropp/topics/Mycotoxins/Pages/EconomicImpact.aspx,
[5] http://www.allaboutfeed.net/Process-Management/Feed-Safety/2013/3/EU-Aflatoxin-contaminated-feed-1196477W/
[6] VKM, Risk assessment of mycotoxins in cereal grain in Norway. Norwegian Scientific Committee for Food Safety (VKM) http://www.vkm.no/dav/eee04d10c4.pdf, 2013
[7] EU (2014) The Rapid Alert System for Food and Feed 2013. Annual Report. RASFF 2013: p. 45.
[8] http://www.allaboutfeed.net/Process-Management/Feed-Safety/2014/7/Mycotoxin-alerts-in-EU-have-gone-down-1568614W/
[9] Wu F. & Munkvold GP. 2008. Mycotoxins in ethanol co-products: Modeling economic impacts on the livestock industry and management strategies. J Agric Food Chem. 56: 3900-3911.
[10] Jovanović M. *et al*. 2015. An outbreak of fumonisin toxicosis in horses in Serbia. World Mycotoxin J. Accepted for publication, DOI 10.3920/WMJ2014.
[11] Vardon PJ. *et al*. 2003. Potential economic costs of mycotoxins in the United Sates. CAST report: Mycotoxins; Risks in Plant, Animal and Human Systems. Task Force Report no. 139, CAST, Ames, Iowa. USA

conducted in Thailand (Lubulwa *et al.* 2015)[12] revealed that aflatoxins in maize can be expected to cause losses for the livestock sector between 7 and 110 million US$ per annum. These estimates, though limited to aflatoxin and fumonisin in 2 countries only, show that on a worldwide basis and taking account of other commodities and other mycotoxins as well, the losses in the livestock sector will be multiple.

There is a pressing need to mobilise the wealth of knowledge that exists from the mycotoxin research in Europe and internationally conducted over the past 25-30 years and to perform cutting-edge research where knowledge gaps still exist. This knowledge needs to be integrated into **practical and affordable tools that can be used by farmers and food processors along the chain** in order to reduce the risk of mycotoxin contamination of crops, feed and food, to prevent losses and waste along the food chain, and to provide traceability information to the supply chain and consumers using mainstream ICT technology. Exisiting knowledge combined with **novel findings** need to be adapted and integrated to provide the vehicle needed to practically implement this knowledge into tools for use along the chain: **this is the mission of the MyToolBox project**.

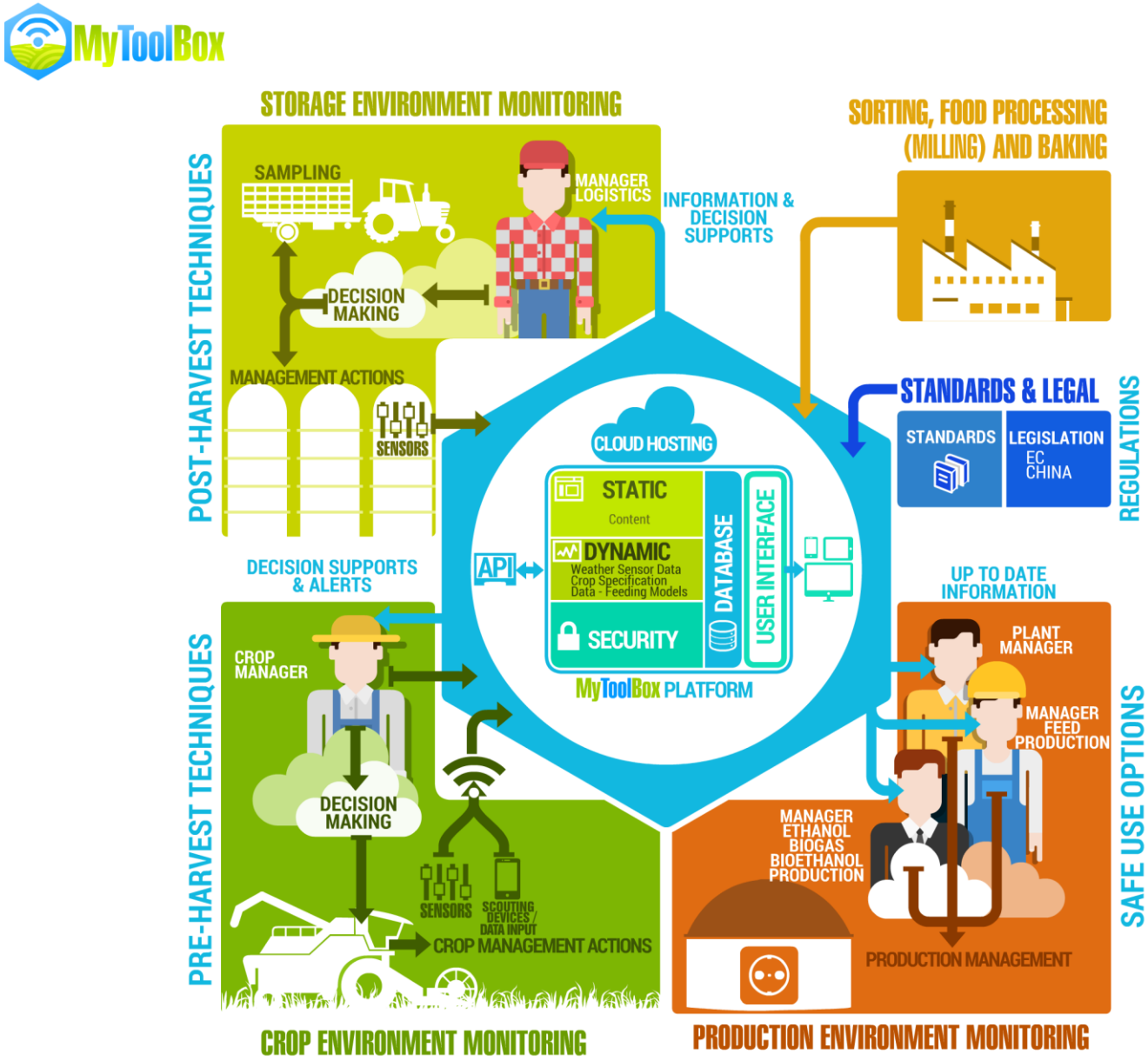

Figure 1- Overview of the MyToolBox project approach

[12] Lubulwa GAS. *et al.* 2015. Estimating the burden of market loss due to aflatoxins in maize: methods and estimates for Thailand. World Mycotoxin J. In press, DOI 10.3920/WMJ2014.1836

Objectives

The overall objective of the MyToolBox project is to **develop a series of integrated measures, which will enable a significant reduction in losses of crops** due to fungal and mycotoxin contamination. Depending on the commodity and the type of intervention, the achievable reductions range from 20-90%. MyToolBox will not only pursue a field-to-fork approach along the food and feed chain, but will also consider safe use options of mycotoxin contaminated batches such as microbial energy conversion to efficiently produce biogas and bioethanol**.** The consideration of the **entire chain (*soil-field-crop-food processing-waste management-alternative energy)*** to ensure food & feed security and safety within a sustainable economic environment, is a major motivation behind MyToolBox. The project will specifically address biological contamination from fungal infection and subsequent events leading to the formation of mycotoxins. The innovative pre- and post-harvest interventions resulting from the research undertaken within the MyToolBox project will **build on and integrate the wealth of knowledge resulting from the substantial EU- and nationally and internationally funded research** previously conducted on mycotoxin prevention and control**. Our goal** will be to implement exisiting knowledge and novel findings and technologies developed in the MyToolBox project into a comprehensive series of tools and applications. These will provide the end users along the whole food and feed chain with the recommended measures for mycotoxin reduction and will be accessible via a dynamic web-based MyToolBox e- platform (MyToolBox-figure 1). Information and decision support tools will be developed for each level of the chain and **will be integrated into the ergonomic and secure web-based MyToolBox platform that will also be accessible over all mobile platforms**. As such, the MyToolBox platform will guide the end user to the most effective measure(s) to reduce biological contamination in crops, and will provide the necessary intelligence to ensure these measures take into account the prevailing conditions such as geographical location, meteorological conditions, land-use, crop management, storage and intended end use with relevance to specific crops. The mycotoxin commodity combinations that will be addressed are the most prevalent ***Fusarium* mycotoxins** (DON, T-2/HT-2 toxins, ZEN and fumonisins) in wheat, oats, maize and animal feed chains, ochratoxin A in wheat and aflatoxins in maize, peanuts and dried fruit (figs). We have mobilised a **comprehensive multi-actor approach** behind a Pan-European-Sino effort towards reducing risk due to mycotoxin contamination in crops all along the feed and food chains, involving pre-harvest interventions undertaken at farm level, together with post-harvest interventions undertaken from storage through to food processing into the finished food or feed product. Within this multi-actor approach **partners from industry (43% of consortium) including 5 end users** from the farming community (including individual farmers), agronomists and professionals working in agriculture and food manufacturing have already been **involved in the project planning and will collaborate closely with the Technology SMEs and partners from academia** at all stages of the various work packages.

As demanded in the Call SFS-13-2015, the MyToolBox project will also **anticipate legislative implications** of the project outputs and work with relevant bodies such as EFSA and DG Santé for **standard setting.** The project will also aim to develop a sound scientific basis for **standard settings in China for authorisation of mycotoxin detoxifying feed additives**, and consequently improve market access of relevant EU products. The variety of **complementary approaches** to be pursued in the MyToolBox project **address all the specific challenges mentioned in the EU-Call SFS-13-2015** and will **increase the productivity and competitiveness of the EU-agri-food and feed sector** whilst **securing safe food crops at minimum mycotoxin levels.** In order to achieve our overall objective, the following **specific objectives** will be fulfilled over the course of our **48-month project**:

Pre-harvest objectives

- To implement **alternatives to conventional fungicides** (suitable for organic farming) for small grain cereals to achieve comparable or better performance to conventional fungicides in reducing levels of biological contamination and mycotoxins by 50% (WP1).
- To integrate **cultural control protocols targeting the *Fusarium* inoculum** within crop debris using biofumigation and accelerated biodegradation combined with minimum tillage (50% reduction, WP1)
- To achieve a 70-90% **reduction of aflatoxin contamination in EU maize** through a) the selection and application of resistant plant cultivars in field conditions (70%) and b) the use of atoxigenic *Aspergillus* strains to outcompete toxigenic *Aspergillus* strains (90%) (WP1).
- To develop **forecasting models** to predict potential fungal contamination of small grain cereals at an early growth stage so that pre-emptive action can be taken to reduce both infection and mycotoxin formation (WP3).

Post-harvest objectives

- To develop **real-time post-harvest environmental monitoring** systems for storage of cereals and peanuts to facilitate effective integrated control of moulds and mycotoxins and provide an early warning of risks, enabling effective prevention/remedial actions to be implemented (WP2).

- To develop **non-invasive real time sorting of dried fruit (figs)** for mycotoxins using visible and NIR spectral imaging and appropriate algorithms to achieve significant improvements compared to manual sorting with 5% or less probability of the batch average concentration of aflatoxin B_1 failing to meet the EU limits (WP2).
- To integrate novel down-stream processing approaches such as **innovative (pre-)milling** for accurate separation of grain tissues with different mycotoxin levels and to minimise the mycotoxin content in the final wheat based food products (up to 50% reduction) (WP2).
- To develop **novel baking procedures at industrial scale** to reveal the thermal processing factors which are relevant for the reduction of mycotoxins and their transformation into modified forms (incl. conjugated toxins) (WP2).
- **Utilising a novel generation of mycotoxin degrading enzymes** for environmentally sustainable use of contaminated batches using fermentation technologies for the **production of biofuels (bioethanol and biogas)** (WP2).

Throughout the food & feed chain

- To design, build, test and validate an **integrated, dynamic, MyToolBox e-platform** for decision support in mycotoxin management, which provides users with information, tools and guidance concerning actions that can be taken to reduce fungal contamination and lead to reduction in levels of mycotoxins in the food and feed chains. The web-based MyToolBox e-platform will be accessible to farmers, food processors and other actors along the chain through all mainstream interfaces (PCs, tablets or smartphones) (WP3).
- **To maximise uptake of the web-based MyToolBox e-platform by the stakeholders** in the project and beyond, the intent of actors will be determined by their perceptions of risks of mycotoxin contamination (= perception of probability consequences) and their perception of the possibilities to reduce these risks in the most reliable and cost-effective way by using the MyToolBox e-platform (WP3).
- To draw up **Good Agricultural Practice (GAP) and Hazard Analysis Critical Control Point (HACCP)** based protocols for prevention and control of mycotoxins that can be applied throughout the entire food & feed chain (WP3).
- To adapt and integrate **cost-effective sampling and monitoring strategies** for mycotoxin management over the entire chain (WP3).
- To establish international cooperation in research and innovation for mycotoxins and in particular with partners from China across all WPs including the initiation of a formal **EU-China dialogue through an EU-China Mycotoxin Forum** as part of the MyToolBox activities (WP5).

Dissemination activities

- To create a **project website including a public dissemination part** and a portal for inter-consortium sharing of information, databases and knowledge management (WP5).
- To **disseminate information** about the project and its **results to actors and stakeholders** along the food supply chain, to opinion leaders/regulators, the media, farmers, professionals working in agriculture and food manufacture, educators, and other multipliers, food and drink industries (including SMEs), consumer organisations, the scientific community and the general public (WP5).
- To **facilitate collaboration, information exchange and interaction between different actors** and organisations involved in the control and management of mycotoxins in the food supply chain through **training**, consultations and webinars (WP5).
- To **develop exploitation plans, including a business model, to ensure the use of the project outputs** among the farming and industry end users, opinion leaders, decision makers, etc. and to establish and verify a robust business model for the MyToolBox e-platform (WP5).

1.2 Relation to the work programme

MyToolBox relates to H2020 topic '*SFS-13-2015 Biological contamination of crops and the food chain*' and **addresses all the specific challenges in the Call**, **as well as the full scope of this Call as explained in table 1**:

Table 1: Relation to the Work Programme

SFS-13-2015	MyToolBox
*The occurrence of **biological contamination in various crops** is of major concern ….. …effectively **monitor …. the incidence of mycotoxin contaminants.***	MyToolBox addresses the issue of **biological contamination of crops** arising from both pre- and post-harvest fungal infection. Pre-harvest infection of crops causes large reductions in yields and losses due to poor quality. During drying and subsequent storage, further fungal infection can occur, which can lead to further losses. **MyToolBox focuses on prevention and minimising pre-harvest fungal infection** at different stages in the plant growth cycle, through selection of fungal resistant crop varieties, minimising reservoir sources of fungal infection for the growing plant, and applying novel biopesticides. **MyToolBox will monitor crops during storage**, measuring real-time parameters such as CO_2 levels, which are indicative of fungal growth enabling remedial action to be taken **to reduce losses from biological contamination**. Moreover, **improved forecasting models to predict contamination** of wheat and maize with fungi and mycotoxins at early growth state will be developed.
*Proposals should aim at **reducing the risk of mycotoxin contamination** in crops and all along the feed and food chains.*	Besides the economic losses from fungal infection, there is also the potential for mycotoxin contamination of crops with risks to human and animal health. However, mycotoxin formation is not inevitable, even if crops are infected with toxigenic fungal strains. To predict mycotoxin formation, **MyToolBox** will develop models based on historical data, which will utilise measurement of parameters in real-time that will be used to model when fungi are most likely to produce mycotoxins. Such **predictions** will be used to **optimise harvesting times or optimise the use of biopesticide treatment to reduce contamination**. Competitive **biocontrol measures** using non-toxigenic fungi isolated locally will be developed for use in the EU, based on well-established techniques already in use in the US and Africa. **Sophisticated sorting techniques** using vision imaging will be implemented **to selectively remove contaminated units** from **dried fruit**. Innovative **(pre-) milling** techniques will result in reductions in levels of mycotoxins in wheat. **Innovative baking strategies** will be optimised to **study the fate of mycotoxins** in finished products.
***Integrated approaches**… are required to effectively control the incidence of mycotoxins in crops and reduce contamination throughout the feed and food chains.*	**MyToolBox recognises the substantial progress already made worldwide** in the area of prevention and control of mycotoxins and thus **builds on this existing knowledge**. Several recognised agricultural strategies including crop rotation, tillage practices, resistance breeding, fungicide application and less susceptible plant cultivars will be integrated to reduce the impact of fungal infection of crops and subsequent mycotoxin formation. As none of these strategies by themselves are able to reduce the impact of fungal infection, **an integrated approach will be pursued, which also takes climate change factors into account**. Pre- and post-harvest interventions will be assembled in a web-accessible MyToolBox platform in a way that will guide actors in the food and feed chains to identify and implement the most relevant strategies. The web-based MyToolBox platform will also **integrate HACCP and GAP protocols** specifically **addressing wheat, oats, maize, peanuts, dried fruit and animal feed**.
*Reducing the risk of mycotoxin contamination ….. bringing **technical, management and organisational solutions, incl. HACCP**.*	**MyToolBox will develop a series of decision support tools** which will be able to **forecast** potential fungal infection and subsequent mycotoxin occurrence, enabling actions such as **early harvesting or fungicide spraying to be undertaken at the optimum time** during the cereal growth cycle. A number of factors relevant to fungal infection and mycotoxin formation are already well-known and Good Agricultural Practice (GAP) combined with **Hazard Analysis Critical Control Points** (HACCP), have already been developed in some food and feed chains. Some of the technical innovations will require changes to existing standards to be able to fully benefit from the developments, and such changes will be sought as part of the project.
*… …. and also deal with the **safe use of contaminated batches.***	Existing EU Regulations, set maximum limits or guideline values for mycotoxins in **animal feed**. Contaminated batches above these limits must be destroyed and are an economic loss. In MyToolBox we will test the performance of specific mycotoxin-degrading microorganism strains and enzymes, on the **process-efficiency** and **degradation of mycotoxins** during **biogas** and **bioethanol** production. This will allow the **safe use of contaminated batches** for both, the efficient production of **biofuels** and of safe **DDGS** (Dried Distillers Grains with

SFS-13-2015	MyToolBox
	Solubles), to be used as a protein rich source in animal feed.
*Proposals should …. fall under the concept of **'multi-actor approach** and allow for adequate involvement of the farming sector.*	Innovative science alone will not achieve the desired outcomes, as it also requires the commitment of all actors along the food and feed supply chain to implement change. **The multi-actor approach in MyToolBox is exemplified by involvement of five prominent end users** representing the food industry (Barilla, Italy), food and crop energy (Südzucker, Germany), grain industry (Agro LV, Ukraine with operations in Poland), Dried fruit industry (TARIS, Turkey) and the maize based feed industry (D.O.O. Agrocentrum Becej, involving also **individual farmers from Serbia**). Hence, through our consortium **end users** from the **farming community**, agronomists, actors along the food & feed supply chains and professionals working in agriculture and food manufacturing **will collaborate closely with the Technology SMEs and with our partners from academia**. Formal evaluations of specific interventions will be conducted as well as beta testing of the web-based MyToolBox platform at an early stage of development to ensure user-friendly and fit-for-use design. Furthermore, **economists** will be involved **in assessing the cost-effectiveness of specific interventions**, and the **involvement of sociologists in assessing potential barriers to acceptance**. This multi-disciplinary project uniquely involves a diversity of actors along the entire food and feed chains essential to ensure that the practical objectives of substantial reductions in biological contamination and reductions in mycotoxin levels are achieved.
*Proposals should benefit **conventional and organic supply chains**….*	**Technical interventions**, primarily developed pre-harvest, **will be applicable to both conventional and organic supply chains**. Novel biopesticides such as salts, botanical extracts, plant defence elicitors and biocontrol agents, also suitable for use in the organic supply chain, will be investigated as alternatives to conventional triazole fungicides. **Novel cultural control mechanisms** will focus on studying crop debris management, applicable in **organic as well as conventional production** systems. This will include microbial agents to colonise and exclude *Fusarium* inoculum from plant debris and the use of biofumigation of crops as a novel cultural control strategy. **The conventional and organic farming community will be involved as partners, stakeholders and end users**. MyToolBox will connect with appropriate organic food certification bodies such as the 'Soil Association' to ensure that proposed interventions are organic-farming compliant.
…contribution to legislation…	MyToolBox will **anticipate legislative implications of the project outputs** and work with relevant bodies such as European Food Safety Authority (EFSA) and DG SANCO for standard setting (see 2.1. Expected Impacts). Moreover, the MyToolBox project aims to contribute to the standard setting for **authorisation of mycotoxin-detoxifying feed additives in China** in cooperation with FEFANA, the Chinese Ministry of Agriculture (MOA) and CAAS-FRI as an important aspect for the registration of mycotoxin detoxifying feed additives from the EU.
*In line with the objectives of the EU strategy for **international cooperation in research and innovation** and in particular with the implementation of the **EU-China dialogue**, proposals are encouraged to include third country participants, **especially those established in China**.*	MyToolBox has EU partners from **Austria** (Coordinator), **Netherlands**, **Germany**, **United Kingdom**, **Spain** and **Italy**, from associated countries of **Norway**, **Serbia, Turkey** and **Ukraine** (with additional operations in Poland) and two partners from **China**: The **Chinese Academy of Agricultural Sciences (CAAS)** a national, integrative agricultural scientific research organization with over 5,000 professional employees and the **Academy of State Administration of Grain (ASAG). From CAAS two institutes will be involved as partners**: The Feed Research Institute (CAAS-FRI) and the Institute of Agro-Products Processing Science and Technology (CAAS-IAPPST). Especially the latter will link its current research projects dealing with mycotoxin formation and its prevention and control in grains and cereals with MyToolBox. **CAAS, ASAG and their partners from the Chinese grain and peanut industry will contribute to the overall delivery of the MyToolBox**, as well as making specific technical contributions to three work packages (WP1, 2 & 3) and to the practical implementation and standard setting in WP4. Some technologies (e.g. CO_2 sensing during the storage and DON forecasting) developed in MyToolBox will be evaluated by these partners in cooperation with **end users in China**, which will ensure that the MyToolBox e-platform meets international aspirations in terms of scope and ease-of-use and may be "transplanted" to China. Chinese partners have extensive networks within their research community, and we propose to build on these networks to **form an EU-China Mycotoxin-Forum** (WP5). Moreover, through the **Project Advisory Board and partner ICC**, numerous organisations and experts from **third countries**, such as USA and Tanzania are involved in MyToolBox.

1.3 Concept and approach

Overall concept underpinning the project

Based on a thorough screening of existing measures to reduce moulds and mycotoxins in the feed and food chain, a consortium with leading experts from 23 partner organisations representing 11 countries (including 2 from China) and 43% industry participation (farmers, technology SMEs, food/feed industry, incl. 5 end users) have recognised significant gaps in the current knowledge and management strategies at different points of interventions for various food & feed commodities and have jointly developed the MyToolBox proposal. Previous studies have shown the combination of more than one control strategy in an integrated system to have a synergistic positive effect (i.e. the combined effect is greater than the sum of the parts)[13]. **By combining the developed control strategies with one another,** in combination **with other known control mechanisms** in a fully integrated mycotoxin management system will allow for maximum reductions in mycotoxins in food and feed. As a result, MyToolBox is not only based on a complete field-to-fork approach ranging from biocontrol and novel cultural control protocols (suitable for organic farming) at a **farmer´s level** to novel thermal processing procedures performed by a major end user from the food industry. **MyToolBox goes beyond the field-to-fork concept** and also considers safe use options of mycotoxin contaminated batches, such as microbial energy conversion to efficiently produce biogas and bioethanol in combination with safe DDGS. **The consideration of the entire chain *soil-field-crop-food processing-waste management-alternative energy* to ensure food & feed security and safety within a sustainable economic approach**, is a major motivation behind MyToolBox. The project will not only implement and develop a series of innovative integrated measures, which will enable a significant reduction in losses of crops due to fungal and mycotoxin contamination. Depending on the commodity and the type of intervention the achievable reduction in losses ranges from 20-90%. The **main idea of MyToolBox, distinguishing this proposal from previous efforts in the area mycotoxin reduction**, is to provide the recommended measures to the end users along the whole food and feed chain in a **web-based MyToolBox platform**. This unique e-platform will be an internet-based decision tool that provides users with access to static and dynamic information (see figure 1) concerning actions that can be taken to reduce fungal contamination and lead to reduction in levels of mycotoxins in the food and feed chains.

These actions will be based on existing knowledge as well as on the novel information/results obtained within the MyToolBox project. The ergonomic and user centred MyToolBox e-platform will be accessible through a PC, tablet and all commonly used mobile interfaces. Through a dedicated and experienced smart systems development team an interactive user-friendly platform will be designed to meet the needs of all actors along the food chain, whether these be farmers, processors, feed suppliers or manufacturers from the food and feed industry. Sound management of the innovation cycle will ensure that end users and stakeholders are mobilised in a co-creation approach, to ensure the MyToolBox e-platform is tailored to meet the needs of each and every end user by establishing a specific set of individual circumstances rather than simply providing generic advice. **MyToolBox is dedicated to the cereal (wheat, oats, and maize) and dried fruit (fig) supply chain in Europe and the peanut supply chain in China**. By step-wise interrogation of the end user, the crop of interest, the relevant stages in the food & feed chain, geographical location, storage conditions, transport and end-use will all be established. The web-accessible MyToolBox will prioritise possible actions (interventions) that can be taken, giving priority to recommending those actions which give the maximum possible reductions in mycotoxin levels at the lowest cost, i.e. focussing on the most cost-effective. The developed web-tool is not an endpoint but will be flexible and will continuously be improved.

MyToolBox will make use existing safety and agricultural models, Codex Alimentarius CoP, HACCP and GlobalGap. These models are based on science, are well established and, moreover, are already accepted by actors in the food and feed supply chain. HACCP and GlobalGap evolve in time and will thus allow the MyToolBox e-platform to stay up-to-date. The implementation of the well-established and accepted PDCA Deming cycle (plan-do-check-act) approach will allow the continuous improvement of the MyToolBox e-platform even beyond the duration of the project. To secure continuity and user acceptance, the platform will be designed for compatibility with all commonly used mobile device and the most popular operating systems (android, iPhone, windows). Access to the platform will be freely available to end users during the lifetime of the project, and its sustainability will be **achieved through a robust business model** that will be developed during the project (that could involve levying a nominal membership fee to users) and maintenance through the participating commercial ICT-related enterprise.

Based on preliminary **market analysis** and dialogue with our industrial partners, especially with end users and stakeholder associations, we know that there is strong interest among the majority of actors along the food and feed

[13] Blandino *et al*. 2012. Integrated strategies for the control of *Fusarium* head blight and deoxynivalenol contamination in winter wheat. Field Crops Res 133:139-149.

chains in a platform of **scientifically validated tools** such as those proposed by MyToolBox, as increasingly all actors in the food supply chain are aware of and dedicated to the reduction of mycotoxins. There is an increasing culture of cooperation and information sharing along the food/feed supply chain in the context of addressing the mycotoxin challenge.

The MyToolBox web-platform will be used on-site by all partners in the supply chain. This means that scientific information must be translated to clear unambiguous instructions. The **transdisciplinary challenges** will be tackled by the project team including scientific groups from the field of both agronomy and social sciences. Furthermore, **end users (farmers, food industry, etc.) and stakeholders will be involved in each step of the project**, which will be enabled through a professional communication campaign **including training in cooperation with major stakeholder associations** that will be designed in order to mobilise and secure the 'buy-in' of the necessary targets. The project partners have access to relevant data on crop growth, food processing, microbiology, climate, economics, human-decision-making, and communication in the European setting. Pictograms, photographs, diagrams and other imagery will be used extensively in the MyToolBox e-platform, both as an aid to achieving a 'user-friendly' product, as well as minimising the needs for extensive translation i.e. to open up access to a community that is not always necessarily English speaking.

Overall approach and methodology

Our ground breaking research activities will be undertaken within 6 Work Packages (WP). The mycotoxin-commodity combinations that will be addressed are the most prevalent ***Fusarium* mycotoxins** (deoxynivalenol (DON), T-2 and HT-2 toxins, zearalenone (ZEN) and fumonisins) in **wheat, oats, maize** and **animal feed chains** (incl. **DDGS**) and **aflatoxins** in **maize, peanuts and dried figs.** However, the project will also impact on other mycotoxins, such as **ochratoxin A in wheat** and **emerging mycotoxins**, such as enniatins. The overall approach, depicted in figure 1, involves pre-harvest interventions undertaken at farm level, together with post-harvest interventions undertaken from storage through to food processing into the finished food or feed product each with a set of specific objectives. Due to our comprehensive integrated approach and the required field trials, a **project duration of 4 years** is considered appropriate. The chosen approach will integrate scientific knowledge, translate and communicate it, apply it in cooperation with the end users and incorporate it in the simultaneously developed web-based MyToolBox platform, for use by all actors in the food and feed supply chain for the reduction of mycotoxin contamination in food and feed. This approach requires the input of scientific data, (anticipation of) legal requirements, but also the involvement of sociologists in assessing potential barriers for acceptance of the developed measures incl. the web-based My-ToolBox by the stakeholders.

The central part of this project is the development of the **MyToolBox e-platform: an interactive, internet based** tool that will facilitate optimal decision making for mycotoxin management (prevention, control, monitoring) in European feed and food chains. This decision support web platform is intended for use by various actors in the chain, including farmers, millers, processors and feed/food producers, as well as governmental authorities, whereby it provides these users with multiple information, tools and guidance concerning actions that can be taken to prevent and reduce fungal contamination and lead to reductions in levels of mycotoxins in the food and feed chains. MyToolBox will be configurable to the activities of each particular actor within the chain, and when there is no other option will enable the actor to direct contaminated batches to safe-use options (see figure 1).

In a co-creation approach, MyToolBox will be **designed, built and tested in close collaboration with the end users including farmers and the multiple stakeholder groups** involved in the project through ICC and the Advisory Board, such to ensure it fulfils all their needs. **Via an interactive User Interface, which will be accessible overall all mainstream interfaces (PCs, tables and smartphones), each of these actors can easily navigate and access the required information and decision support**, as well as input and output information from the system.

The web-based MyToolBox will be highly flexible and scalable so that it can be extended to other mycotoxins and other food and feed chains beyond the scope of the project. Over the course of the project, a business plan for future exploitation of the MyToolBox platform will be developed in order to arrive a sustainable business model.

The **MyToolBox platform** itself (figure 2) will integrate currently available information and make it readily accessible to the user. In relation to its architecture, the **core elements will consist of**:

a) **web platform back-end** (or engine), which consists of a cloud database (Postgres 9.3) housing both static content and dynamic content, as well as a security layer;
b) **web-based User Interface** (coded in HTML5 and JQuery -Javascript) that will be accessible over a standard PC and all mobile platforms (tablets, smartphones- iphone, android, windows…);

c) **Application Programming Interface (API)**- (PHP5) that enables data from the tools and applications developed in other work packages and tasks, as well as other external sources, to be integrated in the MyToolBox platform.

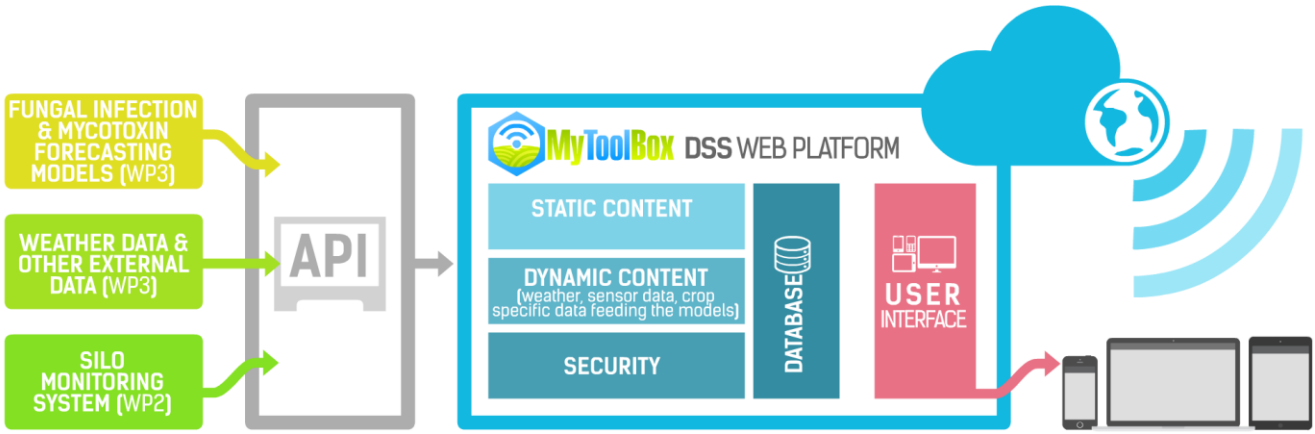

Figure 2: MyToolbox e-platform architecture

As can be seen, the MyToolbox platform consists of static and dynamic components as follows:

Static parts:

- Synthesised existing guidelines as e.g. published by the FAO for Good Agricultural Practice (GAP), Good Manufacturing Practices (GMP), as well as protocols for Hazard Analysis Critical Control Point (HACCP), sorting, milling and baking for mycotoxin management (prevention and control) in small grain cereals, maize and figs (WP3), both pre-harvest and post-harvest.
- Potential corrective actions and recommendations for alternative safe use in case certain action levels or EU legal limits are exceeded.
- Guidelines for cost-effective sampling and monitoring strategies for mycotoxin management by actors in the entire chain. These includes points in the chain at which samples need to be taken, amounts of samples, extent of pooling of samples, and types of measurement (dipstick, ELISA, chromatographic methods) to be used (WP3).

Dynamic parts (via the API):

- Optimal storage conditions, by using real-time sensing DSS (WP2&3).
- Forecasting models to predict fungal and mycotoxin contamination of small grain cereals and maize at early growth stages, for optimal application of biofungicides and optimal harvest time (WP1&3).
- Information from external sources, such as weather stations.

During the first year of the project, the web-based MyToolBox will make use of existing official information, guidelines and data, and will present this to the user in an easy-to-understand, interactive and synthesised format. As the results from WP1-4 become available during the project, they will be added and hence the MyToolBox e-platform will be updated. MyToolbox has an **easy to set up User Administration page**, whereby the user identifies himself, selects the crops and stage of chain, and the required information he/she wants to obtain from the system. The MyToolBox engine consists of static and dynamic components. The former includes **GAP/GMP guidelines, HACCP protocols**, and any other relevant information, presented in a practical and engaging format, by using images, visuals and videos whenever possible. Minimal text will be used making the system user-friendly irrespective of different language constraints. The latter relates to the dynamic real time (or very recent) data that is being entered into MyToolBox via an application programming interface (API) that will be built to allow MyToolBox to accept data from the other project developed applications- i.e., real time post-harvest storage management systems (WP2) and **forecasting models** for crop management (WP3), as well as their party systems, e.g. for obtaining up-to-date environmental data.

In the design and development of the MyToolBox, the so-called 'problem of implementation' of available information and the **agricultural decision support systems** (**DSSs**) (i.e., under-utilization due to both technical limitations of the DSSs and to user's attitude towards DSSs) will be specifically addressed by a **very robust approach to Innovation Management** (discussed in Section 3.2) to ensure that we work effectively with our end-user and stakeholder groups in order to deliver user centric and market-oriented results, whereby we will be:

(i) Focusing on the important problems of the complete chain with a systems approach;
(ii) Using automation and integration in data collection, and supporting flexible input efforts by the user;
(iii) Developing and validating fit-for-purpose, mathematical models;
(iv) Designing a user-friendly interface with navigation facilities within the web-based MyToolBox functionalities (e.g., decision trees) and providing complete and easy-to-understand information;
(v) Delivering MyToolBox through the Web and thereby enabling both continuous updating by the provider and flexible access by the user;
(vi) Cloud databases with the ability to handle large volumes of data and to enable interfacing with the MyToolbox platform over all mobile interfaces
(vii) Designing the MyToolBox e-platform with the goal of assisting the decision maker (by providing necessary information) rather than replacing the decision maker;
(viii) Involving potential users during the MyToolBox development and testing so as to obtain insight into how users make decisions (standard methodologies to engage with the users will be used, such as Focus Groups, Usability Tests, Card Sorting and Accessibility Analysis). This end-user involvement, and in particular farmers, in different stages of design, building and testing of MyToolbox is part of WP4;
(ix) Communicating the benefits of the web-accessible MyToolBox platform via seminars and demonstration;
(x) Involving all the potential stakeholders; and
(xi) Developing a two-way communication mode with the end-users, i.e., by combining "push" and "pull" systems.

Finally, in relation to our development methodologies, the software development of the MyToolbox e-platform will be following agile **Scrum and Test Driven Development methodologies**. Scrum is a highly powerful methodology for keeping the end-user engaged and with influence over the software development process and in particular to ensure user experience and interaction with the system is guaranteed. The Test Driven Development method is used for efficiently developing "almost-bug-free" software platforms because it is a programming methodology oriented to code "reliable-by-design" software.

Positioning of the project

In relation to where our project is positioned on the Technology Readiness Levels (TRLs), table 2 below provides a summary of the starting TRL of each of the different core project components and the expected TRL advance come the end of the action.

Table 2: TRLs advancement for the major knowledge/technology components of the project

Description	WP	TRL at project start	TRL at project end	Explanation
Crop management using biopesticides	1	TRL4	TRL6	Crop management advice for the use of biopesticides based on large field trials
Crop management using biofumigation	1	TRL3	TRL5	Crop management advice for the use of biofumigation based on large field trial
Atoxigenic *Aspergillus* strains for crop management in Serbia	1	TRL3	TRL5	Atoxigenic *Aspergillus* strains isolated from the area, available for use in maize
CO_2 silo monitoring solution	2	TRL4	TRL6	An integrated platform will be field validated in silos
Innovative fig sorting	2	TRL3	TRL5	Sorting technique validated in pilot plant
Innovative milling solution	2	TRL3	TRL5	Milling techniques validated in pilot plant
Novel enzymes in biofuel production	2	TRL4	TRL6	Mycotoxin degrading enzymes and microorganisms tested for efficient production of safe DDGS and biofuels
Web-based cloud platform-MyToolBox DSS	3	TRL5	TRL6/7	The MyToolBox e-platform will be built and scaled up from alpha to a beta version that will be validated with the target group under real conditions
Forecasting models available for mycotoxin prediction in cereals (including maize)	3	TRL3	TRL5	Forecasting models validated with data collected in project

Linkage with national and international research

The MyToolBox consortium will not work in isolation. **The project will build on the positive achievements of the substantial EU-funded research previously conducted** on mycotoxin contamination in the food and feed chain.

Several of MyToolBox consortium members have coordinated previous EU mycotoxin-related projects or led major WPs- in projects such as MYCORED (http://www.mycored.eu), QSAFFE (www.qsaffe.eu), EMTOX[14] and MYCOSPEC (http://mycospec.eu/de/). The consortium has also got **strong linkages** in research and innovation with many other national, European and international projects **around the world**. Some **40 countries** are represented through the participation of partner 3 (ICC) and via the linkage to the **International Society for Mycotoxicology (ISM)**, of which the **MyToolBox Coordinator** is secretary and 2016 President-Elect.

Thanks to the **numerous national research and innovation activities the MyToolBox consortium members** are currently involved in, the project will benefit from valuable input. E.g. to identify key components of Good Agricultural Practice to reduce mycotoxins in primary crops (UK national funding) and to determine field trial settings for *Aspergillus* isolates on the basis of Serbian agroclimatic conditions (Serbia national funding); through the Austrian project ***FUSARIUM*** (led by BOKU) state-of-the-art multi-toxin-methods for verification of the efficacy of the developed interventions at a mycotoxin level across WP1-3 will be available and new insights into wheat-*Fusarium*-interactions using novel Genomics and Metabolomics tools (WP1) will be provided to MyToolBox; insight into the metabolism of mycotoxins in plants and animals and the bioavailability of masked/conjugated mycotoxins will be considered for the studies on conjugation and reduction of mycotoxins through thermal processing and safe-use option fermentation processes in WP2 (Austria, national Christian Doppler Foundation through BOKU).

The **Project Advisory Board** (AB) will include various **international organisations (who already agreed to join the MyToolBox AB – see 3.3 and LOIs in the Annex) representing the European food and feed industry** (e.g. Copa-Cogeca, FEFAC, EUFETEC, INC, FEFANA, Global Food Safety Partnership) as well as **international experts** on mycotoxin prevention and control (e.g. from USDA, EFSA, FAO, CODEX, DG SANTE, ISM). The AB will link this project to other international networks facilitating further scientific and practical advice to MyToolBox. e.g. advisory board member Peter J. Cotty (USDA & University of Arizona) will allow us to access USDA´s expertise on atoxigenic strains of *A. flavus* as biological control agents and its projects funded by the Bill & Melinda Gates Foundation, relevant to WP1; **CAAS** and its Institute of Agro-Products Processing Science and Technology (CAAS-IAPPST), will link the results of their current research projects (~€8 Mio national funding), dealing with mycotoxin formation and its prevention and control in grains and cereals, to the research carried out in WP1 and 2. New relations and collaborations will be forged with other relevant projects over the course of the project with the view to maximising synergies and establishing an enduring scientific relationship with China.

Positioning of the project

The proposed innovative pre- and post-harvest reduction strategies along with the developed measures throughout the entire food and feed chain will **be tested and validated in the field by end users**. Moreover, the web-based decision support system (DSS) MyToolBox Platform will be deployable on-line and accessible through all mobile interfaces. It will be accompanied by precise procedures for the stakeholders on how to maintain and update the content of the MyToolBox e-platform in future. A major strength of the web-accessible MyToolBox is that it is not designed as an end-point but as a flexible system based on the 'continuous improvement' approach. This means that relevant new data can be included at any time to ensure that the system makes use of the newest insights. By the end of the project, the MyToolBox e-platform will provide all actors in the food/feed supply chain with solutions on how to control mycotoxins and on safe use options for contaminated batches.

Gender analysis

Gender issues are important in that women's work and women's lives all over the world have long been intertwined with food. To this end, in relation to business behaviour, gender analysis will also be important, especially in view of growing trends and commitments to see more women occupy high level and decision-making positions in industry. Consequently, **sex and gender analyses will be addressed throughout our work plan**, especially in relation to co-creation opportunities and ensuring females are included in the innovation cycle to give input and feedback to shape the research and its outputs where required. More specific attention will be paid to gender issues in WP2 in case of introduction of automated sorting techniques for figs in Turkey. Traditionally this work is carried out by hand by women and the required work circumstances, in the dark under UV light, is potentially harmful to their health. If we are successful with automated sorting techniques it would improve the health and safety of the female workforce. However, automation might push women out of work and, therefore, specific attention will be paid to **involving women in the operating of the automatic sorting systems.**

[14] http://www.moh.gov.cy/moh/sgl/emtox.nsf/emtox01_en/emtox01_en?OpenDocument

Moreover, we understand that inventions and developments that incorporate the smartest aspects of gender can open new markets and enable innovation in products, processes, services, or infrastructures. Indeed, products that meet the needs of complex and diverse user groups enhance global competitiveness and sustainability[15]. Inspired by how taking gender into account has yielded new research results and sparked creativity, opening new avenues for future research for those who give it serious attention[16]. During the work plan, gender analysis will be addressed in relation to the User Interface of the MyToolBox e-platform, whereby graphic designs of the interface will be consulted with both men and women during the design phase in WP3, and posterior alpha and beta versions will be trialled by both sexes. Females are also key decision makers in relation to household consumption patterns in many households, and as such communication activities targeting the general public and consumers will need to consider gender aspects. It is important to highlight that within the consortium itself, there is a **balanced mobilisation of women and men across the work packages**, working both in senior and early career positions. Moreover, a number of key management positions in the consortium are occupies by females: Scientific Manager, IPR Manager, Dissemination Manager, WP Leaders and Task Leaders. Gender issues will also be addressed during recruiting of new staff and in regard to career progression.

1.4 Ambition

Ground-breaking nature of concept and approach

The MyToolBox project will not only pursue mycotoxin reduction in the **field-to-fork approach,** specifically addressing wheat, oats, maize, peanuts and dried figs, but will also go beyond that by considering safe use options of mycotoxin contaminated batches such as microbial energy conversion to efficiently produce biogas, bioethanol and safe DDGS**.** Up to now, different agricultural strategies including crop rotation, tillage practices, fungicide application and the use of resistant plant cultivars have been used to reduce the impact of fungal infection of crops and subsequent mycotoxin formation. As none of these strategies by themselves have been able to efficiently reduce the impact of fungal infection, a novel integrated approach will be pursued, which also takes **climate change** into account. Consequently, MyToolBox will combine a series of integrated pre- and post-harvest measures (see "beyond state-of-the-art" below), which **shall enable a 20-90% reduction (depending on the type of commodity and intervention)** in losses of crops due to fungal and mycotoxin contamination. The consideration of the **entire chain *(soil-field-crop-food processing-waste management-alternative energy)*** to ensure food & feed security and safety within a sustainable economic environment, is the major ambition behind MyToolBox. The outcomes of the project including novel interventions will be assembled in **a web-accessible MyToolBox e-platform that** provides multiple information to **support decision-making in mycotoxin management** by all actors along the food chain (farmers, processors, feed suppliers or manufacturers from the food and feed industry). The **constant loop back** to the **MyToolBox end users** representing **SME and large agricultural industries** (Agrocentrum Becej and Agro LV., resp.) as well as the **food** (Barilla) and **feed** sector (end user & partner Agrocentrum and FEFANA as member of the Advisory Board) and the **crop energy industry** (Suedzucker) accompanied by an established **world-wide communication platform** (ICC) will ensure maximum interaction and practical applicability of the developed techniques and measures.

The web-based MyToolBox platform is based on the latest ICT technologies and platforms, as well as computational tools, and makes it possible to: a) integrate multiple and asynchronous information/data sources, both dynamic and static; a) perform complex analyses based on these multiple sources and advanced modelling techniques; and c) transform this complexity into simple and intuitive information to final users to support the decision-making process. The **first dynamic component of the platform** is based on **crop monitoring at the single plot level**. The crop environment is continuously monitored by sensors (e.g., weather and soil sensors) that send the environmental data to the remote MyToolBox server. At the same time, growers and advisors (i.e., the users) use handheld devices (such as tablets or smartphones) to monitor the status of the crop and look at the outbreaks of relevant pest and diseases. These data, together with plot-specific information (such as the variety, the previous crop, etc.), are used as inputs for mathematical models capable of predicting *Fusarium* Head blight (FHB) development and potential mycotoxin contamination. The model outputs are provided in real-time to users via an easy-to-use web-based user interface for informed decision-making about the optimal use of external inputs (e.g., fungicides) in such a way to preserve farm income and reducing the negative impacts on the environment. Prediction of mycotoxin contamination also provides information for the dynamic design of optimised sampling plans for: a) determination of mycotoxins in grain batches at the reception in the storage centre; and b) managing the post-harvest handling of grains accordingly.

[15] http://ec.europa.eu/research/science-society/gendered-innovations/index_en.cfm
[16] Schiebinger L. 2008. Gendered Innovations in Science and Engineering. ISBN: 978080475814.

The **second dynamic component** also involves the post-harvest handling of grains. This component is based on **real-time monitoring of the environment in storage facilities** by means of a wireless sensor network (e.g., temperature, humidity, CO_2). The CO_2 production is a more sensitive early indicator of the grain moisture content (moisture content [m.c.], wet weight basis) than temperature and intergranular relative humidity. Where grain/nut moisture contents are above the safe levels (>15%; >8.0% m.c. respectively) CO_2 will increase due to grain, mould and/or pest activity where hygiene is poor. The measured environmental data are transmitted to the MyToolBox central server and used to feed mathematical models predicting the potential dynamics of mycotoxin contamination under the specific storage conditions. These predictions are available for the user so that correct actions can be implemented to reduce the potential risks.

Interventions based on GAP and HACCP derived CCPs (as well as action limits and corrective measures for these CCPs) are the main component of the **static part of the MyToolBox plaform**. Information from multiple external sources, such international standards, and the relevant EU and China legislation is also part of the static component. By using MyToolBox, users can all easily access recommendations on the best choice of preventive and corrective measures, as well as useful information for decision-making about the management of specific problems such as, for instance, the potential corrective measures in case action levels or legal limits are exceeded, or recommendations for alternative safe use in case of mycotoxin contamination above certain limits. MyToolBox will also contain modelling **software that will calculate potential financial savings for the end user.** Access to the MyToolBox decision support e-platform will be freely available to end users during the lifetime of the project, and its sustainability will be achieved through a robust business model that will be developed during the project (that could be that of levying a nominal membership fee to users) and maintenance through the participating commercial ICT-related enterprise.

MyToolBox advance beyond state-of-the-art

The advances provided by MyToolBox are beyond the state-of-the-art pre- and post-harvest interventions as explained in table 3:

Table 3: MyToolBox - Advance beyond the state-of-the-art

Status quo	MyToolBox – Advance beyond the state-of-the-art
Minimum tillage, to minimise energy use and soil erosion, **increases *Fusarium* inoculum** within cereal rotations resulting in increased mycotoxin levels in harvested grains compared to ploughing[17].	More sustainable crop production using minimum tillage while maintaining a high organic matter within the soil. **Most promising available novel cultural control strategies** will be explored targeting the *Fusarium* inoculum within crop debris with a) *Brassica* for **biofumigation**[18] and b) **accelerated biodegradation**[19] by using an extract or competitive fungi[20].
No adequate prevention of *Fusarium* infection and the risk of fungicide resistance when using (**triazole**) **fungicides,** and the risk for endocrine disruption effects in humans[21].	The current strategy will be replaced by a **combined approach for the sustainable use of recently developed biopesticides**[22] (permitted within organic production systems) combined with agro-management tools such as water and soil conservation.
Appropriate biocontrol agents to reduce aflatoxin contamination in maize **are not** available for local use in South-East Europe.	To face **climate change** and thus **aflatoxin contamination** in EU maize, a) **appropriate genotypes of maize** will be selected with various traits (**drought resistance,** early mature plant cultivars) in field conditions that

[17]Edwards SG. 2004. Influence of agricultural practices on fusarium infection of cereals and subsequent contamination of grain by trichothecene mycotoxins. *Toxicology Letters* 153, 29-35; Bernhoft A. *et al*. 2013. Risk Assessment of Mycotoxins in Cereal Grain in Norway; Opinion of the Scientific Steering Committee of the Norwegian Scientific Committee for Food Safety: Oslo, Norway; Ulen B. *et al*. 2010. Soil tillage methods to control phosphorus loss and potential side-effects: a Scandinavian review. *Soil Use and Management*, 26:94-107.

[18] Ngala BM. *et al*. 2015. Biofumigation with *Brassica juncea, Raphanus sativus* and *Eruca sativa* for the management of field populations of the potato cyst nematode *Globodera pallida*. *Pest Management Science* 71:759-769; Perniola OS. *et al* 2014. Biological control of *Fusarium graminearum*: use of *Trichoderma* spp. and biofumigation with aerial part of Brassica juncea. *Revista De La Facultad De Ciencias Agrarias* 46:12.
Hofgaard IS. 2012 Reduction in the inoculum potential of *F. graminearum* on residues after treatment with Biowish crop or Trifender after sowing in spring (unpublished).

[20] Matarese F. *et al* . 2012. Biocontrol of *Fusarium* head blight: interactions between *Trichoderma* and mycotoxigenic *Fusarium*. Microbiology 158: 98-106; Palazzini JM. Et al. 2013. Biocontrol and population dynamics of *Fusarium* spp. on wheat stubble in Argentina. Plant Pathology 62: 859–866.

[21] Under Regulation 1107/2009 on the Placing of Plant Protection Products on the Market; Klix MB. et al. 2007. Comparison of the declining triazole sensitivity of *Gibberella zeae* and increased sensitivity achieved by advances in triazole fungicide development. *Crop Protection*, 26:683-690; Edwards SG & Godley NP. 2010. Reduction of *Fusarium* head blight and deoxynivalenol in wheat with early fungicide applications of prothioconazole. *Food Additives and Contaminants Part A*- 27:629-635; Hillocks RJ. 2012. Farming with fewer pesticides: EU pesticide review and resulting challenges for UK agriculture. *Crop Protection*, 31:85-93.

[22] Forrer HR. *et al*. 2014 *Fusarium* Head Blight control and prevention of mycotoxin contamination in wheat with botanicals and tannic acid. *Toxins*, 6:830-849.

Status quo	MyToolBox – Advance beyond the state-of-the-art
Due to **global warming** there is also a **need for drought resistant maize**[23] in South-East Europe. Very limited information is available for European maize genotypes.	are likely to reduce aflatoxin contamination and b) **atoxigenic *Aspergillus* strains** will be used to outcompete toxigenic *Aspergillus* strains. This will be a **novel adaptation within Europe** of proven technology developed in the USA and Africa and aided by collaboration with the international experts from those continents.
At present only temperature, and sometimes relative humidity (RH) are measured in silos periodically or in some cases remotely as an aid to **post-harvest management** of stored grain and other commodities[24].	The development of a **real time post-harvest Decision Support System** (DSS) based on a sensitive indicator of CO_2 (+ temp and RH) will provide silo managers, for the first time, with **accurate information to make informed decisions** immediately based on this platform technology. This component will be linked to the ICT dynamic part of the MyToolBox e-platform and will become an effective post-harvest tool to **minimise losses from mycotoxins.**
Sorting of dried figs to remove contaminated units is traditionally carried out **by hand**[25]. The required work circumstances, in the dark under UV light, is potentially harmful to the health of the employees[26].	**Non-invasive real time sorting of dried figs** for mycotoxin contamination, using visible and NIR spectral imaging and appropriate algorithms on lab-scale, will be refined and up-scaled in practice to achieve significant improvements compared to manual sorting with 5% or less probability of the batch average concentration of aflatoxin B_1 failing to meet the EU limits. It will also improve work circumstances dramatically.
Milling strategies, such as debranning and micronisation are efficient but **individual tissues of the grains are often not homogeneous and not well separated**. **By-products**, usually intended for animal feed but also exploitable for wholegrain food products, contain about 15% proteins, 40% starch and 40% fibre with levels of mycotoxins which can exceed legal limits[27].	**Innovative pre-milling and milling strategies,** will be combined, up scaled and implemented in MyToolBox and will allow **more accurate separation of grain tissues with** characterized (different) mycotoxin contamination levels. The **synergistic potential of debranning, micronisation** and **dry- or wet-turbo separation** will minimise the mycotoxin concentration in raw materials and the final wheat products destined for the consumer. We hypothesize a **decontamination degree of about 50%** for each component that could then be safely employed to produce any kind of cereal-based foods. Bread, pasta or biscuits could therefore be added with increased fibre content, ranging from 1 to 10% according to the technological requirements.
EFSA published terminology on **masked/ modified** mycotoxins in 2014 and the **need to obtain further data on the fate of these forms** during food and feed processing[28]. Most studies conducted on the fate of mycotoxins, are on **lab-scale with artificially contaminated material** at high levels that do not reflect practical conditions[29].	The fate of mycotoxins (incl. modified forms) in **naturally contaminated raw materials** (with simultaneous co-occurrence of different metabolites) will be monitored during thermal food processing at an industrial scale (at end user Barilla) to a) obtain **structural information of altered chemical structures** (using newest LC-HRMS and metabolomics techniques[30] and to b) **develop novel bakery production protocols,** controlling for instance fermentation, temperature/time and recipes, to develop a sound basis to establish process factors for risk assessment as part of a post-harvest intervention strategy.

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[28] European Food Safety Authority (EFSA) 2014. Opinion of the Scientific Panel on Contaminants in the Food Chain on a request from the Commission related to the presence of modified forms of certain mycotoxins in food and feed. The EFSA Journal, 12(12), 3916

[29] Valle-Algarra FM. *et al.* 2009. Changes in ochratoxin A and type A and type B trichothecenes contained in wheat flour during dough fermentation and bread making. Food Addit Contam Part A. 26:896-906; Voss KA. & Snook ME. 2010. Stability of the mycotoxin deoxynivalenol (DON) during the production of flour-based foods and wheat flake cereal. Food Addit Contam Part A. 27:1694-1700; Meca G. *et al.* 2013. Beauvericin degradation during bread and beer making. Food Control 34:1-8.

[30] Kluger B. et al. 2014. A novel stable isotope labelling assisted workflow for improved untargeted LC–HRMS based metabolomics research 10:754-769.

Maize based **DDGS** can be contaminated with high levels and a broad spectrum of mycotoxins[31] resulting in **losses in feed production**. Micro-organisms can degrade mycotoxins but the adequacy for a safe use to efficiently produce **biogas[32] and bioethanol[33] by fermentation of mycotoxin contaminated batches** must still be given.	The performance of **specific mycotoxin-degrading microorganism strains**[34] and **novel enzymes** provided by partner Biomin, on the detoxification of DON, ZEN and fumonisins (Commission Implementing Regulation (EU) Nr. 1016/2013) during biogas and bioethanol production will be evaluated. Recently available, **novel enzymes** will be used in **technical scale fermentation processes to study the use of mycotoxin contaminated batches in biofuel production** (partner: end user Suedzucker) and to investigate the effect on the (decreased) mycotoxin content in the **DDGS** by-product.
Previous projects and standards (GAP, HACCP) in the area of crop production[35] and mycotoxin reduction provided **either isolated solutions rather than integrated approaches** or **have not involved end users** representing the major food/feed/crop energy/ fruit and grain industry as full partners.	**A fully integrated multi-actor approach with end user involvement is envisaged:** The outcomes of the MyToolBox project with novel intervention strategies will be assembled in the **web-accessible MyToolBox decision support system (DSS)** or e-platform based on new ICTs, which provides multiple information to support decision-making in mycotoxin management by all actors along the food chain and enables to integrate multiple and asynchronous information/data sources, both dynamic and static. The incentives of all actors in the food/feed supply chain will be studied and used to build MyToolBox to maximise the opportunities for practical use of the platform by the stakeholders in and beyond the project.
Weather-driven **forecasting models** have been developed in some countries[36], **mostly to predict *Fusarium* head blight** and/or DON contamination in wheat. Most of these models are empirical, and few are mechanistic[37]. Some models for mycotoxin contamination in **maize** were also developed within the EU-project MYCORED on which MyToolBox will build on. DONCast is available in Europe but only accessible to end users linked to one specific agrochemical company (www.doncast.eu/). These models were validated to a limited extent, mainly in the regions where they have been developed[38].	For DON in wheat and aflatoxins in maize, available forecasting models will be evaluated for performance, and extended such they cover the most important wheat and maize growing areas in Europe. **Enlargement of existing models** to include additional regions, cereals (oats), fungi and mycotoxins. **Effective integration of both existing empirical and mechanistic models (see this table)** and their extensive cross-validation to provide the end users with a real time decision support system based on a **multiple-model platform** and **novel monitoring** techniques. This forecasting system will allow both short term decision regarding fungicide application and harvest timing, as well as long term planning.

Innovation potential

[31] Zachariasova M. *et al*. 2014. Occurrence of multiple mycotoxins in European feedingstuffs, assessment of dietary intake by farm animals. Anim Feed Sci Tech 193:124–140; Renewable fuels association (2014) URL: http://www.ethanolrfa.org/ (accessed 20.11.2014); Wu F. and Munkvold G.P. (2008) Mycotoxins in ethanol co-products: modeling economic impacts on the livestock industry and management strategies. J Agric Food Chem 56:3900-3911; Zhang Y. and Caupert J. (2012) Survey of mycotoxins in U.S. distiller's dried grains with solubles from 2009 to 2011. J Agric Food Chem 60:539-543; Schaafsma, AW et al. 2009 Mycotoxins in fuel ethanol co-products derived from maize: A mass balance for deoxynivalenol,. J.Science of Food and Agriculture. 89: 157-1580 .

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[33] Klosowski G. 2010 The influence of raw material contamination with mycotoxins on alcoholic fermentation indicators; Bioresource Technology 101: 3174-3152.

[34] Vekiru E. *et al.* 2010. Cleavage of Zearalenone by Trichosporon mycotoxinivorans to a Novel Nonestrogenic Metabolite. Applied and Environmental Microbiology, 76, 2353-2359.

[35] Coceral/Cogeca. 2010. European Good Hygiene Practices Guide for the collection, storage, trading and transport of cereals, oilseeds and protein crops, 95 pp.; Codex Alimentarius CAC/RCP 51-2003 Code of Practice for the Prevention and reduction of mycotoxin contamination in cereals, including Annexes on ochratoxin A, zearalenone, fumonisins and tricothecenes; EU Recommendation (EU) No 583/2006 on the prevention and reduction of *Fusarium* toxins in cereals and cereal products; HGCA. 2002. Practical guidelines to minimise mycotoxin development in UK cereals, in line with forthcoming EU legislation, using the correct agronomic techniques and grain storage management. Project report No.289, 65 pp.; Van der Fels-Klerx HJ. & Booij CJH. 2010. Managing mycotoxins in the cereal supply chain using geographic information. J Food Prot. 73:1153-1159.

[36] Van der Fels-Klerx HJ *et al*. 2012. A model to estimate deoxynivalenol contamination of wheat in north western Europe for use in climate change impact assessment. J Food Prot 75:1099-1106; Van der Fels-Klerx HJ. *et al*. 2010. Descriptive modeling to predict deoxynivalenol in winter wheat in The Netherlands. Food Add. Contam. Part A 27:636-643; Rossi V. *et al*. 2003. A model estimating the risk of *Fusarium* head blight on wheat. Bull. OEPP/EPPO Bull. 33:421–425; Landschoot S. *et al*. 2012. Ordinal regression models for predicting deoxynivalenol in winter wheat. Plant Pathology 62:1319-1329. Hooker DC. *et al*. 2002. Using weather variables pre- and post-heading to predict deoxynivalenol content in winter wheat. Plant Disease. 86:611–619; Schaafsma AW. & Hooker DC. 2007. Climatic models to predict occurrence of *Fusarium* toxins in wheat and maize. Int. J. Food Microbiol. 119:116–25.

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[38] Prandini A. *et al*. 2009. Review of predictive models for *Fusarium* head blight and related mycotoxin contamination in wheat. Food Chem. Toxicol. 47: 927–31.

The MyToolBox project has high innovation potential and will produce new products and services that can be marketed and patented, as it will lead to:

- A user-friendly and sustainable multi-actor decision support system (DSS) to reduce mycotoxin contamination in the food and feed supply chain, that takes into consideration the user requirements and desired user experience of our target end-users (that will include farmers, grain silo managers, food and feed production managers, among others)
- Effective production and application of atoxigenic fungal strains to use for biocompetitive exclusion of toxigenic strains in the field in the EU.
- Application of "next generation" advanced models to forecast fungal growth and mycotoxin formation in the field.
- Procurement of alternatives to conventional fungicides (bio-pesticides), to effectively combat fungal growth in the field.
- Availability of real-time environmental monitoring systems in storage facilities for post-harvest mycotoxin control.
- Applicability of innovative sorting, milling and baking strategies to increase the safe use of mycotoxin-contaminated batches for food and feed.
- Insight into the fate of mycotoxins during industrial milling and baking operations to facilitate science-based decisions about possible derogations and fine-tuning of current EU-legislation on mycotoxins.
- Optimised processes for mycotoxin reduction in DDGS, to enable better valorisation of these materials for feed use and efficient production of biofuels.
- Updated standards for Good Agricultural Practice and safe food and feed storage and production.

Exploitation of the project results is the most direct way of gaining commercial benefits, such as new products/services, increased profitability and industry growth, as well as socio-economic benefits such as sustainability, jobs and competitiveness. **A strong impact-oriented approach to our project** ensures that from the outset activities are in place to plan for the future use and sustainability of the project's outputs. As such, **a sound business model will be developed** in order to ensure the viability of the proposed platform as a sustained MyToolBox e-platform that will be used for many years to come in the European crop production, food and feed supply chains and this will be fundamental for ensuring the long-term impacts of the project.

To conclude, **MyToolBox is clearly an ambitious proposal** with many goals set at different points of possible interventions at the pre- and post-harvest levels and through-out the food chain. It is our **intention to co-operate with** and **to avoid duplication of research carried out in running major mycotoxin reduction programmes** (especially in the area of resistance breeding against *Fusarium* Head Blight) and to come up with a proposal in cooperation with major end users and SMEs which addresses commonly identified scientific and **technical gaps in mycotoxin reduction** in various stages within the food and feed chain **as identified by the end users**. **We are aware of the ambitious nature of our project goals but are convinced that all the goals defined in the proposed project MyToolBox are realistic for 3 major reasons**: 1) The multi-actor based consortium involves the relevant international experts from farming, industry and academia, 2) the current state-of-the-art and the knowledge gained in recent (inter-)national projects will be taken into account and cooperation will be realised with other ongoing activities in the area of mycotoxin reduction and 3) for numerous objectives set in MyToolBox we will not have to start from scratch but **will build on existing knowledge and achievements of the project partners being involved in MyToolBox** as e.g., a pre-selection of biopesticides (partner Axeb), significant expertise and high-performance industrial instrumentation in pre-milling (Barilla) or novel enzymes to be employed in biofuel production as a safe option of contaminated batches (Biomin, Südzucker) – **all being available to MyToolBox from the very beginning of the project.**

2. Impact

2.1 Expected impacts

MyToolBox mobilises a **multi-actor partnership** (farmers, technology SMEs, food and feed industry, policy stakeholders and academia) to develop novel interventions **aimed at achieving a 20-90% reduction in crop losses due to fungal and mycotoxin contamination**. MyToolBox will not only pursue a field-to-fork approach but will also consider safe use options of contaminated batches, such as the **efficient production of biofuels**. A major component of MyToolBox, which also distinguishes this proposal from previous efforts in the area mycotoxin reduction, is to **provide the recommended measures to the end users** along the food and feed chain **in a web-based MyToolBox**. Cutting edge research will result in new interventions, which will be integrated together with existing measures in the

MyToolBox that will guide the end user as to the most effective measure(s) to be taken by farmers to reduce crop losses. We will focus on small grain cereals, maize, peanuts and dried figs, applicable to agricultural conditions in EU and China. **Crop losses using existing practices will be compared with crop losses after novel pre-harvest interventions** including investigation of genetic resistance to fungal infection, cultural control, the use of novel biopesticides (organic-farming compliant), competitive biocontrol treatment and development of forecasting models to predict mycotoxin contamination. Research into post-harvest measures including real-time monitoring during storage (on farm grain silos, silo stores in food processing plants, silo centres, etc.), innovative sorting of crops using vision-technology and novel milling technology will enable cereals with higher mycotoxin levels to be processed without breaching regulatory limits in finished products. Research into the effects of baking on mycotoxin levels will provide a better understanding of process factors which influence both the mycotoxin levels and the formation of conjugated and other modified forms of these toxic secondary metabolites. Involvement of leading institutions from China is aimed at establishing a sustainable cooperation in mycotoxin research between the EU and China.

The MyToolBox project will contribute to each of the expected impacts of *SFS-13-2015* as follows:

a) A toolkit (plant resources, management tools, technologies) to effectively monitor and reduce the incidence of mycotoxin contaminants in crop production as well as in the food and feed supply chains

MyToolBox mobilises a multi-actor and multi-disciplinary partnership **throughout the food and feed chain** (with 43% industry participation including 5 end users and Chinese institutions who collaborate closely with farmers and other relevant stakeholders from the industry) to develop novel interventions aimed at achieving an **increased volume of safe food and feed in the EU** by a **reduction in crop losses** due to biological (fungal) and mycotoxin contamination, of at least **20% and up to 90% - depending on the type of commodity and point of intervention-** due to biological (fungal) and mycotoxin contamination. Cutting edge research will result in new and adapted interventions, which will be integrated together with existing measures in a web-based MyToolBox platform. It will represent a **one-stop-shop MyToolBox web-based platform** and will consist of information and advisory modules, management tools, as well as technologies that equip farmers and all other actors in the food and feed supply chain with systematic/pro-active, **cost effective, affordable approaches for effective monitoring and reducing the incidences of mycotoxin contaminants in crops**, food and feed, and reducing the dependence on expensive, wasteful end-point testing/segregation. Since MyToolBox will be a sustainable web-based platform that will be readily accessible by all actors in the EU food and feed chain beyond the project, this **impact will last for many years to come** in the EU and will ensure the long-term impacts of the project.

The **pre-harvest impact** will be measured by comparing the achieved reduction in mycotoxin content and increase in yield in field trials between the conventional agricultural systems to the new designed systems. The use of **biopesticides** and **biofumigation** or **debris treatment** in EU field trials should **each reduce the deoxynivalenol level in wheat by 50%**[39], the combined effect is yet unknown. In EU field trials the **aflatoxin content of resistant maize cultivars** will be compared to the aflatoxin content of susceptible cultivars, which is expected to be lower than the **70%** achieved by others[40]. The aflatoxin contents of maize (EU) and peanuts (China) after treatment with atoxigenic fungal strains will be compared to non-treated crops in field trials. The aim is to achieve a **90% reduction in aflatoxin contamination**[41].

The **post-harvest impact will** be measured by comparing the achieved reduction in mycotoxin content of cereals in trials in grain silo's using real-time sensors compared to conventional practices. This would contribute to approx. 15% savings (~ 100 Mio Euro/year). While it is impossible to accurately determine the reduction in costs, it is estimated that **cost for monitoring will be reduced by at least 30%** in all stages of the food and feed supply chain. This impact will be measured by comparing the estimated current cost for monitoring by EU governments and industry to the new designed monitoring models.

The novel non-invasive **real time sorting of dried fruit (figs)** will achieve a 5% or less probability of the batch average concentration of aflatoxin B_1 failing to meet the EU limits as compared to conventional sorting. The **developed innovative milling and baking techniques** will result in, resp. **50 and 15% mycotoxin reductions** in the finished food products. The effect of the use of **enzymes on mycotoxin reduction** during biofuel production will be assessed in trials with and without adding the enzymes. Impact will be a reduction of the DON and fumonisin content to 4% of the mycotoxin level (DON) achieved by conventional production (Project ENERCYCLE).

b) Contribution to legislation and standard setting

[39] Paul P.A. *et al.* 2008 http://apsjournals.apsnet.org/doi/abs/10.1094/PHYTO-98-9-0999.
[40] Windham G.L. and Williams W.P. (1998) http://apsjournals.apsnet.org/doi/abs/10.1094/PDIS.1998.82.3.281.
[41] Dorner J.W. 2004. http://informahealthcare.com/doi/abs/10.1081/TXR-200027877.

At a European level, MyToolBox will **anticipate potential legislative implications of the project outputs** and therefore work with relevant bodies such as EFSA, DG SANTE and the Chinese Ministry of Agriculture (MOA) for standard setting. Compliance assessment will be undertaken of maximum levels in the feed/food chain for selected examples which will lead to identification of points to be changed in legislation/standards in order to implement the MyToolBox e-platform and analysis/discussion of benefits and drawbacks and making regulatory impact assessment.

Our pan-European approach will have impact on European cohesion, in particular in raising food safety standards in New Member States, as well as in candidate Countries, such as **Serbia** which faced severe food safety and social problems due to aflatoxin contamination in maize in 2013. **Our pre-harvest research on maize will help Serbian farmers to conform to EU legal limits, and hence aiding Serbia's aim to increase food and feed safety and to facilitate entry to the EU.**

The project will also provide new information on the presence of mycotoxins and their modified forms that are formed during food processing which will be forwarded to EFSA as a relevant contribution for the overall risk assessment of mycotoxins in foods.

Together with our partner CAAS-FRI from the Chinese Academy of Agricultural Sciences and the **Chinese Ministry of Agriculture (MOA)**, under the auspices of FEFANA, we aim to contribute to the standard settings for authorization of mycotoxin-detoxifying feed additives in China. **Current guidelines (e.g. from EFSA) for the registration of detoxifying feed additives in the European Union will serve as examples for possible adaptions of Chinese legislation. As a basis** *in vivo* experiments will be performed to assess the efficacy of detoxifying feed additives under local Chinese conditions based on the guidelines established in the EU. Thus, our activities **will help European producers** of mycotoxin-degrading feed additives to gain broader access to the Chinese market as well as to contribute to an improvement of Chinese legislation.

c) Safe use options for contaminated batches

Implementing MyToolBox will reduce the number of incidents and crises in EU related to mycotoxin contamination in crops, food and feed, thus limiting crop losses because it enables all actors in the farming, food and feed supply chain to effectively monitor contamination and use the **decision support tools** to take immediate actions to prevent further deterioration and to **direct contaminated batches** to appropriate use. Use of MyToolBox in countries exporting to the EU will lead to less border rejections in EU, thus saving food and feed, besides from improving food and feed sustainability in the country of export. Despite increased crop production with an increased part of food and feed quality, a certain volume of crop will remain unfit for consumption. Part of this volume can be brought to food or feed quality by applying the **novel sorting and milling techniques** developed in MyToolBox. The impact will be measured by comparing the volume fit for food/feed between the conventional and novel processing techniques. The still remaining volume of mycotoxin contaminated grains unfit for further physical processing for food/feed will be subjected to **innovative bioethanol production using e.g. novel mycotoxin degrading enzymes** to allow the DDGS by-product still to be used as feed component. Impact of this measure will be a **10-30% improved efficiency of the biofuel production** due to the improved fermentation process which will at the same time lead to low levels or even **mycotoxin free DDGS**. The outcome of these MyToolBox studies could also **positively contribute to the current food vs fuel debate** as mycotoxin contaminated batches could then be safely employed to produce biofuels on the one hand and high quality DDGS on the other hand.

d) Reduced occurrence of mycotoxin contamination in crops in order to improve productivity and competitiveness of European agriculture and food sectors

Maintaining the competitiveness of the industry at European-level is critical for economy, growth and jobs in many rural locations that are dependent on a healthy agro-food industry. Increasing the competitiveness in the food and agricultural sector is of key strategic interest to the EU, which in 2010 invested €43.8 billion into Agricultural expenditure and direct aid[42]. Moreover, **by 2050, food production will need to be 70% higher** than at present to satisfy the world demand, whilst faced with a limited increase in arable land and other natural resource constraints[43]. Productivity therefore needs to be increased. Currently, the economic consequences of mycotoxins in terms of food and feed losses, increase of the cost of inspection and analyses, etc., are considerable, and lead to costs in billions of € annually to the European and global food supply chain[44,45,46].

[42] http://ec.europa.eu/budget/library/publications/budget_in_fig/dep_eu_budg_2010_en.pdf.
[43] http://www.ecpa.eu/event/food/outlook-eu-agriculture-causes-constraints-and-challenges-productivity
[44] Waller M. and Williams G. (1996) Mycotoxin economics. Feed Manag 47: 23-25.
[45] USDA (1999) Grain fungal diseases and Mycotoxin. Reference pp 6-28.

Without doubt, the pan-European deployment of the outputs of the MyToolBox project will have a marked impact on **improving the productivity** of our EU agriculture and food and feed sector, by providing pre-harvest interventions undertaken at farm level, together with post-harvest interventions undertaken from storage through to processing into the finished food or feed product to prevent yield and monetary losses due to mycotoxin contamination, through prevention, reduction and reuse strategies. In economic and job creation terms, should the deployment of MyToolBox achieve a reduction of 5% of the annual EU losses due to mycotoxins (estimated at €2 bn), **we could save the industry and the economy €100 million per year, the equivalent of 830 jobs directly** (based on average capital intensity in the industrial sector of €120k per job per annum[47]). To this end, from 2020 (when the MyToolBox platform will be widely deployed) to 2030, the project could have contributed to reductions in EU mycotoxin losses of €1 bn, safeguarding over 8300 jobs. In relation to wealth and new job creation attributed to growth, should the EU be able to add even 1 cent per ton to the value of the cereals it produces, this would generate €28.5 M per annum, which would yield €285 M by 2030, leading to the creation of some 2375 new jobs (based on the 2010 volumes of cereals produced- 285.2 million tons of cereals).

If we consider, for example, that **wheat and maize** account for 68% of cereals grown within the EU with an annual production of over 200 million tons. With annual losses due to mycotoxins estimated at 5-10%, this accounts for a loss of €1.2-2.4 billion for wheat alone. **A reduction of losses** of even 1% would save €12-24 million annually, but would also provide an additional 66-133 thousand tons of wheat, which would have major implications for improved productivity, food security, as well as reduce the carbon footprint of EU agriculture. With average **reduction by 20-50%** of current losses in wheat due to *Fusarium* mycotoxins alone, which is the **impact achievable through MyToolBox for wheat,** yielding cost savings in the multi-millions € per year. The effects as a result of unexpected contamination of maize with **aflatoxins** can be even more severe: the long term costs for aflatoxins contaminated maize based feed are estimated to cost Serbia hundreds of millions € per year. It is expected that **post-harvest losses**, due to mould spoilage and mycotoxins, pests in food/feed chains, in the EU will be **reduced by at least 15%,** representing value of £75M-100. In addition, the achievable impact of MyToolBox could **reduce the border rejections by at least 30%** due to prevention of moulds/mycotoxins during shipping transport (current mould/mycotoxins represent 30-40% of all EU border rejections, representing a losses of ca. 30-40 Mio€ per annum).

In 2013 about 5.2 million m³ of **bioethanol** were produced in Europe (and 88.8 million m³ worldwide), with almost exclusively grains as a feedstock, resulting in a production of about **4.2 million tons of DDGS** in Europe. We estimate the **losses in feed production due to mycotoxin contaminated DDGS to be in the range of €15-20 million annually for the swine industry alone. The major end user,** Suedzucker, one the world´s greatest producers of sugar and operating one of Europe´s largest bioethanol production plants, **participates in MyToolBox based on the expected** impact of reduced waste by **a**) significantly lowering mycotoxin levels in DDGS, **b)** 10-30% improved efficiency of the bioethanol production **c)** increased market value (>10%) of mycotoxin free DDGS. In short, the potential for reducing mycotoxin-related yield- and economic- losses in the EU agriculture and food sector **will directly impact on productivity and competitiveness and will open up the potential for further investments and innovation** in these sectors as a result of increased profitability.

e) Reduced risks for human and animal health, and thus increase consumer confidence in agro-food products

The impact of MyToolBox will be a decreased **risk for human and animal health in the EU,** due to a **reduction of the mycotoxin levels in food and feed** and, at the same time an **increase in volume of safe food and feed** produced in the EU**.** The benefits of MyToolBox for **consumer and market confidence in agro-food products** will be high due to **effective communication** of the results of this project to the general public, retailers, policy makers and stakeholders. With between 25 and 50% of global food and feed sources affected by mycotoxin contamination, this is a major food safety hazard. As food safety is the key criteria by which consumers judge quality[48], it needs to be safeguarded from farm-to-fork.

Impacts of MyToolBox will be **the reduced number of incidents** and **more transparent supply chain**, due to improving pre- and post-harvest measures and a more reliable monitoring of crops for mycotoxin contamination and, most important, better communication among the EU-28. The impact will lead to a reduction in RASFF notifications thus saving a lot of effort and money on control and issue management. Moreover, **it will improve consumer trust** in imported food and feed. EFSA will be member of the advisory board to safeguard the interests of the consumer.

[46] Swamy H.V.L.N. *et al.* 2004. Effects of feeding blends of grains naturally contaminated with *Fusarium* mycotoxins on growth and immunological parameters of broiler chickens. Poultry Sci 83: 533-543.
[47] Apel R. 2004. http://larouchepub.com/other/2004/3135germany_jobs.html.
[48] Anonymous 2005. http://www.foodproductiondaily.com/Safety-Regulation/Consumer-trust-vital-says-Czech-food-safety-head.

If we consider the negative impacts for **animal health**, and indeed the directly related economic losses, the US pork industry alone is estimated to sustain an annual loss of 147 Million USD from reduced weight gain as a result of swine feed that is contaminated with AF, ZEN, DON, FB and OTA[49]. Estimates of the annual economic impacts of economic impacts of livestock losses in the developing world (Philippines, Thailand and Indonesia) as a consequence of aflatoxins alone are ~200 Million USD[50]. Beyond these economic losses, there are also the negative animal welfare impacts of poor animal health. MyToolBox will contribute to reducing the economic losses and increasing the wellbeing of livestock.

f) Contribution to long-term collaboration with China on food safety

As part of our long-term China collaboration strategy, we will exchange information and share experiences on the production, formulation and application of biocontrol agents (partner CAAS-IAPPST), forecasting models and on the use of on-line sensing for environmental factors during the storage of grains and peanuts. This will contribute to the impact of creating transparent supply chains for food and feed both in the EU and in China. In cooperation with the Coordinator, our Chinese partner Prof. Liu Yang (CAAS-IAPPST), who chairs the Division of Mycotoxicology of the Mycological Society of China will **set up an EU-China-Mycotoxin Forum as a dialogue platform** between China and the EU to provide building-blocks for further EU-China dialogue across a wider range of food safety topics. The EU-China Mycotoxin-Forum will be chaired by Prof. Yang and the MyToolBox coordinator Prof. Krska and will receive support from ICC and its current president Prof. Fengcheng Wang (Partner 3). Cooperation will be sought with the International Society for Mycotoxicology (ISM)- the MyToolBox-Coordinator is General Secretary of ISM - and the Division of Mycotoxicology, Mycological Society of CAAS. It will draw together research achievements on mycotoxin prevention and control, and subsequently provide building-blocks for further EU-China dialogue on mycotoxin prevention. Measures will be implemented to ensure that the EU-China Forum is sustainable beyond the lifetime of the project. Our EU-Sino partnership will also **contribute to the standard settings for authorization of mycotoxin-detoxifying feed additives in China,** whereby current EU guidelines for the registration of detoxifying feed additives will serve as examples for possible adaptions of Chinese legislation. This is important as currently, ~475 million pigs are farmed in China, representing nearly 60% of the world´s total pig population. Since up to 24% of complete feed for pigs have exceeded even the maximum limits set for DON levels by the Chinese authorities[51], the potential of mycotoxin-degrading feed additives on the Chinese market is clearly recognizable. **A need for more than 54,000 metric tons of detoxifying agents can be estimated in China.** The EU-China partnerships established through the MyToolBox project will contribute to creating transparent supply chains for food and feed imported in the EU and exported to China, which will have far reaching implications for food safety and consumer confidence.

Improving innovation capacity and the integration of new knowledge

Innovations in pre- and post-harvest strategies and safe use options will be developed in the MyToolBox project. These innovations and new knowledge will be combined with recently developed innovations, into new innovations and insights that are translated to clear instructions in the MyToolBox decision strategies. This strategy will be applicable beyond the project and to other areas where contaminants play a role.

MyToolBox makes innovations for mycotoxin prevention available to all actors in the food and feed chain, thus **strengthening the competiveness and growth** of the companies while meeting the needs for EU and global markets. The increase in available high quality food and feed will result in more jobs available in the EU and increased export of food and feed. Moreover, the export of mycotoxin binders is expected to increase thus leading to world-wide improvement of food and feed.

Key to improving the innovation and the integration of new knowledge in the EU crop and food production and supply chain in order to better equip farmers and the industry to reduce crop and food losses due to mycotoxin contamination is our **post-project strategy for the dissemination and exploitation** of all the results generated during the project. As will be discussed in the next section, the MyToolBox consortium has a clear vision and strong capacity to bring the project innovations to the market. The consortium has strong industrial European partners from all relevant parts of the food and feed chain. Furthermore, the members of the scientific advisory board and the stakeholders are strong representatives of the food and feed sector. MyToolBox will develop a sound **business case** to take the MyToolBox e-platform to the market beyond the project period. The aim is to develop a sustainable business model that enables the deployments, maintenance and further development and expansion of the MyToolBox e-platform in future. The MyToolBox consortium has good participation of industrial representatives from along the value chain, who will ensure

[49] Wu and Munkvold, 2008. J. of Agric. Food Chem 2008.
[50] Lubulwa and Davis, 1994. Cited in Wu 2004.
[51] Li X.Y. *et al.* 2014. Occurrence of mycotoxins in feed ingredients and complete feeds obtained from the Beijing region of China. J Animal Sci Biotechnol 5: 37-44.

that the MyToolBox and the business plan are relevant and viable. The involvement of both the industrial partners and the stakeholders will guarantee quick acceptance of the MyToolBox by other partners beyond the project. The project fully addresses the market and societal needs and will significantly contribute to further implementation of safe use options in regulatory context in EU as well as in China.

Barriers and obstacles, and any framework conditions (such as regulation and standards), that may determine whether and to what extent the expected impacts will be achieved

The MyToolBox consortium recognises the difficulties of acceptance of new agricultural practices and detoxification procedures by farmers and regulators and the complexity of multiple stakeholder dialogues. To this end, within WP4 (T4.1) we will identify and engage with farmers, **retailers, regulators, quality managers of multiple food and feed suppliers and consumers in order to establish their needs and constraints**. Besides assessing potential strength, weaknesses opportunities, and threats (SWOT), from the stakeholders' perspective, the level of understanding and acceptability of different developed safe use concepts and perceived information needs will also be explored, as will any regulatory impacts. A SWOT analysis will help to understand the barriers and will identify the conditions for maximal acceptance. Importantly, MyToolBox already benefits from an existing cooperation with DG SANTE and the European Food Safety Authority and a number of partners have experience in relation to the application process for seeking regulatory approvals. Furthermore, WP4 will be fully dedicated to the **involvement of all stakeholders** to ensure acceptance by the proposed farmer and industrial partners in the food and feed chain. The achievement of the expected impacts will be monitored throughout the project. The impacts have been clearly defined with measures for monitoring.

It is noteworthy that the MyToolBox results will be an enabler to the adhere to the EU crop and food production and supply chain to comply with EU Regulations and Recommendations 1881/2006, 1126/2007,165/2010 and 165/2013 that set maximum levels for mycotoxins. This should contribute favourably to numerous EU policies and Directives that outline requirements guaranteeing the standards of all European products. Moreover, this project will contribute to Rural Development, Food Security and Sustainability policies, and it will support an emerging strategy: a move from food security to self-sufficient food security, one of the long terms goals of the Treaty of Rome. To ensure wide dissemination of the results, and to plan for post-project exploitation, work package (WP5) is aimed at targeted dissemination of MyToolBox results to the relevant international discussions and committees, including China, and to developing a sustainable business model to ensure the continuity and growth of the MyToolBox e-platform.

2.2 Measures to maximise impact

Dissemination and exploitation of results and outline of dissemination plan

The MyToolBox project will develop a series of integrated measures that will be combined in the decision model, which enables a significant reduction in losses of crops due to fungal and mycotoxin contamination. The exploitation potential of the project results is very promising. As such, our project is backed with very sound strategies for **the dissemination and the exploitation of the project results**, to maximise the societal, innovation and competitive impacts.

Our dissemination plan encompasses a comprehensive series of actions to diffuse the non-confidential results that are generated during the project to as wide and relevant an audience as possible in order to heighten awareness about the results, to contribute to knowledge building in Europe, as well as to ensure that **as many actors from the crop and food chains and consumers/citizens as possible** have access to these results. This includes farmers/crop producers, collectors, food processers, retailers, consumers, as well as scientists, policy makers, mass media (TV, radio and the written press) and multipliers for addressing the general public, early stage researchers, including young people. In keeping with the general profile of each target audience, **the most appropriate dissemination tools and channels will be selected for targeting each and the expected impacts of our dissemination activities have been defined (table 4).**

Table 4: Outline of the MyToolBox Dissemination Plan

Target Audience	Tools & Channels	Expected Impacts
Farmers/Crop producers	Project corporate identity tools (logo, label)	Generation of awareness about MyToolBox and its benefits for reducing crop and food losses due to mycotoxin contamination among EU farmers, crop and food production and supply chain and other industry stakeholders in order to stimulate interest in the project results and to
Crop collectors/storage	Project Website	
Food processors	Project Leaflets	
Retailers and consumers	Project Posters	
	Video clip (for YouTube, project website, etc.)	
	Powerpoint Presentations	

Industry Multipliers-Sectoral Associations (e.g. Copa Cogeca, also member of the stakeholders group, Standardisation bodies and policy stakeholders, national food safety authorities, EFSA, EC, national governmental bodies, ICC, AACCI, CEN, ISO, WHO, FAO, CODEX	Press releases	develop novel interventions to increase volume of safe food and feed in the EU.
General public and Mass media	***Trade fairs and exhibitions***: Alimentaria (ES)); Anuga (DE); The Cereals Event (UK); SIMA (FR); GrainTech (EG), Agrotech (PL); Growtech (TR), GFIA (AE). Agritech 2017 (IL); Agritech 2016 (IN);	Effective knowledge sharing among the research and scientific communities, which will contribute to the ERA, and to the EU recovery package and the EU 2020 strategy for smart, sustainable and inclusive growth.
Research and academic stakeholders	***Conferences, seminars and workshops***: ICC International conferences; Conferences of the International Society for Mycotoxicology; MoniQA International Conferences; European *Fusarium* seminar, Annual World Food and Innovation Forum; Nordic Baltic *Fusarium* Seminar; World Mycotoxin Forum, European Association of Agricultural Economics; Annual World Food and Innovation Forum; European Commission Mycotoxin Forum; IUPAC Symposia; Gordon Conference on Mycotoxins and Phycotoxins; Food Mycology Workshops; International Mycological Congres; British Mycological Society; Association of Applied Biologists; British Society of Plant Pathology; in the course of 2016-2019 in China: Chinese cereals and oils association annual congress; China International Food Safety & Quality (CIFSQ) Conference; China International Food Safety Technology & Innovations Expo: participation in organising, scientific committees, with poster, paper, presentation, or chairing.	Sound communication towards policy-makers to increase the influence of the project and its results in policymaking (which will be reinforced by the presence of our expert partner in standards setting, BfR).
Early stage researchers		Communication to retailers and consumers about the efforts being made along the chain to increase food safety and quality and to reduce biological contamination
Government & policy makers Innovation programs		The general public should also feel they are getting value for money from public funded research that will ultimately benefit them as consumers, by understanding how MyToolBox contributes to food safety, food security and environmental benefits, .
Educational stakeholders and students	***Magazines***: Milling & Grain; Miller Magazine, Farmers Weekly, Grain Journal; ***National magazines***: VMT (NL), De Molenaar (NL).	
Other cross-cutting H2020 projects and international/ national projects	***Open Access Journal Articles***: J. of Food Engineering; J. Agric. Food Chem., J. of Stored Products Research; J. of Environmental Monitoring and Assessment; PlosOne; Food Additives and Contaminants; World Mycotoxin Journal; International J Food Microbiology; Journal of the Chinese Cereals and Oils Association (Journal of Food Safety and Quality Science and Technology of Cereals oils and Foods)	More young people will be attracted to scientific and research careers, their curricula will be enriched by knowledge from the project. Early stage researchers inspired by the results, will be motivated to remain in EU- research and possibly work in areas that are beneficial for society, sustainability and innovation.
Other innovation programs and initiatives (e.g. COST, LIFE, etc.)	***Involvement in networks/clusters***: Mycotoxin Stakeholder Groups (UK); Food4Life; EIP-AGRIHome-Grown Cereals Authority events; NABIM (UK and Irish Millers Association); GAM (European Millers Association	New jobs generated by the industrial partners as well as EU suppliers due to the development of new business opportunities.
	Features in Newsletters and e-bulletins All About Feed- www.allaboutfeed.net; ICC News Alerts	Special attention will be paid to dissemination in China. At least one of the stakeholder meetings will be held in China, preferably with a dissemination to involve Chinese stakeholders. Scientific meetings will be held in China following the networks to form an EU-China Mycotoxin-Forum.
	Visibility on Web portals e.g. Food Production Daily	
	Inclusion in EU dissemination resources such as the EU Horizon Magazine and the EIP Agri Website http://ec.europe.eu/eip/agriculture	
	Mass media, e,g. Euronews	
	Social media	
	Programme meetings	
	Policy roundtables	Sustainable EU-China dialogue on mycotoxin occurrence, detection and reduction of mycotoxins and new business opportunities the area of pre- and post-harvest techniques for EU-partners in China

Within the framework of *Task 5.4- Dissemination to a wide audience,* the partners will build on and execute this dissemination plan following a process of analysing: **Why** – the purpose of the dissemination action; **What** will be communicated – the message; **To whom** – the audience; **How** – the method; **When** – the timing. Performance indicators will be established as part of the upgraded plan, such as number of citations (for publications), number of hits on web, number of downloads (for electronic documents/materials), number of industry consultations (from tradeshows/exhibitions), number of policy-maker engagements, etc.

Importantly, as part of our dissemination strategy, the "**MyToolBox Roadshow**" of events will be organised, which will consist of an agenda of events that will be planned for the 48 month duration of the project, where in order to maximise resources and to bring mutual benefit to events being organised by third parties, we will schedule out the trade fairs, exhibitions, workshops, seminars, conferences, round tables, etc. of relevance in Europe and internationally that MyToolBox consortium members can attend (e.g. Alimentaria (ES); Anuga (DE); The Cereals Event (UK); SIMA (FR); GrainTech (EG); Agrotech (PL); among others). Our target will be to have MyToolBox present in **at least 40 events in Europe and internationally** over the course of the project. A philosophy of 'not wasting' any dissemination opportunities will be instilled in the consortium and partners will be encouraged to capitalise on the relevant marketing and dissemination events they are attending or organising to plan a dissemination activity for MyToolBox. In addition, a number of dedicated MyToolBox events will be organised (it is envisaged 1 per year), of which one will be in China, whereby the culmination of the Roadshow will be the Final MyToolBox Workshop/Conference in conjunction with a major Mycotoxin Event, such as the IUPAC conference on mycotoxins that will be organised to coincide with the end of the project.

Our dissemination channels (which will be complemented by a **professional communication campaign** discussed below) will cover all the traditional channels such as dissemination at conferences (papers and posters), publications in peer-reviewed journals, dissemination in industry magazines, newsletters and on-line platforms, as well as more vanguard channels, such as social media platforms, project blogs and videos, etc. An important dissemination platform for the project will also be the **project website** that will be built during T5.3.2, which will serve as an enduring internet site for promoting the project and housing all the downloadable public deliverables and outputs of the project, and it will also house our 'MyToolBox Green Channel' for open access to the peer-reviewed publications of the project. It will be widely disseminated across Europe and worldwide to generate visitors and traffic to the site. It is noteworthy that we have already registered the www.mytoolbox.eu domain name.

In addition, our project dissemination activities will link with the broader dissemination strategies of the European Technology Platform, *Food For Life*, as well as with those of Meetings of EIP-AGRI, EU Arable Farmers network, International Association for Cereal Science & Technology, EU Crop Protection Association, and other initiatives, etc. to contribute to efficient coordination and dissemination of the project outputs across the relevant European, Chinese and international communities to promote the project. Moreover, the dissemination strategy will also consider how it can best be used to **maximise the future exploitation of the results of the research**.

In order to protect the commercial opportunity, a **confidentiality policy will underpin our dissemination activities**. The consortium experience in collaborative projects points towards the most effective default position being that all information is confidential unless otherwise agreed as non-confidential. This is the clearest starting point. However, that said, the **academic participants in the consortium aim to publish results** (publications being considered very positive success indicators by all the consortium, also serving as peer reviewed documents towards the industry, standards bodies and policy stakeholders as discussed above), and the industry partners and market actors are also eager **to showcase and exhibit results at tradeshows and exhibitions and through other industrial channels** (technical/sectoral magazines.). As such, the **IPR Manager** (Dr. Mirta Pinilla from IRIS) will work closely with the **Dissemination Manager** (Dr. Michaela Pichler from ICC) to manage the identification of non-confidential publishable/discloseable information that can be available for dissemination activities.

Initial business and exploitation plans

Over the course of the project we will integrate existing and new knowledge in relation to technical, management and organisational solutions that are effective for mycotoxin reduction, into practical and affordable management strategies that can be used by farmers and food processors along the chain in order to reduce the risk of mycotoxin contamination of crops, feed and food, to prevent losses and waste along the food chain, and to provide traceability information to the supply chain and consumers using mainstream ICT technology. As such, the project outputs are highly relevant for future market exploitation.

Critically, all the expected impacts from this project as discussed in Section 2.1 can only be realised through a robust post-project exploitation strategy. **Our intention is to realise all the exploitation pathways for the results:**

a) **Commercial exploitation-** the industrial use of the project knowledge and outputs for increased competitiveness and growth will deliver high innovation and growth impacts.
Our initial plan for the post-project exploitation of the MyToolBox web-based platform is to use it for the development of a new data service for the value chain, whereby the proposed Business Model (which will be further developed and validated over the course of the project) will be a subscription model (typical for Software as a Service platforms) for advanced use of the platform (access to the dynamic components), the pricing of which will be studied during the project. A **freemium model** will allow the static component of the platform to be readily used for free by farmers, food and feed producers and all stakeholders. A detailed guideline will accompany the MyToolBox platform for the periodic reviewing of the content and incorporation of new knowledge. This will ensure that as a wide a community of actors from the value chain can benefit from the results of this project for high competitive and social impact in the industry.
However, in order to ensure that the results that have a high commercial value can be commercially exploited, a **subscription model** will be studied to contribute to the economic and business impacts of the project. During Subtask 5.5.1 Business and exploitation plans, we will ensure that we arrive at a robust pricing strategy that is based on market research among our target groups (farmers, grain store operators, food and feed producers, etc.) in order to establish the value they perceive for the different dynamic components of the platform and to establish the price tolerance to ensure we deliver a high value for money subscription model. It is noteworthy that IRIS has expertise in the development of SaaS (software as a service) marketing plans and commercialisation strategies.
Striking this balance between free dissemination of the results and capitalising on those that have commercial value is a winning formula for sustainably tackling societal challenges.
In addition to the business model for exploiting the MyToolBox platform, for each of the commercially exploitable outputs of the project (Table 5 provides a starting point), we will draw up an individual business plan.

b) **Education and training**- the use of the new knowledge generated for upskilling and for enriching curricula in European education will have a high impact for raising the European mycotoxin knowledge base among the farming sector, as well as along the food and feed supply chain.

c) **Further research-** planning future spin-out research and development projects especially in areas that will contribute to the transferability of the project results to new applications, such as for example the large-scale integration of mycotoxin degrading enzymes in biofuel plants and the metabolomics approaches optimised in MyToolBox to study the effect of novel baking procedures on the fate of mycotoxins to other areas of food and feed processing.

d) **Contribution to standards-** the project results can contribute to improved standards in relation to the prevention and control of mycotoxins in the Europe and internal food and feed chains. The project will also aim to enable standard settings in China for authorisation of mycotoxin detoxifying feed additives produced in the EU and consequently improve market access of relevant EU products. As has been discussed, during the project we will work with relevant bodies such as EFSA and DG SANTE for standard setting, and we will continue these efforts in a post-project phase. Moreover, we aim to contribute to the standard setting for authorisation of mycotoxin-detoxifying feed additives in China in cooperation with FEFANA, the Chinese Ministry of Agriculture (MOA) and CAAS-FRI as an important aspect for the registration of mycotoxin detoxifying feed additives from the EU. These are medium to long-term ambitions that will be followed through as part of our exploitation plans.

Each partner has summarised their broad exploitation expectations (table 5). These statements showcase the exploitation ambitions of each of the partners considering the contribution of the results to patents, to new or existing products, services, infrastructures, knowledge bases and standards, to ensure the project innovations are delivered to the market. During the project, and through *Subtask 5.5.1 Business and exploitation plans* in-depth plans will be drawn up, whereby partners will collectively and individually (based on joint or individual ownership of the different exploitable aspects of the project results) create business plans (or exploitation plans) for how they intent to use the results (based on the summarised intentions outlined in table 5).

Table 5: Summary of exploitation intentions of the MyToolBox partners

nº	Short Name	Intention for use of project results
	ALL	Promote the use MyToolBox to governments and stakeholders in the respective countries for reduction of losses due to fungal infection and mycotoxins.
1	**BOKU**	Use of knowledge from Task 2.5 for intensifying know-how in the field of inhibition processes during biogas and bioethanol production and degradation of contaminants by microbiology will directly be exploited for industrial applications. This includes the use of mycotoxin contaminated batches in biogas and bioethanol production, the production of safe and high quality DDGS and the efficient production of biofuels.
2	**DLO-RIKILT**	Use the expertise collected and gained through the project in interaction with national government and food safety authority, as well as private companies, through the running of national projects on related topics. Use for international activities, in and outside EU, e.g. training given by DLO-RIKILT personnel.
3	**ICC**	Use of project knowledge and results to develop internationally-accepted standard methods to facilitate the trade of grains and grain-based food products within and between countries. Use of the project knowledge to offer training activities to future stakeholders and end users. Dissemination of the gained knowledge worldwide.
4	**HAU**	Use the knowledge from Task 1.1, Task 1.2 and Task 3.1 on alternative and cultural control of *Fusarium* and in the development of an integrated management strategy to reduce *Fusarium* and mycotoxins in UK cereals.
5	**FLI**	Feedback from end-users and stakeholders (Task 4.1) will be used to progressively improve the design of MyToolBox and ensure that it meets end-user needs and is therefore more likely to be widely employed along the food and feed chains. The results of experimental work undertaken with feed additives will be used as a basis for establishing criteria by authorities for market acceptance in China.
6	**BARILLA**	Use of the knowledge from Task 2.1 to test the efficiency of a real time monitoring system for improving storage management solutions for mycotoxins in stored wheat and from Task 3.3 to validate the DSS system with different stored cereals as a management tool; Use of the knowledge from Task 2.2 to validate an innovative screening technology on suitable commodities; Use of the knowledge from Task 2.3 and Task 2.4 for the development of pre-/milling and food processing (especially baking) techniques to mitigate mycotoxin contaminations in the finished food products; Use of the knowledge from Task 3.1 and Task 4.1 to synthesise information to provide guidelines (GAP, GMP, HACCP protocols) to control mycotoxin contamination evaluating at the same time the real application on the field of the MyToolBox e-platform along the entire chain up to the end-user level.
7	**CRAN FIELD**	Use of platform technology in Task 2.1 and Task 3.3 of the DSS post-harvest management tool to potential user stakeholders via Home-Grown Cereals Authority events; NABIM (UK and Irish Millers Association); GAM (European Millers Association), Soil Association, presentations at National Cereals Symposia, Research aspects via Symposia organised by British Mycological Society, Association of Applied Biologists and British Society of Plant Pathology and at the Biannual World Mycotoxin Forum and Gordon Conference on Mycotoxins and Phycotoxins. Task 3.1 on integrated HACCP tools will have wider benefits at both a national (HGCA) and European level and be of benefit as components of the static platform for industry and for agricultural and food related organisations (FSA (UK); EFSA (EU); European Commission; ILSI Europe).
8	**IRIS**	Use of the knowledge from Task 2.1 for the development of a real time monitoring system for improving grain storage management solutions for mycotoxins in stored cereals and nuts; Use of the knowledge from Task 3.5 for the production and deployment of a SaaS DSS platform for mycotoxin reduction in conjunction with HORTA
9	**UNS**	Use the knowledge from Task 1.3, Task 1.4 and Task 3.1 in the development of an integrated management strategy to reduce *Fusarium* and mycotoxins in maize in Serbia and Italy. Use the knowledge in training purposes in academic courses on integrated crop management.
10	**HORTA**	Use of results from Task 3.2 for improvement of the small-grain cereals and maize models inserted in the HORTA's web platform for farmers in Italy and used in the MyToolBox as external dynamic component (i.e., a component residing in the HORTA's platform and linked to the MyToolBox via web services for exchanging numerical inputs and outputs); Use of the DSS developed in Task 3.5 for future exploitation based on the agreements developed within the Consortium during the project.

nº	Short Name	Intention for use of project results
11	**BfR**	Knowledge gained in studying food processing factors and in new developments coming from work on milling and food processing from WP2 will be used to provide a sound scientific basis for setting of mycotoxin limits in processed foods and refining existing limits where appropriate.
12	**SDU**	Designs of novel sorting system will be patented and exploitation will be pursued from pilot scale sorting of figs to full commercialisation through an existing SME enterprise previously established by SDU. The developed sorting system will also be extended to other dried fruits, nuts, chili peppers and pistachios.
13	**WU**	Knowledge gained from the survey on the end-users' potential motivational barriers to implement MyToolBox (Task 4.2) and from the cost-effectiveness analyses (Task 3.4 and Task 4.2) will be used in running socio-economic research projects (national and international) on related topics as well as for training purposes in academic courses on Decision Science and/or Risk management in food supply chains
14	**BIOFORSK**	Use the knowledge from Task 1.1, Task 1.2 and Task 3.1 in the development of an integrated management strategy to reduce *Fusarium* and mycotoxins in Norwegian cereals
15	**CAAS-IAPPST**	Use the knowledge from MyToolBox in the development of an integrated management strategy to reduce *Fusarium* and mycotoxins in cereals, maize and peanuts in China.
16	**BIOMIN**	BIOMIN intends to secure foreground-IP of Task 2.5 by supporting a quick and reasonable patent filing to not hinder scientific communication of the MyToolBox project results. BIOMIN intends use these results-IP, especially the developed methods and protocols to reduce mycotoxins in the bioethanol or biogas production process for increased process performance and DDGS safety, to enlarge the existing BIOMIN business segment of mycotoxin deactivation from the feed additive industry towards the biogas and bioethanol industries worldwide. That will allow organic economic growth of the company and strengthen especially the EU BIOMIN entities as being the major producers of the mycotoxin deactivating product portfolio. Furthermore, existing labour force will be secured and, in addition, new jobs in research, development and marketing (>30) will be created within BIOMIN as well as by its distributors. To globally spread the benefits of these results within a contemporary manner, BIOMIN is willing to out-licence the generated foreground-IP, preferably to the relevant industry partners of MyToolBox, so that it may become a standard in industry.
17	**ASAG**	Use of the knowledge from Task 2.1 for the development of a real time monitoring system for improving grain storage management solutions for preventing and reducing mycotoxin contamination in stored cereals and peanuts in China.
18	**SÜD-ZUCKER**	Use of knowledge to improve management of contaminated grains in bioethanol and animal feeding production. This will add to the companies sustainability aim to development new, sustainable production processes and products and better resource management to bring by-products on the feed market, an area where currently the US is the leading nation. By implementing the use of novel, effective enzymes and processes, the EU could become the new market leader. The existing labour force will be secured and, in addition, hundreds of new jobs could be created in the EU as well.
19	**AXEB**	Use of the knowledge of WP1 and Task 3.1 (pre-harvest reduction strategies for mycotoxins) for an expansion of our commercial portfolio with specific products for pre-harvest control of mycotoxins in cereals.
20	**AGRO-LV**	Advice to the farmers to implement debris management technologies and use of MyToolBox to increase high quality crop production.
21	**TARIS**	Pilot scale sorting equipment will be critically compared in terms of efficiency and performance with existing manual sorting under UV light. Providing cost-effective manufacture is achieved uptake will be consider on a commercial scale.
22	**AGROCENTRUM**	Implement crop management strategies for the growing of cereals (except rice), leguminous crops and oil seeds in Serbia.
23	**CAAS-FRI**	Use the gained knowledge on mycotoxin detoxifiers and the related legal issues for proper standard setting in cooperation with the EU and to develop strategies for safe use options for contaminated batches in China.

How the proposed measures will help to achieve the expected impact of the project

Our **multi-actor** and **vertical integration approach (i.e. including the whole value chain)** supported by collateral implementation channels (such as socio-economic and legal analyses, etc.) will contribute to the achievement of the expected impacts of the project. A fundamental strategy for success, is that of enabling actors and stakeholders from along the farming/crop and food production and supply chains to enter the research-development-innovation cycle linked to mycotoxin reduction to ensure more, safer/higher quality food and feed, social innovation is fostered, improved technical, management and organisational solutions are exploited, developed and implemented, as well as the development of new business models. To realise this there is a **strong innovation dimension/cycle** in our approach with a view to **promoting early stakeholder participation, adoption of the project results, and continuity of the MyToolBox initiative/movement for impact.**

In short, sound management of the knowledge generated during the project (discussed below), active planning in relation to post-project exploitation activities for industrialisation of the most market ready results, and further research work for escalating along the TRLs, as well as a solid communication strategy to disseminate the non-confidential aspects of the result to the European, Chinese and international industry from the crop and food production and supply chain, as well as to market actors and multipliers, and policy makers, will all contribute to achieving the expected impacts of the project. It is worthy of mention that partner ICC, will provide a pathway to implementation that is not available through the normal commercial (market) channels, thus providing a **global footprint for MyToolBox.**

How the participants will manage the research data generated and/or collected during the project

The MyToolBox project will participate in the Open Research Data Pilot initiative. Effective exploitation of data depends upon proper data management and curation and as such as Data Management Plan will be drawn up and managed through *Task 5.2- Data Management*. A formal **data plan** will be drawn up and implemented during the project that will include: which data will be generated during the research; metadata, standards and quality assurance measures; plans for sharing data; copyright and intellectual property rights of data; data storage and back-up measures; data management roles and responsibilities; costing or resources needed. Indeed, data management will be embedded in the wider knowledge management of the project, and it will follow the Guidelines on Open Access to Scientific Publications and Research data in Horizon H2020.

Strategy for knowledge management and protection

To pave the way forward for commercial exploitation and dissemination of the project results, the consortium will implement an integral strategy for the effective **management of this knowledge**.

Background IPR and access rights

Background- information and knowledge (including inventions, databases, etc.) held by the participants prior to their accession to the EC grant agreement, as well as any intellectual property rights which are **needed** for carrying out the project or for using foreground[52] held by the consortium, or indeed by third parties- is defined in the proposal preparation phase (pre-Grant Agreement). The current status of the Background being brought to the project by the consortium has been included in each participant's profile description in section 4.1, whereby each beneficiary has provided a background statement (which will also be included in the Consortium Agreement that will be signed by all partners in the eventuality of a successful grant application). **Access to Background** that is, or will be found to be, necessary for the implementation of the project will be granted **royalty free** and **background needed to use the results of the project** will be provided either **royalty free or under fair and reasonable conditions**.

A comprehensive review of "prior art" and patents (*Section 1.4*) has revealed that to date there is **no relevant Background held by third parties** that is needed for implementing the research or using the project results. In order to ensure that the intended MyToolBox R&D effort does not trespass into the owned knowledge space of a third party, and to mitigate against expensive infringement claims, during *Task 5.1- Knowledge Management Activities,* up-to-date patent reviews will be performed periodically to confirm the novelty of the proposed research and the freedom to exploit the results.

Results definition, protection, ownership and access rights

The expected results from the project are listed in table 6 below, and will be continually assessed in terms of **novelty, patentability and protection** as part of Task 5.1, as part of an effective knowledge management strategy. Results shall be owned by the participant generating these[52]. However, the consortium realizes that it is unethical to withhold

[52] REGULATION (EU) No 1290/2013 Of The European Parliament And Of The Council of 11 December 2013 laying down the rules for participation and dissemination in "Horizon 2020 - the Framework Programme for Research and Innovation (2014-2020)" and repealing Regulation (EC) No 1906/2006

innovations or management practices that can provide more and safer food from all actors in the food chain. To avoid doubt or confusion over ownership, a protocol of procedures will be defined, which will be followed by each partner to clearly show the development of the generation of knowledge or results. In cases where the results are generated jointly (i.e. where the separate parts of some result cannot be attributed to different participants), it will be **jointly owned**. The allocation and the terms of exercising this joint ownership will be agreed in Joint Ownership Agreements that will be established during our knowledge management activities in T5.1. However, an overriding principle will be that the **participating industry partners are always provided with the access that they need to the results to exploit them in keeping with the needs and strategy of their business.** As such, access rights to results for implementing the project will be granted on a royalty-free basis. Access rights for use purposes will be granted either **royalty-free**, or on **fair and reasonable conditions** to be agreed in *Exploitation Agreement(s).* Table 6 provides an overview of the expected scenarios of results ownership and access rights.

Table 6: Expected Results Ownership and Access Rights

WP	Description of the Expected Result	Envisaged IPR Tool	Owner(s)	Partners to which access rights will be granted and conditions
WP1	Real time monitoring system for improving grain storage management solutions for mycotoxins in stored cereals and nuts	Patent or Know How secured through confidential design	**IRIS, CRANFIELD**	Preferential use to BARILLA and ASAG (early access and cost price in first units purchased)
WP1	Biofumigation product/s for mycotoxins control in cereal crops	Know How secured through confidential design	**AXEB**	Access rights to partners that require access to be negotiated under Fair and Reasonable conditions
WP1	Biopesticide for *Fusarium* control	Patent or know How secured through confidential formulation	**HAU, BIOFORSK, AXEB**	Access rights to partners that require access to be negotiated under Fair and Reasonable conditions
WP1	Atoxigenic biocontrol agent	Patent	**UNS, AGROCENTRUM**	Preferential use to partners that require access under Fair and Reasonable conditions
WP2	Innovative (pre-) milling strategies for accurate mycotoxin separation and for minimization of their content in final wheat based food products	Patent or Know How secured through confidential design	**BARILLA**	Preferential use to partners that require access under Fair and Reasonable conditions
WP2	Novel food processing procedures to develop a sound basis to establish process factors for risk assessment	Patent or Know How secured through confidential design	**BARILLA, BOKU**	Preferential use to partners that require access under Fair and Reasonable conditions
WP2	Final risk assessment of mycotoxins (free/modified forms) during food processing through "in-process stable isotopic labelling"	Patent or Know How secured through confidential design	**BARILLA, BOKU**	Preferential use to partners that require access under Fair and Reasonable conditions
WP2	Real time monitoring system for improving grain storage management solutions for mycotoxins in stored cereals and nuts	Patent or Know How secured through confidential design	**IRIS, CRANFIELD**	Preferential use to Barilla and ASAG (early access and cost price in first units purchased)
WP2	Product compositions including product formulations -IP suitable for reducing the toxicity and / or	Patent Trade secret, if patenting does not	**BIOMIN**	Access to University partners for scientific use only. No access rights will be provided to industry partners for commercial use. No

WP	Description of the Expected Result	Envisaged IPR Tool	Owner(s)	Partners to which access rights will be granted and conditions
	the amounts of mycotoxins within the biogas and / or production bioethanol processes including DDGS. Methods and protocols for applying above mentioned products-IP within the biogas and / or production bioethanol processes.	seem to be possible		rights for scientific use outside the project. Distribution or licensing agreements will be negotiated under Fair and Reasonable conditions
WP2	Pilot scale automated sorting system for removal of aflatoxin contaminated figs.	Patent or Know How secured through confidential design	**SDU**	Preferential use to Barilla and TARIS
WP3	Models for to predict fungal infection and mycotoxin accumulation in wheat, barley and maize.	None	**DLO-RIKILT, HORTA**	Free access to all partners via the MyToolBox Platform
WP3	MyToolBox web-based platform	Know-how through software development secured by code	**HORTA, IRIS, DLO-RIKILT**	Free access to all partners.

Open access to peer-reviewed scientific publications

The two ways authors can provide open access are (1) by self-archiving their journal articles in an open access repository, also known as 'green' open access, or (2) by publishing in an open access journal, known as 'gold' open access.[53] It is envisaged that **green open access** will be the preferred option in this project, whereby the partners will publish in peer-reviewed scientific publications and then self-archive a version of the article for free public use in their institutional repository, in a central repository, or on some other open access website. The project website (or Internet Portal) will house a 'MyToolBox Green Channel' for open access to the peer-reviewed publications of the project. In relation to expected publications from the project, these include and are not limited to: "Alternative control of Fusarium mycotoxins in small grain cereals" to be published in *Quality Assurance and Safety of Crops and Foods; "*Conjugation and reduction of mycotoxins during thermal processing" in *Food Additives and Contaminants;* DSS System post-harvest: Real time monitoring systems for improving grain storage management solutions for mycotoxins in stored cereals and nuts in the *Journal of Agriculture and Food Chemistry;* and Development of a mycotoxin mitigation strategy exploiting food processing effects to be published in the *World Mycotoxin Journal*, etc.

Communication activities

Communication activities are considered to be an important aspect of this project, to ensure that we effectively promote our project and its outputs, as well as its benefits and impacts to the stakeholders that we want to reach. It is relevant that citizens are aware of the development and its potential societal impacts, especially in the context of its positive impacts for food security, food quality and safety (and ultimately consumer safety). Indeed **effective social engagement is particularly important** in the case of this project given that consumer confidence has been negatively impacted in recent years as a result of quite a number of highly publicised failings in the food supply chain to deliver high quality and safe products. Efforts to increase consumer trust in the food supply chain would be very positive, especially as consumers have a central and crucial role to play as a key engine for future prosperity. Strengthening consumer confidence in the safety of the products they buy is fundamental for strengthen the European food supply and value chains. Moreover, it is important to communicate the societal benefits as the project is benefiting from tax-payers contributions. Through our communication activities we have an opportunity to ensure that **citizens are informed** and feel like they are getting value for money.

Additionally, as one of the key's to successfully gaining impact **is through our ability to secure stakeholder 'buy-in',** i.e. the cooperation or positive participation of our key stakeholders, they will also be the target of our

[53] Jeffery, Keith G. 2006. "Open Access: An Introduction". Ercim News, 64, January 2006.

communication activities- a) farmers; b) food and feed producers and food processors; c) multipliers (Collectives, Associations, etc.); d) OEMs and engineering companies (who install food processing plants); e) policy makers, etc.

Through *Task 5.3 Communication activities and multi-actor engagement* we will plan out our communication activities and develop the necessary communication tools (project website, leaflet, posters, social media accounts, video, etc.) building on and complementing table 4 above, whereby communication activities will be effectively intertwined with the project dissemination effort to ensure that we engage, stimulate the interest and support, and motivate the collaboration of actors along the value chain in order to benefit our action and to increase its impacts for the benefit of consumers and the value chain.

As such, we will develop a communications plan, which will consist of the following steps:

1. **Identify and list all possible stakeholders**- the target audiences of our dissemination activities will be used as the starting point (in addition to any other stakeholders that might be considered important).
2. **Organise our stakeholders-** following a stakeholder analysis, the target audience of our communication activities will be categorised into "communities of support" for helping MyToolBox to realising its ambition, goals and objectives.
3. **Development of the MyToolBox brand –** in order to help us in engaging with our stakeholders the principles behind the MyToolBox brand (visually identified through its logo and a tagline, which will both be trademarked) will be further elaborated and documented.
4. **Design and implementation of effective communication campaigns (and supporting tools and materials)-** our aim will be to generate understanding within the different communities of support as to what MyToolBox is seeking to achieve. We will ensure regular communication to our stakeholders through these campaigns, which will involve the use of emailings, face-to-face meetings (in the case of policy stakeholders), leaflets, posters, videos and virals, the website and social media, etc. The impacts of these campaigns will be regularly assessed to enable any modifications or improvements to be made to ensure our communication efforts are effective.

3. Implementation

3.1 Work plan — Work packages, deliverables and milestones

Brief presentation of the overall structure of the work plan

A highly structured and logical **48 month work plan** has been drawn up with the input of all partners including end users, such as farmers and food processors (see Section 3.3). Five **integrated** research and innovation work packages (WP) have been envisaged (see Figure 3 below), and an overarching project management work package. The work packages with corresponding milestones and deliverables are described in detail in table 3.1a. Milestones are designed as the point where the first available information from the WP will be handed over to the MyToolBox e-platform for implementation. The content and usability of this information will be assessed by key-users during the project and finally fine-tuned in the last phase of WP3.

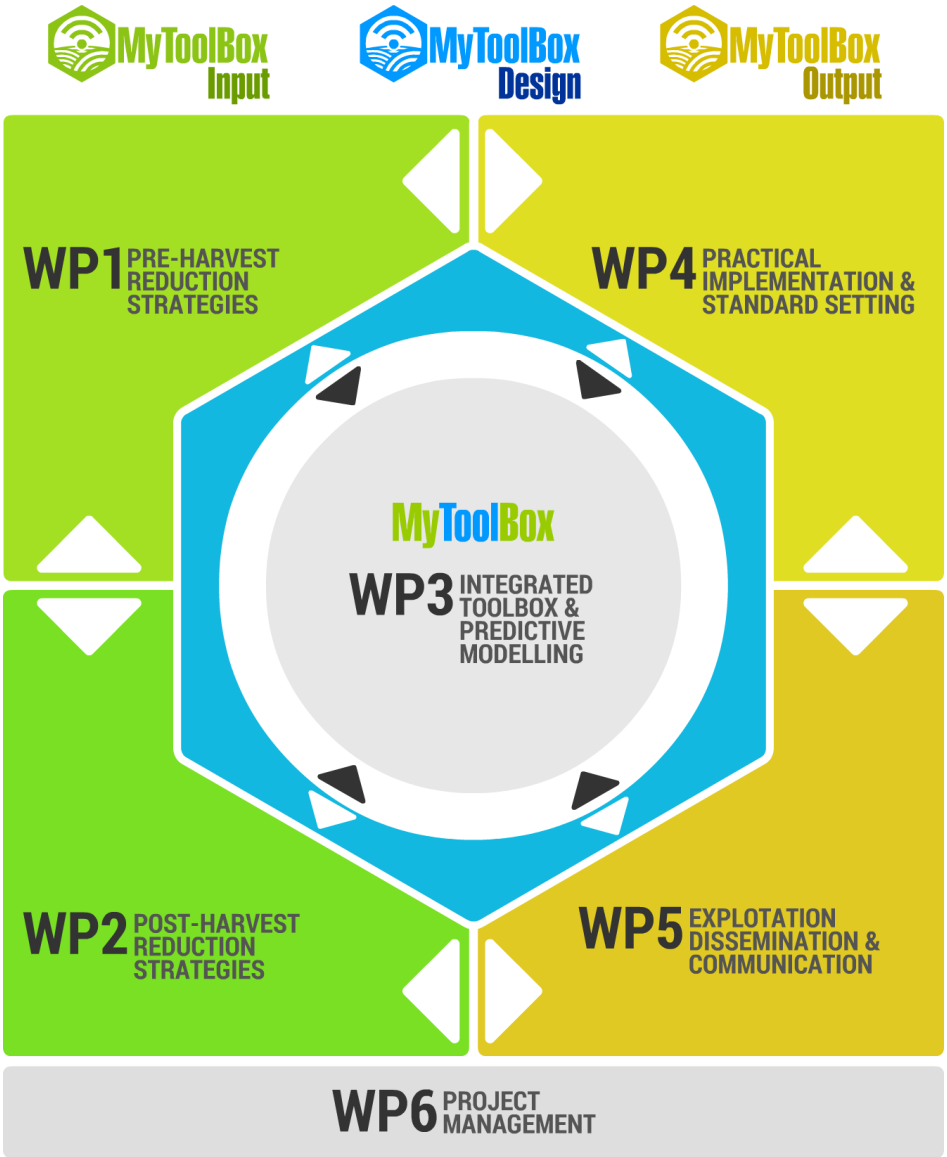

Figure 3: The MYTOOLBOX Work Packages

In close cooperation with the end-users in our consortium, crop yield and mycotoxin contamination after current crop management will be compared with crop yields and mycotoxin contamination after using the novel **pre-harvest interventions (WP1)**. These include investigation of genetic resistance to fungal infection, cultural control, the use of novel biopesticides (organic-farming compliant), and competitive biocontrol treatment. **Field trials will be carried out in** North-West and South Europe and the results will be translated into clear and easy to use management strategies included in MyToolBox.

Research into **post-harvest measures (WP2),** including real-time monitoring during storage to be carried out in 3 countries including China, innovative sorting of crops, novel milling technology and efficient biofuel production, will enable raw cereal materials with higher mycotoxin levels to be processed without breaching regulatory limits on mycotoxins in finished products. To overcome problems in a relation to user acceptance of the web-based MyToolBox decision support e-platform, our multi-actor approach and well managed innovation cycle ensures that stakeholder groups are involved at all stages of the **development of MyToolBox e-platform (WP3)** to ensure that their feedback is considered in the iterative development approach. Their involvement will be crucial to translate the key components of HACCP, GAP and the Decision Support System (DSS) for pre- and post-harvest treatments together with legal aspects and socio-economic drivers of the actors into clear, unambiguous advices to prevent mycotoxin contamination and the safe use of contaminated batches. Hence, the individual measures will be **implemented in the MyToolBox decision support e-platform (WP4), followed by the evaluation** of the tool from various angles (mostly face-to-face interviews with key-users and use of questionnaires) through a multi-actor approach with end users and stakeholder groups at relevant points in the food chain in different countries, and this assessment will be conducted feeding

outputs into WP4 to aid the socio-economic analysis. The **cost-effectiveness of the developed** intervention strategies in combination with the MyToolBox e-platform to effectively **monitor and reduce the incidence of** mycotoxin contaminants in crop production and in the food and feed supply chains will be verified **a**) for *Fusarium* mycotoxins in wheat in the UK and **b**) for aflatoxins in maize in Serbia and **c**) in cooperation with the Chinese partner, for the peanut supply chain in China.

Early in the project, a business plan will be developed in order to draw up a robust business model for the **commercial exploitation (WP5)** and maintenance of the MyToolBox outcomes after the project: this is the key to the **sustainability of the measures developed in the project and of its web-based MyToolBox platform** and it will be centred on a strong value proposition to the end user community. Within the MyToolBox project we will also implement a **communication campaign including training (WP5)** to reach the **targeted stakeholders including farmers** and their associations and to disseminate information about the project and its results to actors and stakeholders along the food supply chain, to opinion leaders/regulators, the media, professionals working in agriculture and food manufacture, educators, and other multipliers, food and drink industries (including SMEs), consumer organisations, the scientific community and the general public. A compliance assessment will be carried out to identify critical issues in legislation/standards in order to implement MyToolBox and to **anticipate legislative implications of the project outputs. Project management (WP6)** will closely follow the actions and developments initiated by all partners with the aim to reach all objectives within the time and budget limits.

Gantt Chart: Timing of the different work packages and their components

| WP. No. | WP and Task Titles | Del. & Milest. | Months in the project | | |

Table 3.1a: Work package description

Work package No.	WP1	Start date or starting event				M1			
Work package title		**Pre-harvest reduction strategies**							
Particip. No.	01	04	09	10	14	15	19	20	22
Particip. short name	BOKU	**HAU**[54]	UNS	HORTA	BIOFORSK	CAAS-IAPPST	AXEB	AGRO LV	AGROCENTRUM
PM per participant									

Objectives

This WP will use novel technologies with proven effects in related scenarios to expand the range of tools available for the mitigation of pre-harvest mycotoxin formation. Throughout the entire project period, the farm companies (AGRO LV and AGROCENTRUM) will provide advice on the enduser suitability of mitigation strategies. Results will be incorporated with known mitigation strategies to develop integrated mycotoxin management tools within WP3 to be used by end users and to be incorporated into the web-based MyToolBox.

O1.1 Implement **alternatives to conventional fungicides** (suitable for organic farming) for small grain cereals to achieve comparable or better performance to conventional fungicides in reducing levels of biological contamination and mycotoxins by 50%.

O1.2 To integrate **cultural control protocols targeting the *Fusarium* inoculum** within crop debris using biofumigation and accelerated biodegradation combined with minimum tillage (50% reduction)

O1.3 To achieve a **70-90% reduction of the aflatoxin contamination in EU maize** through a) the selection and application of resistant plant cultivars in field conditions (70%) and b) the use of atoxigenic *Aspergillus* strains to outcompete toxigenic *Aspergillus* strains (90%).

Description of work (broken down into tasks), and role of participants

T1.1 Alternative control (incl. control suitable for organic farming) of *Fusarium* mycotoxins in small grain cereals (M1-M40)

Lead: HAU PM; Partners: BOKU 4.5PM; BIOFORSK 3.5PM; AXEB 1.25PM; AGRO LV 0.8PM;

HAU and BIOFORSK will conduct field experiments in the UK and Norway on the reduction of mycotoxins in small grain cereals through the application of biopesticide technology. Current chemical control of *Fusarium* mycotoxins is limited to a single chemical group, the triazoles. This creates a large selective pressure for fungicide resistance to develop and there are currently safety concerns with triazole fungicides as they are classed as endocrine disruptors, and as such they may be removed from the market. **This task will look at a number of alternatives to triazole fungicides**. This will include alternative chemistry as well as novel biopesticides such as salts, botanical extracts, plant defence elicitors and biocontrol agents. Several recent studies conducted at HAU, BIOFORSK and other institutes have shown reductions using various forms of biopesticides, for example preliminary experiments at BIOFORSK have shown significant reduction in *Fusarium* and its mycotoxins in wheat after treatment with a foliar fertiliser. This study will allow a comprehensive comparison of different categories of biopesticides to be evaluated within the same experiments. Due to their nature several of these alternatives are suitable for organic as well as conventional production systems. **An identical matrix of up to 18 biopesticides will be tested on wheat in the UK and on oats in Norway** in comparison to untreated and triazole (prothioconazole) treated controls in year 1. In the growing seasons of years 2 and 3 a selection of biopesticides showing most promising activity will be optimised for dose and timing. HAU **will use artificial inoculation with *Fusarium graminearum* and mist irrigation t**o generate high disease pressure whilst BIOFORSK plan to use natural infection at sites with high disease pressure. Both institutes have used these strategies successfully for many years to test the efficacy of products against *Fusarium* infection and subsequent mycotoxin contamination of harvested grains. Field experiments will be conducted according to EPPO guidelines. In order to verify the the efficacy of the plant protection products, the harvested grains will be quantified for deoxynivalenol using immunoassay (ELISA) technology by HAU and BIOFORSK. In addition, and to verify the impact of these biopesticides on mycotoxins beyond Deoxynivalenol, targeted analysis of all relevant *Fusarium* mycotoxins including their masked (conjugated) forms will be carried out by BOKU using a sophisticated and fully validated LC-MS/MS based multi-analyte method, which covers more >300 secondary metabolites of fungi. BOKU will also ensure accuracy and comparability of the ELISA measurement results through LC-MS/MS reference analyses and by providing laboratory- and certified reference materials. In

[54] Partner underlined is WP leader

Europe, EPPO guidelines now constitute a reference in Commission Regulation (EU) No 545/2011 of 10 June 2011, implementing Regulation (EC) No 1107/2009 regarding the data requirements for plant protection products. AXEB will advise on the biopesticide technology and take forward the commercialisation of potential biopesticide products (see also WP5). AGRO LV will advise on end-user suitability of the biopesticide technology and conduct demonstration trails of successful products for growers in year 4.

T1.2 Cultural control of *Fusarium* mycotoxins in small grain cereals (M1-M40)

Lead: HAU 55PM; Partners: BOKU 4.5PM; BIOFORSK 9PM; AGRO LV 0.2PM;

Cultural control mechanisms will be investigated focusing on the study of the **crop debris management** in this Task. This **will include microbial agents** to colonise and exclude *Fusarium* inoculum from plant debris and the use of biofumigation crops. These biological control strategies are suitable for organic as well as conventional production systems. BIOFORSK will conduct laboratory and field studies on the efficacy of chemical and biological control agents to reduce the inoculum of *Fusarium* in crop debris for oats in Norway. This will involve an **initial screen of up to 20 products in laboratory studies** followed by **optimisation and validation of selected products from year 1 in field experiments in the growing seasons in year 2 and 3.** Preliminary experiments at BIOFORSK have shown significant reduction in *Fusarium* inoculum after treatment with Biowish (fertiliser that stimulates microbial activity) and some reduction by Trifender (*Trichoderma asperellum*). HAU will conduct studies in glasshouse on the efficacy of biofumigation technology to reduce *Fusarium* inoculum in crop debris followed by optimisation and validation in the growing seasons in years 2 and 3 for wheat and maize rotations in the UK. Within this task *Fusarium* inoculum will be quantified using qPCR technology developed and validated in previous projects at HAU and BIOFORSK. AGRO LV will advise on the end user suitability of debris management technology. As described in T1.1, targeted analysis of the *Fusarium* mycotoxins will be performed by a novel and fully validated LC-MS/MS based multi-analyte method developed by BOKU, which covers all relevant myocotoxins including their masked (conjugated) forms.

T1.3 Reduction of aflatoxins in maize through genetics (drought resistance, early maturity) (M1-M40)

Lead: UNS 15.6PM; Partners: BOKU: 0.5 PM; AGROCENTRUM 4PM; HORTA 11.5PM

UNS and HORTA will conduct **field experiments in Serbia and Italy** to determine the resistance of local and European maize hybrids to aflatoxin contamination at harvest. A range of **maize hybrids will be screened for aflatoxin resistance** based on known genetic traits that impact on aflatoxin contamination (namely drought tolerance and rate of maturity). Hybrids will be selected based on local and European origin as well as the screening of newly developed hybrids. An initial screen will be conducted of 50 genotypes with a single sowing date and location (Serbia) in year 1. **In the growing seasons in year 2 and 3, 20 selected hybrids will be tested at 2 locations (Serbia and Italy)** and with 2 sowing dates (early and late). Experiments will be conducted according to EPPO guidelines and harvested grains will be quantified for aflatoxin using ELISA tests. BOKU will ensure accuracy and comparability of measurement results through complementary LC-MS/MS reference analyses and by providing laboratory- and certified reference materials. In addition to the individual farmers involved through Agrocentrum the **results of the field trials for ear rot resistance** will be presented at the annual meeting organised by Agrocentrum (´field day´) for breeding companies, agricultural organisations and grain producers, to obtain feedback from end users and other relevant agricultural institutions.

T1.4 Biocontrol for aflatoxins in maize (M1-M40)

Lead: UNS 36.4PM; Partners: BOKU 0.5 PM; AGROCENTRUM 8PM; CAAS-IAPPST 2PM

Local (Serbian) atoxigenic isolates of *Aspergillus flavus* have been identified and will be tested within field experiments. UNS and AGROCENTRUM will conduct studies of **2 atoxigenic isolates on 20 maize hybrids in 12 field experiments** (6 non-irrigated and 6 irrigated experiments) split between 2 sites and conducted in years 1-3. Experiments will be **conducted according to EPPO guidelines** and harvested grains will be quantified for aflatoxin using ELISA testing. BOKU will ensure accuracy and comparability of the measurement results through the measures described in T1.1 und T1.3. Task 1.4 will be a novel adaptation within Europe of proven technology developed in America, and currently being adopted in Africa and Asia, aided by the collaboration of international experts from those continents (Dr Peter Cotty, USDA and member of the MyToolBox Advisory Board; Dr Martin Kimanya, The Nelson Mandela African Institution of Science and Technology; Prof Liu Yang, CAAS-IAPPST). In particular Prof Liu Yang will provide detailed information on the production, formulation, application, standardisation and long-term monitoring of **biocontrol agents within crops based on on-going projects within China.** AGROCENTRUM will advise on enduser suitability of the maize hybrids and atoxigenic biocontrol strategies as well as demonstrate experiments to growers.

Deliverables		
D1.1	Pre-selection completed of the most promising a) biopesticides, b) cultural control agents and methods and c) atoxigenic *Aspergillus* strains in consultation with end users and Advisory Board	M16
D1.2	Guidelines for biopesticide use to reduce mycotoxins in European cereals	M40
D1.3	Guidelines for cultural control of mycotoxins in European cereals through management of crop debris	M40
D1.4	Guidelines for resistant cultivars to reduce mycotoxins in European maize	M40
D1.5	Guidelines for use of atoxigenic biocontrol to reduce mycotoxins in European maize	M40

Work package No.	WP2	Start date or starting event:				M1				
Work package title	**Post-Harvest Reduction Strategies**									
Particip. No.	01	05	06	07	08	12	16	17	18	21
Particip. short name	BOKU	FLI	**BARILLA**	CRAN-FIELD	IRIS	SDU	BIO-MIN	ASAG	SÜD-ZUCKER	TARIS
PM per participant										

Objectives

The approach in this WP is to provide stakeholders involved in post-harvest and processing from key food chains with innovative robust management tools that offer practical integrated solutions for effective mycotoxin minimisation. Objectives are:

O2.1 To develop **real-time environmental monitoring systems for storage of cereals and peanuts** to facilitate effective integrated control of moulds, mycotoxins and pest activity for integration into a **Decision Support System (DSS)** of the web-based MyToolBox for post-harvest storage managers;

O2.2 To develop **non-invasive real-time sorting of dried figs** for aflatoxins using visible/NIR spectral imaging and appropriate algorithms for achieving significant improvements compared to manual sorting, and with a 5% or less probability of the batch average concentration of aflatoxin B_1 to meet EU limits;

O2.3 To integrate novel downstream processing approaches such as **innovative (pre)milling** to accurately separate grain tissues with different mycotoxin levels and to minimise the mycotoxin content in the final wheat-based food products (with a 50% reduction);

O2.4 To draw up **novel baking procedures** controlling, e.g., fermentation, temperature, time and recipes to develop a sound basis for establishing process factors for risk assessment;

O2.5 To exploit a novel generation of **mycotoxin degrading enzymes** for environmentally sustainable use of contaminated batches using fermentation technologies for the production of biofuels (bioethanol and biogas).

Description of work (broken down into tasks), and role of participants

T2.1 Real time monitoring systems for improving grain storage management solutions for mycotoxins in stored cereals and nuts

T2.1.1 Design and testing of measurement device for real-time monitoring of key abiotic parameters using ATEX compliant sensor platform components (M1-M18)

Lead: IRIS *Partners: CRANFIELD ; BARILLA ASAG*

The designs of the measurement device will be drawn up. The device will include an **IR CO_2 sensor, a Temperature (T) and Relative Humidity (RH) sensor**. To this end, we will optimise the range of infra-red (IR) CO_2 sensor detection required in different grain types: studies will be conducted with the CO_2 sensor platform to identify the range of concentrations which need to be measured for wheat/barley, oats and peanuts under different moisture content for real-time monitoring in farm stores and grain silos depending on relative grain bulk. In addition, a review of commercially available T and RH sensors will be performed in order to select the most optimal for integration into the sensor node required.

The **sensor node** (into which the sensor components will be integrated) **will be designed**, as will the specifications for its encapsulation in order to ensure ruggedness, water and air tightness. Each node will be coded with the source code for its intercommunication ensuring no noise and interference with other systems, low consumption and seamless communication at all times. It will be ensured that the node can accurately monitor in the range required for different grain silo sizes. During the design stage we will ensure that we fufill all the criteria for ATEX compliance based on an "ATEX-by-Design" approach.

The **integrated CO_2/RH/T sensor platform will be tested in real grain silo conditions**. The number of sensor nodes required to adequately cover a standard commercial silo will be studied. BARILLA and ASAG (China) will

provide input with regard to the number of sensors and placement considerations for later practical trials. The fabricated nodes will then be installed in a pilot scale silo, and in-situ functionality and **performance assessment tests** will be carried out, in order to verify: (a) the ability to withstand pressures and long term placement in grain stores; the ability to provide rapid, accurate and repeated readings of CO_2/HR/T; to offer sufficient range to ensure measurements from the whole silo can be obtained; (b) correlation with existing CO_2 respiration results of mycotoxigenic moulds in different commodities which are available from CRANFIELD.

Following an analysis of the results of the pilot trials, the necessary modifications and improvements will be made to the sensor platform and designs of an optimised sensor will be drawn up, from which production of the final improved sensor platform will be performed.

T2.1.2 Biotic mould spoilage and mycotoxin models (M1-M18)

Lead: CRANFIELD *Partners: IRIS ; BOKU ; ASAG*

The work in T2.1.2 is complementary to T2.1.1 as it will provide data on rates of CO_2 production in relation to **fungal activity in cereals at different moisture contents and temperatures**. This will inform the design and development of the sensor platform and will be combined with the existing knowledge base at CRANFIELD for some mycotoxigenic moulds which is already available. **ASAG and CAAS-IAPPST will provide peanut samples** and corresponding biological data for these studies. Boundary temperature x moisture content models will be developed for growth and mycotoxin production in cereals/peanuts specifically for those toxins which are not available: zearalenone in wheat; T2/HT-2 in oats; aflatoxin in peanuts. The temporal production of CO_2 by mycotoxigenic fungi under different good, marginal and poor storage conditions will be obtained and this will be related to relative potential risks of mycotoxins exceeding the EU legislative limits. Analyses of the various mycotoxins will be carried out by BOKU using a novel **LC-MS/MS multi-toxin-method** as described in T1.1.

T2.1.3 Integration of abiotic data with biological data (M18-M36)

Lead: CRANFIELD *Partners: IRIS ; BARILLA ; ASAG*

Real-time abiotic data on CO_2, T and RH from pilot scale silos will be integrated with the biological growth/toxin boundary models using process engineering solutions. This will be done by linking the rates of CO_2 production under different temperature and RH conditions to these boundary models and the relationship to exceeding the EU legislative limits for the toxins in each grain type. The integrated system will include (a) commodity type and (b) toxin of interest. Trials will be run to test the prototype system in small scale bin systems in farm stores and finally in large grain silos (including silos in China, see WP3) for accuracy and for remote downloading by store managers. This will be utilised in the development of the **final DSS system** tool for post-harvest management that will be developed in WP3.

T2.2 Innovative drying and sorting of figs (M12-M36)

Lead: FLI *Partners: BARILLA ; SDU ; TARIS*

Labour-intensive, **manual screening** with potential health risks to operators, will be replaced by an automated machine-system using multiple cameras in the **visible and near-infrared regions** (NIR) of the spectrum in real time, combined with novel transparent conveyor belts. Both single and hyperspectral bands will be used to detect 'hot-spots' of aflatoxin contamination visible on the surface of dried figs. Intelligent learning algorithms will be developed to automatically detect the contaminated food items and to eliminate those from the conveyor belts through a jet of air.

SDU has experience in the development and validation of improved sorting technologies and a PhD project will be focused on this aim. FLI envisage sourcing and providing suitable quantities of dried figs as well as sourcing aflatoxin contaminated materials. FLI will also provide support in experimental design **to assess the efficiency of the newly developed sorting technology** and in evaluation of the results. Work will be conducted with dried figs (provided by TARIS) with the technology but could in the future (beyond the project scope) progressively be extended to other dried fruit (apricots), chilli peppers and tree nuts such as pistachios. End-users (TARIS and BARILLA) will be closely involved in provision of suitable commodities for screening and in establishing the efficiency of the screening technology in factory environments.

T2.3 Innovative pre-milling and milling of grains (M1-M36)

Lead: BARILLA

Within pre-milling and milling strategies, debranning, micronization and dry- or wet-turbo separation are efficient, proven and versatile, cost effective methods for companies to achieve high quality particle size reduction results and to allow more accurate separation of grain tissues with characterized (different) mycotoxin contamination levels. The approach to be employed in T2.3 involves the development of the **synergic potential of these pre-milling and milling strategies**, in order to detail the composition of the internal tissues and to minimise the mycotoxin concentration in raw materials and the final wheat products destined for consumers. By-products, usually intended

for animal feed but also exploitable for wholegrain products, contain about 15% proteins, 40% starch and 40% fibre with levels of mycotoxins which can exceed legal limits. Basically, all these fractions could be decontaminated and directed to human consumption, leading to a **significant increase in yields**: we hypothesise a decontamination degree of about 50% for each component that could be easily employed to produce any kind of cereal-based foods. The potential improved recovery from milled fractions means that EU limits on **unprocessed cereals** as well as on by-products intended for animal feed could be raised, i.e. a greater proportion of contaminated grain and other fractions obtained during milling as bran could go for human consumption: this will be a focus of BfR in with respect to adapted standard setting in WP4. Bread and pasta could be added with increased fibre content, ranging from 1 to 10% according to technological requirements.

T2.4 Conjugation and reduction of mycotoxins during thermal processing

T2.4.1 Development of a mycotoxin mitigation strategy exploiting food processing effects (M10-M42)

Lead: BARILLA *Partners: BOKU*

Food processing can alter mycotoxins (parent compounds/modified forms). EFSA has published opinions on modified mycotoxins in 2014 and the need to obtain further data on the fate of these forms during food processing. Most studies conducted on the fate of mycotoxins have been performed at lab-scale with artificially contaminated material at high levels that do not reflect practical conditions. BARILLA has the possibility to collect cereals and other commodities through their networks of milling plants and storage centres. Similarly, BARILLA can also obtain naturally contaminated raw material as flours starting from fungal inoculation and incubation in order to obtain raw materials at c**ontamination levels similar to those employed during real processing**, as opposed to artificially contaminated samples through spiking experiments.

Pilot plant/industrial production lines will be exploited for the production of **different bakery products** from the aforementioned raw materials (cereals but also almonds, hazelnuts, nuts, figs and raisins). The experimental models will be based on the Design of Experiments (DoE) approach which is an effective tool for maximising the amount of information and will benefit from the accurate LC-MS/MS multi-analyte measurements enabling the simultaneous determination of free and masked/modified forms of mycotoxins and executed by BOKU. This experimental setup shall reveal which **technological process under real industry conditions can lead to a true mitigation strategy**, i.e. reducing the overall mycotoxin impact from the raw material to the finished products ready for human consumption.

T2.4.2 Evaluation of other (food processing) metabolites of mycotoxins (M28-M42)

Lead: BOKU *Partners: BARILLA*

The release and absorption of the parent mycotoxins as modified forms (assuming that these exert the same toxicity as the parent compound) may contribute to the overall consumer exposure. BOKU has recently developed a novel liquid chromatography-high resolution-mass spectrometric (LC-HR-MS) based **metabolomics approach** using *in vivo* stable isotopic labelling combined with a sophisticated software tool (MetExtract) to extract features originating from true metabolites. Exploiting the same strategy, BOKU will spike blank flour with a mixture of ^{12}C and ^{13}C mycotoxins, starting with deoxynivalenol; the flour will be used for the small scale production of bread, cookies and other bakery products **under real industry conditions in collaboration with BARILLA**. Furthermore, other toxins (e.g. ZEN, FB1, T-2) or even conjugates enzymatically synthesised from ^{13}C toxins (e.g. DON-3-Glc, HT2-3-Glc, NIV-3-Glc) can be assessed. In this way a sort of "in-process stable isotopic labelling" devoted to understanding mycotoxins fate during food processing will be brought to completion.

The relevance of this study will be that for the first time ever not only **masked/modified forms but also other (food processing) derivatives of mycotoxins** (neogenesis) **will be determined** in relevant matrices to obtain structural information of altered chemical compounds. This is the first step towards a true risk assessment of mycotoxins including all their modified forms and degradation products.

T2.5 Safe Use Options for Contaminated Batches (deactivators, biogas, bioethanol)

T2.5.1 Development of a degradation system related to biogas production (M6-M24)

Lead: BOKU *Partners: BIOMIN*

This subtask is related to biogas and in particular to determining the **effect of mycotoxin-contaminated cereals on the degradability of biomass** during anaerobic digestion. While previous studies have shown the degradability of mycotoxins, to date the degradation rates have not been analysed. To guarantee the complete degradation of mycotoxins, **specific established organisms and/or enzymes** will be added to the digesters. Organisms and enzymes will be provided by BIOMIN who have already successfully employed these in mycotoxin detoxifying feed additives. Degradation performance will be analysed and process parameters for anaerobic digestion of mycotoxin (DON, ZEN, Fumonisins) contaminated wheat and maize, respectively, will be developed using laboratory scale biogas reactors (CSTR = Continuous Stirred Tank Reactor) available at BOKU and subsequently transferred to

fermentation experiments at technical scale. **End user input** will be received through the Austrian Competence Center Bioenergy 2020+ where BOKU is area managing institution.

T2.5.2 Development of a degradation system related to bioethanol production (M6-M24)

Lead: BOKU: Partners: BIOMIN ; SUEDZUCKER

The use of even one mycotoxin-contaminated delivery in a bioethanol plant would lead to the contamination of the entire production charge, which is a big financial problem for bioethanol plants especially in view of the mycotoxin enriched **Dried Distillers Grains with Solubles** (**DDGS**), a bioethanol by-product, which is added to animal feed. Mycotoxin free DDGS would **reduce bioethanol production costs** and mitigate the current food versus fuel debate. The performance of mycotoxin-degrading organisms and enzymes provided by BIOMIN, will be evaluated during the fermentation of mycotoxin (DON, ZEN, Fumonisins) contaminated wheat and maize. Economic feasibility and intervention points for the addition of these organisms will be identified and test fermentations at laboratory scale at BOKU and subsequently at technical scale at the site of **end user SUEDZUCKER** will be carried out. This will allow the assessment whether contaminated grain batches could be diverted to the bioethanol fermentation industry and cleaned from mycotoxins by using the selected mycotoxin degrading enzymes.

Deliverables

D2.1	ATEX compliant sensor system for measurement of CO_2/RH/Temp in required sensitivity range	M18
D2.2	Final assessment of systems utilising mycotoxin degrading enzymes/microbes during biogas and bioethanol production	M24
D2.3	Integrated sensor platform for CO_2/RH/Temperature real time measurement including option for downloading of information by farm store managers	M36
D2.4	Final assessment of newly developed sorting technology for figs	M36
D2.5	Guidelines for effective grains micronization/wet extraction milling procedures	M36
D2.6	Report on the presence of mycotoxins and all their modified forms that are formed during food processing	M42

Work package No.	WP3	Start date or starting event:			M1			
Work package title	**Integrated MyToolBox e-platform for mycotoxin management**							
Particip. No.	02	04	06	07	08	10	13	17
Particip. short name	**DLO-RIKILT**	HAU	BARILLA	CRANFIELD	IRIS	HORTA	WU	ASAG
PM per participant								

Objectives

To limit mycotoxin contamination of the final products to the lowest level achievable, all actors in the entire production chain should apply and show utmost adherence to best management practices. This requires integration of all up-to-date pre- and post-harvest knowledge to develop a practical framework for providing solutions to farmers, storage managers and processers. Consequently, this WP aims

O3.1 To **evaluate Good Agricultural Practice (GAP) and Good Manufacturing Practice (GMP) guidelines**, available in different countries, and **Hazard Analysis Critical Control Point (HACCP)** based protocols for prevention and control of mycotoxins, and synthesise them into a user-friendly format for application in MyToolBox.

O3.2 To **develop forecasting models** to predict potential fungal and mycotoxin contamination of small grain cereals and maize at an early growth stage so that pre-emptive action can be taken to reduce both fungal infection and mycotoxin formation.

O3.3 To **test and optimise the final real-time post-harvest environmental monitoring systems for storage** of cereals and peanuts developed in WP2.

O3.4 To **adapt and integrate cost-effective sampling and monitoring strategies** for mycotoxin management over the entire chain, for inclusion in MyToolBox.

O3.5 To design, built, test and validate the **web-based MyToolbox platform** for facilitating decision support for mycotoxin management (prevention, control and monitoring) in feed and food chains of Europe.

Description of work (broken down into tasks), and role of participants

T3.1 GAP, GMP guidelines and HACCP protocols to prevent and control mycotoxin contamination (M1-18)

Lead: CRANFIELD ; Partners: DLO--RIKILT ; HAU ; Barilla

Mycotoxin management (prevention, control, monitoring) should be part of Good Agricultural Practices (GAP) in the primary stage, GMP and HACCP in consecutive stages of the food and feed production chain. To date, application of such guidelines and protocols is mainly restricted to DON or *Fusarium* toxins in small grain cereals, and particularly focusing on the primary production stage. The European Union has published principles to minimize *Fusarium* mycotoxins in cereals, to be used as a starting point by Member States for setting up national guidelines. So far, specific codes of practices have only been developed in the UK, completed by HAU, aimed to reduce *Fusarium* and OTA mycotoxins in cereals.

Available GAP and GMP guidelines and HACCP protocols to prevent and control mycotoxins will be gathered, structured, and evaluated. Also, knowledge gained in previous European projects will be collected and synthesised. This will be done for each of the mycotoxins/chains under investigation, including small grain cereals (wheat, barley, maize, and dried fruit. **Critical Control Points** (CCPs) **will be determined** for each of these chains, as well as action limits and corrective measures for the identified CCPs. Results from WP1 and WP2, which will provide new information for MyToolbox during the project, will be gathered and included in the available guidelines as well. In this way, all available information will be synthesised for *inclusion in the static component* of MyToolBox (T3.3).

T3.2 Forecasting for optimal farm management in the EU and in China (M1-M40)

Lead: DLO-RIKILT ; Partners: HORTA ; ASAG

The **use of predictive models** for mycotoxins enables **better informed decision making** on proper farm management, such as the use of varieties, crop rotation and soil treatment (e.g. ploughing), and use of (bio)fungicides for control. Depending on the aim, forecasting models can be used for short term decisions, like use of fungicides around the critical period of cultivation and optimal harvesting time; for mid-term decisions like areas for more intensive control on mycotoxin contamination in batches; and/or for long-term decisions like changes in farm management, e.g. on crop variety. Currently, operational models are available for forecasting the presence of DON in wheat, either mechanistic or empirical, but **not for other toxins/crop combinations**. Also, these models are restricted to certain European areas, and not dedicated for specific users (actors in the chain).

In this tasks, forecasting models will be collected and further developed for **inclusion in the dynamic component of the web-based MyToolbox platform.** Model development will start from available models, which will be extended and tested for other countries. The task will start with a literature evaluation on available predictive models in small grain cereals and maize, and evaluation of model characteristics. Then models will be selected for extension (to other countries, and to include ZEN in DON models) and further development, and will be tested in other European regions. The final models will be *linked into the dynamic component of MyToolBox* for interactive use by farmers and advisors.

Available models are either empirical or mechanistic, or a combined modelling approach is used. The decision on the modelling type will depend on the available data in relation to user requirements. When multiple models are available for a particular mycotoxin/crop, e.g. in case of DON in wheat, a model envelope approach will be used, in which multiple models will be run, and their outputs combined for instance, on Fuzzy sets.

Models will need input on weather data, agronomic data, and grain samples results for the particular mycotoxin/grain. **Weather data will be collected from national weather stations, open available** (see for instance: http://eca.knmi.nl/ from which weather data from entire Europe can be retrieved**) or from other open sources**. Agronomic data as well as data on mycotoxins in grain samples will be collected via project partners (HAU, HORTA, DLO-RIKILT) and will be further collected in the course of the project.

The focus will be upon:

- Farmer model for *Fusarium* spp and **DON and ZEN contamination of wheat**, for the main wheat growing areas of Europe: available forecasting models for DON in wheat, which are mostly developed on the national basis (e.g. available for France, Belgium, Netherlands, Swiss, Italy, Czech Republic, and one for North West Europe), will be collected and evaluated for integration in MyToolBox. The available models will be used as the starting point for modelling and these will be linked into the MyToolBox platform. However, available models are scattered, regarding regional coverage and modelling approach used (mechanistic vs empirical). Since, not for each relevant country, a model is available (e.g. Germany, Poland), available models will be tested for these other countries. Furthermore, the models will be extended to also include ZEN, which often co-occurs with DON. The farmer's model for DON and ZEN in wheat growing areas of Europe **will help famers to determine the optimal application of fungicides and optimal harvest time**. After extension and testing, models will be validated, which will include collection of data (farm management, weather) and wheat samples at harvest, which will be chemical analysed for DON and ZEN concentrations.
- **Forecasting presence of *Fusarium* spp. and DON contamination in wheat grown in the Anhui and Hebei**

provinces of China (ASAG, DLO-RIKILT). Weather data, including forecasts, will be collected from these provinces as well as wheat samples at harvest. These samples will be analysed for DON content (ASAG). The available data will be used for developing a forecasting model for DON contamination of wheat grown in these provinces. To this end, experiences gained in the previous task, as well as available models will be shared (DLO-RIKILT). The developed model will be **tested by two end-users, located in the Anhui province**, for pre-harvest forecasting (ASAG). Currently, ASAG is also developing its own forecasting models and therefore exchange of expertise between MyToolBox and ASAG in this area would be of mutual interest to improving the forecast of DON levels in wheat. In this area ASAG will also **involve two end users, i.e. the farmers** Luo Jiakui and Lin Qiyin who are located near Mingguang City in the Anhui province, where mostly wheat is grown.

- **Forecasting model for T2/HT2 in oats**, **to be used by farmers**. Available models, developed in Scandinavia, will be provided by BIOFORSK for integration in MyToolBox.
- **Forecasting models for fumonisins and aflatoxins in maize** for central Europe (maize growing areas) (DLO-RIKILT, HORTA).

T3.3 DSS System post-harvest: Pilot and full scale silo trials in the EU and in China with integrated platform (M 36-48)

Lead: IRIS *Partners: CRANFIELD ; BARILLA; HORTA:; ASAG;*

Pilot scale testing and fine-tuning of the developed integrated **real-time postharvest DSS system for grain stores** and silos with different stored cereals as a management tool will be performed in T3.3. Also, the optimal number of integrated sensors required for each grain store/silo will be determined. **Trials will be conducted to test the prototype system in the EU, Turkey and in China**. More specifically, the postharvest DSS system will be evaluated by or via Barilla in the EU for durum wheat (Turkey and Italy) and via HORTA for maize in Italy. With respect to **testing the DSS systems** at an **end users level in China,** ASAG will provide **access to** its **grain silos** (200–7000t) **for wheat** in the Anhui & Hebei province **and for maize** in the Jilin & Hebei province and in the Sichuan province. The involvement of different geographical regions enables MyToolBox to include a variety of food crops and mycotoxins occurring in China, such as DON, aflatoxins and fumonisins. ASAG will provide access to its grain silos to install the real-time sensors to be developed within MyToolBox and to study their functionality and their corrosion resistance especially against the fungicides used in these facilities. Through ASAG, also **two end users from the Chinese grain industry** will be involved in post-harvest monitoring, i.e. Hebei Baoding Qinghuaxufeng Trade of Grains and Oil Co. Ltd. and the China Grain Reserves Corporation, which owns most of the grains and closely cooperates with ASAG. Moreover, ASAG has also got storage facilities for peanuts in the Shandong province where it collaborates with Hualong (Rushan) Foodstuffs Enterprise Co. Ltd., the leading company of Chinese peanut industry both in production and export which has already expressed its interest to serve as end user to MyToolBox. ASAG will coordinate the installation and maintenance of the developed real-time post-harvest sensors in cooperation with these partners from the grain and peanut industry and the MyToolBox participants.

T3.4 Optimization of the mycotoxin monitoring schemes for the entire chain (M6-M24)

Lead: DLO-RIKILT *Partners: WU*

This task aims to optimise mycotoxin sampling and testing in the chain, while minimising the costs, using a modelling approach based on existing information (**no experimental work involved**). **Various types of detection methods**, such as instrumental methods, ELISA, and dipstick methods, will be evaluated. Each type of detection has its own characteristics, such as percentages of false-positives and negatives, and related costs. **Using optimisation modelling, the sampling and testing of mycotoxins will be optimised**, resulting into the number of samples to be taken at each point in the chain, rate of pooling of samples, and analytical method to be used. Input from WP1 and WP2 will be used in this task. The focus will be on *Fusarium* mycotoxins in small grain cereals and maize. **Results will lead to guidelines/advice on sampling and analyses**, including optimal points for testing along the chain, for use by supply chain actors. The guidelines will be implemented in the static component of MyToolBox at the end of year two, and verified in years 3 and 4.

T3.5 Design and building of the MyToolBox e-platform (M1-M48)

Lead: HORTA; *Partners: DLO-RIKILT CRANFIELD, IRIS*

This task encompasses the **design, building, testing and validation of the internet-based MyToolBox e-platform:** an interactive management and decision support tool for all actors within the chain. The MyToolBox e-platform will be developed and integrated in detail for mycotoxins in small grain cereals and maize production chains, and its working principle will be demonstrated for dried fruit. The tool will be very flexible so that it can later on (after the project) be extended to other mycotoxins and chains. **The development will be in close interaction with the stakeholders** (in WP4), which will provide user aims and requirements for the design and development. In

years 3 and 4, they will test the first version of MyToolbox (WP4). MyToolBox will integrate currently available information and make it accessible to the user. Results from WP1-4 which become available during the project will be added. Amongst others, **the forecasting models** of T3.2 **(dynamic part), the optimised monitoring schemes** of T3.3 **(static part),** and **the real time post-harvest storage management systems** of WP2-3 **(dynamic part) will be linked into web-accessible MyToolBox e-platform**. The tool will be specific to the activities of each particular actor within the chain, and will enable the actor to direct batches to safe-use options based on CCPs for mycotoxin contamination in the specific production system. Furthermore, it will provide: guidelines for GAP, GMP and HACCP; potential corrective measures in case action levels or EU legal limits are exceeded; and recommendations on the best choice of preventive and corrective measures (including inspection). For batches with contamination above certain action limits, recommendations for alternative safe use will be provided.

This task is divided into **three subtasks**:

T3.5.1: **System architecture and specifications**.

This sub-task includes the following activities: definition of the MyToolBox architecture, technological cloud infrastructure and cloud hosting; definition of the operative system to be used for the app; definition of inputs and outputs APIs (Application Program Interfaces) and structure of the databases (for crops characteristics, weather and environmental data, etc.); formalisation of the model's algorithms (models from WP2 and WP3); definition of the front end GUI (Graphical User Interface) and adaptive features (access by PC, tablet and smartphone devices); definition of the graphic layout, mock-up development and approval (screenshots or 'mock ups' of User Interface concepts will be made available to our end-users/clients for feedback, discussion and approval prior to front-end coding in sub-task 3.5.2); connection and relationships with the project web-site.

T3.5.2: **Development of the e-platform**.

This sub-task includes the following activities: setting up of the user's authentication module (registration, authentication, profiling); creation of the data flows and configuration of the databases; development of the inside components (dynamic models running in MyToolBox as internal web services; the models formulated within and supplied by WP2 and WP3 will be converted to executable code in the appropriate computer programming language) and the outside components (**dynamic models entering in MyToolBox via web services**); development of the front end GUI for data entry and data viewer for different users with restricted access, i.e., registered users and administrators; interoperability testing among dynamic models (data exchange testing); setting up of the static pages (static contents from WP1 and T3.3, with public access); development of the app to access the MyToolBox authentication module in order to provide easy and fast access to the platform (the app will be developed for compatibility with one of the following standards: android, iOS and windows operating systems); development of info-graphic and visual reporting tool as interpretation guide, ensuring easy to interpret information for users; optimisation work and scale up of the platform.

Sub-task 3.5.3: **Verification and validation**

This sub-task includes the following activities: testing of the data flows and MyToolBox outputs against the original components (expected outputs); testing with partners and with a panel of users in close collaboration with WP4.

Deliverables		
D3.1	Alpha release of the MyToolBox e-platform	M15
D3.2	Guidelines on optimal, cost-effective schemes for monitoring mycotoxin contamination and synthesis of available GAP, GMP guidelines and HACCP protocols for mycotoxin management in small grain cereals, maize and figs	M24
D3.3	Forecasting models (for various combinations of mycotoxins/crops) for optimal harvest timing and use of (bio)fungicides	M40
D3.4	Report on trials with the DSS platform in different silos systems in different regions (including China) by post-harvest grain/nut managers	M40
D3.5	Tested and validated web-based MyToolBox platform for integrated mycotoxin management and decision support tool kit for all actors in the chains of small grain cereals, maize and figs available	M48

Work package No.	WP4		Start date or starting event			M15			
Work package title	Practical Implementation and Contribution to Standard Setting								
Participant No.	2	5	6	10	11	13	16	17	23
Participant short name	DLO-RIKILT	**FLI**	BARILLA	HORTA	BFR	WU	BIOMIN	ASAG	CAAS-FRI
PM per participant	2	15	4	1	13	18	2.5	8	12

Objectives

The objectives of WP4 are focussed on ensuring that the proposed interventions to reduce levels of mycotoxins are acceptable to end-users, barriers to implementation are overcome and that the MyToolBox is user-friendly and meets user expectations. Implications from the project results, in terms of standard setting, will be exploited to improve industry competitiveness.

O4.1 To **evaluate the elements of the MyToolBox** developed in WP1, WP2 and WP3 in terms of their practical viability for **different end-users** and their performance in terms of preventing and reducing levels of mycotoxins.

O4.2 To **evaluate the application of the MyToolBox project (individual measures and web-based MyToolBox platform) for mycotoxin control** over the entire chain, given the variation in risk behaviour of actors along the chain

O4.3 To **develop recommendations to increase compliance** for control and prevention throughout the entire chain including the use of incentives.

O4.4 To **anticipate legislative implications of project outputs** (MyToolBox) and work with relevant bodies such as EFSA and DG Santé for standard setting.

O4.5 To establish the performance of mycotoxin-detoxifying feed additives under local conditions in China in order to **enable standard settings in China for authorisation of** these **feed additives**, and thereby improve market access.

Description of work (broken down into tasks), and role of participants

T4.1 Evaluation of the developed MyToolBox e-platform at an end user's level (M18-M48)

Lead FLI 13 PM *Partners: BARILLA 4PM; DLO-RIKILT 1PM; HORTA 1PM; ASAG 8PM*

Essential to the 'MyToolBox' project is the principle of **maximum involvement of end-users** within each WP at early developmental stages, to ensure that proposed tools (interventions) are viable in practice and are likely to be attractive for ultimate implementation. The **interactions with end-users has already started while setting up the proposal** and will continue at the start of the project. This close cooperation will be part of an iterative process orchestrated by partner 5 (FLI) to ensure that developments in the project MyToolBox are consistently realistic in terms of practical viability and also reflect to perceived requirements of end-users. The design of the MyToolBox e-platform itself (essentially a dissemination vehicle) cannot wait until all the individual tools have been developed and optimised, but will be conceptualised at an early stage. **During conceptualisation of the MyToolBox e-platform, views will be sought from a variety of end-users** through the use of Focus groups, which will feed into WP3 as an aid to improved design. The tools will range from generic advice encompassed in GAP and HACCP, through to practical measures which can only be tested and evaluated after substantial changes to existing practices and capital investment (use of alternatives to conventional fungicides, biocontrol measures, improved sorting, changes in milling of grain, changes in the baking processes). The individual measures will be evaluated in different ways (mostly face-to-face interviews and use of questionnaires) with end-users and stakeholder groups at relevant points in the food chain and in different countries. **Questionnaires and a structured format for interviews will be developed** by partner 5 (FLI), but in close collaboration with partner 13 (WU) to ensure adequate coverage of socio-economic aspects and also ensuring that the feedback covers areas of interest specifically raised by individual partners in WP1, WP2 and WP3. T4.1 will include examination of the potential for the exploitation of the DSS system developed in WP2 as an effective integrated management tool for the stored grain industry post-harvest for minimizing mycotoxins present during downstream processing. In T4.1 partner 6 (**BARILLA) will operate as a food industry end-user providing contacts** inside the company both internationally and at various stages of the food processing chain, partner 3 (ICC) will facilitate interactions internationally with end-users in the cereal industry as well as facilitating training indicated in WP5. Partner 21 (TARIS) will provide feedback specifically as a dried fruit end-user. Partner 17 (**ASAG) will facilitate Interactions with end-users in China** which will include China Grain Reserves Corporation, Hebei Baoding Qinghuaxufeng Grain and Oil Co. Ltd and Hualong (Rushan) Foodstuffs

Enterprise CO., Ltd. the latter who will act as a peanut production end-user. Both partners 15 and 17 will assist with language translation with end-users as appropriate. **The assessment of the MyToolBox measures including the web-based platform will be conducted continuously on** an on-going basis, feeding into T4.2 to aid the socio-economic analysis, which in turn will feed into WP3. The prototype 'MyToolBox' after refinement will be finally re-evaluated with a range of end-users including some involved in early-stage evaluations.

T4.2 Socio-Economic and Stakeholder Analysis (M12-M48)

Lead WU 18PM; Partners: DLO-RIKILT 1PM

T4.2 **will analyse how the risk behaviour of individual actors along a supply chain can affect the cost-effectiveness of proposed control and prevention measures**. This study will use a principal-agent approach in the stochastic estimation of the cost and benefits/effectiveness of the individual actors as well as of the whole chain (by simulation studies), given the defined interventions. As the cost-effectiveness of mycotoxin interventions in the chain depend on the uptake level of the various actors (= principal agent problem), it is of importance to have an estimation on the expected uptake level as well as insight in the factors that will drive the level of uptake (based on output of T4.1 in combination with in depth questionnaires). These insights provide the window of opportunities with respect to the development of incentive mechanisms aimed "to induce the supplier to apply measures as the buyer requests" in an attempt to optimise chain control. **Key elements of incentive mechanisms will be identified** by a literature review on actual and theoretical incentive mechanisms (e.g., quality premiums, discounts, tournaments, production contracts, regulation) aimed at food quality and food safety control. Relevant incentive mechanisms will subsequently be evaluated for their expected impact on costs and benefits/effectiveness by the use of simulation studies. Based on these study results, recommendations will be developed to stimulate the uptake of interventions throughout the chain.

T4.3 Legislation and Standard Setting for foods (M36-48)

Lead BFR 13PM; Partners: FLI 1PM

T4.3 will **focus on analysis of the actual regulated maximum levels of (selected) mycotoxins** in food, at first instance Regulation (EC) No 1881/2006, Directive 2002/32/EC, Commission Recommendation 2006/576/EC) in view of the underlying processing factors and potential transfer rates. Undertaking a compliance assessment of maximum levels in the feed/food chain for selected examples (data evaluation/modelling) which will lead to **identification of points to be changed in legislation/ standards** in order to implement the MyToolBox e-platform and analysis/discussion of benefits and drawbacks and making regulatory impact assessment (Month 12-48). After development of risk-based assessments, meetings will be sought with relevant EU policy-makers **in EFSA and DG Santé** to discuss legislative implications. A minimum of 3 meetings are envisaged between months 36-48.

T4.4 Standard Setting for Feed Additives (M6-M47)

Lead FLI 1PM; Partners: BIOMIN 2.5PM; CAAS-FRI 12PM

T4.4 **aims to contribute to the standard settings for authorisation of mycotoxin-detoxifying feed additives in China** aided by the expertise of stakeholder FEFANA in cooperation with CAAS-FRI and the Chinese Ministry of Agriculture (MOA). To this end, feeding trials to evaluate the efficacy of mycotoxin detoxifying feed additives in target species are crucial. In the last decade, the focus of such *in vivo* studies has shifted from the sole assessment of performance data to the determination of specific mycotoxin biomarkers in biological matrices of exposed animals. The latter allows the demonstration of an increased excretion/reduced deposition of mycotoxins and therefore represents an important aspect for the registration of mycotoxin detoxifying feed additives from the EU.

The efficacy of feed additives to reduce the absorption of dietary aflatoxin B_1 (AFB1) in dairy cows (M6-M15) and absorption of dietary fumonisin B1 (FB1) in pigs (M15-M24) will be investigated with *in-vivo* experiments conducted at the research facilities of the Feed Research Institute at CAAS (CAAS-FRI) **under local conditions in China**. The feeding trial will be performed according to Commission Regulation (EC) No 429/2008 of 25 April 2008 on the preparation and the presentation of applications and the assessment and the authorisation of feed additives. (a) Based on these guidelines a 4x4 Latin square design will be applied, and dairy cows will receive four treatment (control diet, AFB1 contaminated diet, AFB1 contaminated diet [preferably naturally contaminated material will be employed] supplemented with an EU authorised feed additive, AFB1 contaminated diet supplemented with a local feed additive). Due to the absence of respective official or advisory limits in China, maximum dietary mycotoxin concentrations (aflatoxin B_1, AFB_1; fumonisin B_1, FB_1) will be set on the basis of Directive 2002/32/EC and Commission Recommendation 2006/576/EC. Used animal numbers will be defined by statistical sample size calculations. Aflatoxin M_1 concentration in milk will be quantified by a validated LC-MS/MS method to evaluate the efficacy of the feed additives. In addition, hematological, immunological and performance parameters will be

assessed. (b) In a randomised block design, weaned piglets will be assigned to four treatment groups (control diet, FB1 contaminated diet, FB1 contaminated diet supplement with an EU authorised feed additive, FB1 contaminated diet supplemented with a local feed additive). Mainly, the sphinganine to sphingosine ratio in serum will be evaluated by LC-MS/MS. In addition, FB1 and hydrolysed FB1 (HFB1) levels in serum and faeces as well as haematological, immunological and performance criteria will be determined.

Based on these feeding trials and in cooperation with CAAS-FRI and MOA, we **aim to contribute to the standard settings for authorisation of mycotoxin-detoxifying feed additives in China**. Current guidelines (e.g. from EFSA) for the registration of detoxifying feed additives in the European Union will serve as examples for possible adaptions of Chinese legislation. Our activities shall help European producers of mycotoxin-degrading feed additives to **gain broader access to the Chinese market as well as to contribute to an improvement of Chinese legislation**.

Deliverables

D4.1	Report providing end-user feedback to WP1, WP2 and WP3 partners on preliminary assessment of proposed interventions and initial MyToolBox e-platform design	M18
D4.2	Report on preliminary end user evaluation of MyToolBox for application to cereal industries and to the feed supply chain and alternative uses of contaminated commodities	M34
D4.3	Report on standard-setting implications of animal feeding trials with detoxifiers conducted in China	M36
D4.4	Report setting out basis for establishing processing factors for mycotoxins based on transfer rates and overview of end-users assessment of final version of MyToolBox	M45
D4.5	Report covering socio-economic impact of MyToolBox including recommendations for incentives	M45
D4.6	Report with recommendations to regulators resulting from mycotoxin reduction strategies developed in MyToolBox	M47

Work package No.	WP5	Start date or starting event				M1	
Work package title	**Exploitation, Dissemination and Communication**						
Participant No.	01	03	08	15	17	Other core partners	All other partners
Participant short name	BOKU	**ICC**	IRIS	CAAS-IAPPST	ASAG		
PM per participant							

Objectives:

O5.1 To **disseminate information** about the project and its results **to >5000 actors and stakeholders along the food supply chain**, to opinion leaders/regulators, the media, professionals working in agriculture and food manufacture, educators, and other multipliers, food and drink industries (including SMEs), consumer organisations, the scientific community and the general public.

O5.2 **To facilitate collaboration, information exchange and interaction between different actors** and organisations involved in the control and management of mycotoxins in the food supply chain through the presentation of MyToolBox at international conferences (~8) and workshops and the organisation of 4 webinars and 4 training sessions for end user.

O5.3 To **develop exploitation plans, including a business model**, to ensure the use of the project outputs among the farming and industry end users (including farmers and food processors), opinion leaders, decision makers, etc. and to establish and verify a robust business model for the MyToolBox platform.

O5.4 To strengthen international cooperation in research and innovation for mycotoxins and to **ensure a sustainable dialogue with China.**

Description of work

T5.1 Knowledge Management Activities (M1-M48)

Leader: IRIS *Partners: All partners will play a role.*

The strategy for knowledge management and protection that is discussed in Section 2.2 will be implemented, whereby **all knowledge that is generated during the project will be assessed for its novelty** and to enable the most appropriate **IPR strategy to be defined**. All patentable items of the project results will be patented by the owner or joint owner(s) of the results. Internal expertise in relation to patent applications within the consortium will be mobilized under the leadership of the IPR Manager from IRIS. In keeping with the rules outlined in Section 2.2 in relation to IPR ownership and access rights, which **will also be agreed in the Consortium Agreement**, the IPR

Manager will work closely with the knowledge holders to assist in fulfilment of the IPR rules, as well as negotiating any additional agreements (e.g. Joint Ownership Agreements) as necessary in line with developments as they unfold. At M24, a Knowledge Management report will be produced and the status of IP protection and any Joint Ownership agreements will be reported (D5.5).

T5.2 Data management (M1-M48)

Leader: BOKU *Partners: IRIS. All partners will play a role.*

A Data Management Plan (DMP) will be delivered at Month 6, following the H2020 Guidelines. This plan and includes the **participation in the Pilot Open Access Research Data Scheme** (D5.2) and will serve as a blueprint for guiding all the partners in sound data management procedures and will outline: which data will be generated by the project; data standards and quality assurance measures; how data will be exploited and shared; copyright and intellectual property rights of data; data storage and back-up measures; data management roles and responsibilities. The DMP will be co-developed by the Coordinator and IRIS with the support of the WP leaders, followed by a contribution by all partners involved in the activities.

T5.3 Communication activities and multi-actor engagement

This task will ensure the **communication activities** defined in Section 2.2c are built upon, **implemented and their impact continually assessed** to ensure they are effective in informing, engaging, create awareness and promoting information about the MyToolBox project, its aims, its funding source, its outputs and impacts to our target audiences. This task is divided into the following subtasks:

T5.3.1 Communication planning, implementation and reporting (M1-M48)

Leader: ICC *Partners: IRIS will support heavily with their marketing team, which includes digital marketing professionals. All partners will contribute by performing communication activities.*

A Communication plan (D5.3) will be developed (building on the preliminary outline discussed in Section 2.2 b), and appropriate activities will be implemented. Key messages and target audiences as well as appropriate tools and channels (as mass media, social/digital media, demos, pilots, posters, brochures, newsletters and EU dissemination resources such as the EU Horizon Magazine) to meet the needs of the target audiences will be developed at the beginning of the project and updated throughout. This will seek to ensure media interest and coverage from the outset of the project right through, and beyond the project's completion. IRIS will lead the branding and brand development strategy in consultation with all the partners. The communication plan will be implemented and all activities will be reported in D5.6.

T5.3.2 MyToolBox identity, website and promotional materials (M1-M48)

Leader: ICC; *Partners: IRIS, BOKU + all partners will contribute inputs, ideas and expertise.*

The **project website** (D5.1) will be available at the following domain, which has already been registered **www.mytoolbox.eu** . It will be built and maintained by ICC and it will follow the common graphic theme developed by IRIS for all project communications, including the project logo and visual theme in order to build a strong brand awareness for MyToolBox. ICC takes care of hosting the site during the project and a plan will be in place for maintenance of the site after the project's end (defined from T5.5.2). **The public part** will be actively maintained by ICC. It will consist of **well-presented non-confidential project information** including project concept, its partners, core objectives, work areas, project videos and expected impacts. Post-project, the relevant public information and results will remain available on this site.

Project leaflets, posters, newsletters and other promotional materials such as project videos, brochures, etc. will be produced in order to support project communication and dissemination efforts. These materials will be reported and annexed to the Communication Report (D5.6).

T5.3.3 Multi-actor engagement (M1-M48)

Leader: ICC; *Partners: CAAS-IASSPT; ASAG M; All partners*

Maximum involvement of end-users within each WP of MyToolBox will be established at early developmental stages, to ensure that proposed tools (interventions) are viable in practice and are attractive for ultimate implementation as described in WP4. The impact of MyToolBox will be maximised through connecting to ICCs **world-wide network** of associations, cooperatives and industrial partners in the agro-food & feed area in WP5, to establish a direct global dialogue with end-users including those from China. The **EU-China-Mycotoxin Forum** will be established as a dialogue platform between China and the EU to provide building-blocks for further EU-China dialogue across a wider range of food safety topics. It will be chaired by **Prof. Yang (China)** and the MyToolBox coordinator **Prof. Krska (EU)** and co-chaired and supported **by ICCs current president Prof.Fengcheng Wang** who also serves as director of the Research Center for Cereal, Oil and Food of the State Grain Administration of

China. This Forum will be reported in Final Plan for the dissemination and exploitation of the project's results (D5.6). The Advisory Board, which is composed of policy makers, scientists and stakeholder from the industry, will ensure active involvement of key stakeholders and scientific experts to support strategic project decision making and finally Standards development and policy advice for "Better Future Regulations" as described in WP4. Exchanges will be mobilised with other networks, such as **the International Society for Mycotoxicology and the Society for Mycotoxin Research** as well as with projects in similar areas and synergies will be used. The activities performed in relation to multi-actor engagement will be reported in the Communication Report in D5.6.

T5.4 Dissemination to a wide audience (M6-M48)

Leader: ICC; *Partners: IRIS and all partners will be involved in actively disseminating the project results*

The project partners will disseminate the project results and the newly developed tools as defined in the communication plan and with participation in conferences (~8) and organisation of workshops (~4), etc. – targeting the relevant stakeholder audiences and multi-actors, including the research community, SME and industry, policy makers, interest groups as well as the general public (D5.7). The final project meeting will be organised **in conjunction with an internationally established mycotoxin conference such as the IUPAC or ISM conference on mycotoxins**, which has already been chaired twice by the MyToolBox Coordinator. A satellite workshop (D5.9) organised by ICC will enable the wrap up of the project, to disseminate new findings and to demonstrate the newly developed tools, and to outline a Roadmap for a future research and cooperation agenda.

In keeping with H2020 guidelines, partners will also publish their work in peer reviewed publications.

T5.5 Preparation for the post-project use of the results

We will perform a structured series of activities to gear up for the future exploitation and deployment of the project results in the crop and food industry value chains and in the global market. The following subtasks have been planned:

T5.5.1 Business and exploitation plans (M6-M48)

Leader: IRIS; *Partners: All partners will be involved in planning how they will exploit their results.*

As discussed in Section 2.2b, the **proposed exploitation routes for the project results are**: a) Commercial exploitation, whereby we will use the results in the production and commercialisation of a new technologies, tools and services for the global markets; b) **Further research**- especially in the context of research required for the transfer of the developed technologies to other applications; c) **standardisation activities**- for securing standardisation of the MyToolBox technologies and for improving food and feed safety standards through improved mycotoxin management.

We will assess, by means of multi-actor interaction and feedback, the sustainability and implementation possibilities of the newly developed tools in business cases. We will also assess the potential use, impact and policy implications of the new findings and newly developed tools (which will be linked with WP4). We will plan the transition and up-scaling of innovations based on project results. Linking in with T5.1, IPR issues will be assessed, as will open access possibilities, patent and/or commercialization of products and services for sustainability of project results. **Business plans will be developed for outputs that show high potential for sustainability** and/or commercialisation, such as the use of the MyToolBox platform for the development of data services, as well as the silo monitoring solution from WP2 as a new product, whereby IRIS will perform an up-to-date market analysis of the current market offering in relation to grain silo monitoring solutions, as well as to define the Unique Selling Points (USPs) for positioning of our DSS platform in the market. IRIS will draw up a business plan for the exploitation of the DSS system as an effective integrated management tool for the stored grain industry post-harvest for minimizing mycotoxins present during downstream processing. Plans for future research or spin-off projects will also be elaborated. **Post-project dissemination activities** will also be planned in order to ensure continuity and outreach of the projects activities and results to the target audience. This will ensure the non-confidential results and our post-project developments continue to reach a wide audience post-project. Drafts of these plans will be included in the Interim 'Plan for the dissemination and exploitation of the project's results' and final versions in the Final Plan at M48 (D5.4 and D5.6 respectively).

T5.5.2 Technology Watch & Links with other Relevant Research Projects (M1-M48)

Leader: IRIS; *Partners: BOKU. All partners will contribute to technology watch actions.*

This work will take account of any relevant past or on-going research activities from the literature and at EU and national level. A preliminary table (table 3) of projects has been provided in Section 1.3 and additional relevant projects from H2020, FP7, FP6, Eurostars, LIFE, MANUNET, and national projects etc. will be identified and their

compatibility to link into MyToolBox will be analysed. **This action will be on-going throughout the project** to ensure that relevant research developments from third parties, along with technological developments and innovations, are identified analysed and, if relevant, their results can be built on in the project. The partners will aim to mobilise joint activities and manage synergies with relevant research projects. A "Technology Watch" repository will be populated and will be periodically updated with new literature, articles and reports resulting from this on-going task. Major relevant findings will also be documented in a dedicated section of the 'Plans for the dissemination and exploitation of the project's results' (D5.4 and D5.6) and their impact on the future industry adoption of the project outputs will be analysed.

T5.6 Training activities (M12-M48)

Leader: ICC *Partners: CAAS ; All partners will assist in the preparation of the training activities*

Specific topical and **regional training workshops** (at least 4) will be organised, to ensure that there is sufficient knowledge transfer opportunity during the project. Each workshop will be organised by ICC in a different country and will focus on different aspects of MyToolBox (e.g. biocontrol of maize in Serbia and real time post-harvest monitoring of grains in China). Ideally, these meetings will be organised in conjunction with a Project Management Board Meeting or a Steering Committee Meeting. The training, in which up to **50 participants and especially end users**, including farmers, can participate, will focus on the operational and wider socio-economic benefits of the newly developed tools. Four **webinars** will be offered to support the interaction with, and the training of actors of concern, as well as providing demonstration activities of newly developed tools. The training activities will be reported in D5.7.

Deliverables		
D5.1	Project website- public website available at www.mytoolbox.eu	M4
D5.2	Data management plan- standard plan following the H2020 guidelines	M6
D5.3	Initial Communication Plan- written report	M6
D5.4	Interim Plan for the dissemination and exploitation of the project's results- written report with initial Business Plans annexed	M24
D5.5	Knowledge Management Report- written report	M36
D5.6	Final Plan for the dissemination and exploitation of the project's results (with Communication report annexed)- written report with all dissemination and communications materials, articles in magazines, etc. and final Business plans annexed, along with a report on the EU-China Mycotoxin-Forum).	M48
D5.7	Report on Training activities (with materials annexed)	M48
D5.8	Satellite workshop at final conference- Proceedings from the final conference	M48

Work package number	**WP6**	**Start date or starting event:**	**M1**
Work package title	**Coordination and project management**		
Participant number	1	2	3
Short name of participant	BOKU	DLO-RIKILT	ICC
PM per participant:	29	5	2

Objectives

The objectives of WP6 are the overall scientific and managerial coordination of the MyToolBox-project. WP6 will be involved in aligning the outcomes of the work packages to ensure the sustainability of the developed decision support system MyToolBox. WP6 will also deal with the financial, administrative, and ethical management of the MyToolBox-project. Key is time- and cost-efficient implementation of the work plan.

O6.1 To **provide leadership, project management and effective scientific, administrative and technical coordination** to ensure MyToolBox delivers high quality outputs in an appropriate and timely manner and within budget.

O6.2 To provide and apply **methodologies for effective coordination** of all project activities and maintenance of contact with EC representatives.

O6.3 To undertake **quality assessment of project progress, results and impact**.

O6.4 To achieve **timely submission of progress reports and cost statements** to the commission.

O6.5 To ensure the **Description of Action (DoA)** and the **Consortium Agreement** are maintained.

O6.6 To create a **project management intranet** for inter-consortium sharing of information, databases and project knowledge management.

Description of work (broken down into tasks), and role of participants

T6.1 Coordination

Lead: BOKU 10PM *Partners: DLO-RIKILT 3PM*

The coordination task concerns the overall coordination of the technological and scientific developments within the different WPs to ensure alignment of the developed products and scientific findings. The **Project Coordinator** (BOKU) is responsible for contact with the EU representative to communicate all results, scientific and financial, and related issues. The Coordinator (BOKU) will be assisted by a Project Manager (BOKU) and a Scientific Manager (**Deputy Coordinator** from DLO-RIKILT).

The Coordinator and Scientific Manager will monitor the project progress and risks with the **Project Management Board** (PMB) every 6 months and with the **Project Steering Committee (PSC)** (see T6.3) every 12 months and at *ad hoc* meetings when necessary (task T6.6). The Project Manager will assist the Coordinator in administrative issues related to reporting and finance. The Project Manager is responsible for the timely delivery of draft scientific and financial reports to the Project Coordinator. The Scientific Manager will monitor the progress of the milestones and deliverables as agreed upon in the work plan and will (pre)review all project reports.

T6.2 Financial management

Lead: BOKU 6PM *Partners: All partners*

BOKU is responsible for the financial management of the project and for the timely distribution of funds to partners from the EC payments. It will present payment procedures and financial reporting aspects during the kick-off meeting. It will **provide excel-templates and guidance to all partners for preparing the formal financial reports**. It will also prepare and submit the financial reports to the EC. All consortium members are responsible for delivering their financial reports to BOKU complete and in-time. The Coordinator keeps records of the financial management, and informs the EC on any necessary financial transfers.

T6.3 Decision making, conflict management

Lead: BOKU 2PM

Each WP is led by one of the partners (WP leader) with a deputy WP leader from a different partner. This will ensure focus on the work that is agreed upon. The WP leaders take the routine decisions, communicate those with the Coordinator, monitor progress of output according to the work plan, and **prepare all documents and documentations** for their WP. The **decision-making body** will be the **Project Steering Committee,** with 1 representative from each partner organisation and chaired by the Coordinator (see details of composition in section 3.2), each partner has a vote and the ability to steer the project. Major strategic decisions will be made by the Steering Committee, such as change of scope, IPR related decisions, conflicts, budget related decisions, etc. First line of day-to-day decision making will be up to the **Project Management Board** whose membership will comprise the Coordinator and the WP leaders. Matters of a strategic nature are floated to the Project Steering Committee and will aim to be accompanied by a recommendation from the Project Management Board.

The Project Coordinator will handle any cases of fraud, unethical behaviour, and breach of procedures, however, such incidences are expected to be highly unlikely. The Project Coordinator will distribute a list of decisions to be taken to all participants prior to each meeting. Partners may request in writing for additional topics to be discussed, and the updated agenda is distributed before the meeting. All meetings are documented, including the outcome of all decisions and distributed to all partners.

T6.4 Organisation of reporting to the EC

Lead: BOKU; 6PM *Partners: DLO-RIKILT 2PM; All other partners;*

BOKU, the Project Coordinator, will **present reporting procedures during the kick-off meeting**. The Project Coordinator will provide templates for the formal reports and updates during the course of the project when necessary. The Project Coordinator will prepare a brief **6 month report** and distributes it to all participants. Formal project reports will be prepared by the Coordinator in keeping with the contractual reporting periods stipulated in the Grant Agreement. The Coordinator will present this formal report, including finance, IPR, ethics, and gender issues to the EC on time (and during formal review meetings as proposed by the EC). The Coordinator will also prepare a brief non-confidential version of the formal report for public access on the website.

T6.5 Day-to-day management (Internet, helpdesk and back-office)

Lead: BOKU 1PM *ICC: 2PM*

ICC, who will be building the **project website** (www.mytoolbox.eu) in WP5, **will also build a secure project intranet** (D6.2) utilizing an open-source content management system with common web technology, such as HTML 5. This secure internet will be employed **for all project internal communications** and will only be accessible by

consortium members. Different access levels will be incorporated based on user roles. It will provide tools for knowledge management, the development of databases and the mycotoxin control MyToolBox, to share preliminary results and internal documents amongst partners, to discuss and edit pre-publication documents and deliverables. This intranet will be maintained by BOKU. Shared documents, contact data, templates, agreements and links to official guidelines for reference and reports and conclusions of meetings will be stored on the password protected part of the website which will be maintained by BOKU.

T6.6 Organisation of consortium meetings (incl. telephone/video-conferences, web meetings)

Lead: BOKU 2PM

A **kick-off meeting** for all project partners including the Steering Committee (and the Project Management Board as part of the Committee), the Advisory Board and representatives from the EC will be organised by the Coordinator (BOKU. The back-office of BOKU will take care of all aspects of the consortium meetings, as presented in paragraph 3.2, such as agenda, invitations, preparation of materials (hand-outs, programmes, etc.), organisation of rooms and equipment, catering and dissemination activities to the consortium in cooperation with the host of the meeting, i.e. typically the location of one of the WP leaders. BOKU will also organise and host all consortium telephone/video-conferences (regular according to schedule, and on demand). In **total 5 Steering Committee meetings (the second will be organised in China)**, **4 additional Project Management Board** meetings and several ad-hoc meetings, depending on the needs, will be organised. 2 plenary meetings will be organised in conjunction with an end user meeting organised by partner 3 (ICC). The minutes of all meetings will be delivered in a timely manner by the Project Coordinator (D6.1)

T6.7 Maintenance of all legal agreements

Lead: BOKU 2PM

All partners will sign a Consortium Agreement, with detailed matters on access to know-how, intellectual property ownership, and liability in line with the information contained in Section 2.2b as the starting point. During the kick-off meeting all contractual requirements on the Grant Agreement will be reviewed with all the partners to ensure that all partners recap on their commitments.

All partners must report issues relevant to the legal agreements to the Coordinator in a timely manner. The Coordinator will decide on how to proceed with any deviation and will inform the Project Officer.

Deliverables		
D6.1	Minutes from KO, PMB and PSC meetings (M1 and thereafter)- confidential meetings of each meeting	M1-M48
D6.2	MyToolBox Intranet management tool- secure collaboration platform for project management, document repository and internal communication.	M2

The development of the web-based MyToolBox platform in interaction with the other WP activities is depicted in the Pert Diagram below (figure 4):

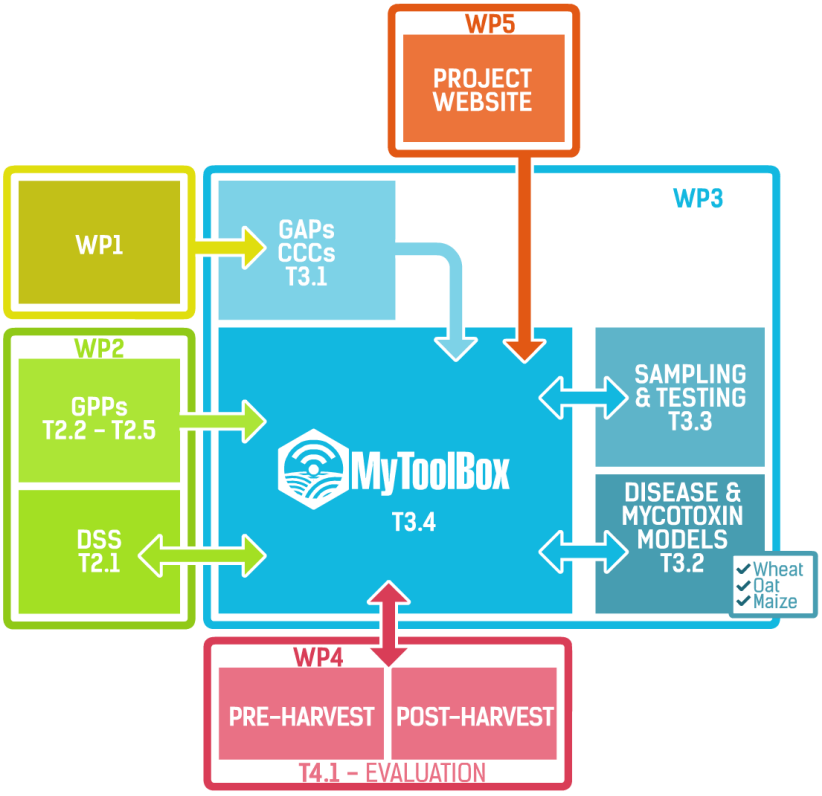

Figure 4: Pert Diagram (GAP - Good Agricultural Practice; CCP-Critical Control Point; DSS-Decision Support System; GMP-Good Manufacturing Practice)

Table 3.1b: List of Work packages

WP No	Work Package Title	Lead Participant		Persons Months	Start Month	End Month
		No	Short			
1	Pre-harvest reduction strategies	04/RTD	HAU		M1	M40
2	Post-Harvest Reduction Strategies	06/IND	BARILLA		M1	M42
3	Integrated MyToolBox e-platform for mycotoxin management (prevention, control and monitoring)	02/RTD	DLO-RIKILT		M1	M48
4	Practical Implementation and Contribution to Standard Setting	05/SME	FLI		M6	M48
5	Exploitation, Dissemination and Communication	08/SME	ICC		M1	M48
6	Coordination and project management	01/RTD	BOKU		M1	M48

Table 3.1c: List of Deliverables55

Delive-rable Nr	Deliverable name	Work package number	Short name lead participant	Type[56]	Dissemination Level	Delivery date
D1.1	Pre-selection completed of the most promising a) biopesticides, b) cultural control agents and methods and c) atoxigenic *Aspergillus* strains in consultation with end users and Advisory Board	WP1	HAU	R	PU	M16
D1.2	Guidelines for biopesticide use to reduce mycotoxins in European cereals	WP1	HAU	R	PU	M40
D1.3	Guidelines for cultural control of mycotoxins in European cereals through management of crop debris	WP1	HAU	R	PU	M40
D1.4	Guidelines for resistant cultivars to reduce mycotoxins in European maize	WP1	UNS	R	PU	M40
D1.5	Guidelines for use of atoxigenic biocontrol to reduce mycotoxins in European maize	WP1	UNS	R	PU	M40
D2.1	ATEX compliant sensor system for measurement of CO_2/RH/Temp in required sensitivity range	WP2	IRIS	DEM	CO	M18
D2.2	Final assessment of systems utilising mycotoxin degrading enzymes/microbes during biogas and bioethanol production	WP2	BOKU	R	CO	M24
D2.3	Integrated sensor platform for CO_2/RH/Temperature real time measurement including option for downloading of information by farm store managers	WP2	IRIS	DEM	CO	M36
D2.4	Final assessment of newly developed sorting technology for figs	WP2	FLI	R	PU	M36
D2.5	Guidelines for effective grains micronization/wet extraction milling procedures	WP2	BARILLA	R	PU	M36
D2.6	Report on the presence of mycotoxins and all their modified forms that are formed during food processing	WP2	BOKU	R	PU	M42
D3.1	Alpha release of the MyToolBox e-platform	WP3	IRIS	DEM	PU	M15
D3.2	Guidelines on optimal, cost-effective schemes for monitoring mycotoxin contamination and synthesis of available GAP, GMP guidelines and HACCP protocols for mycotoxin management in small grain cereals, maize and figs for use in WP3	WP3	CRANFIELD	R	PU	M24
D3.3	Forecasting models (for various combinations of mycotoxins/crops) for optimal harvest timing and use of (bio)fungicides	WP3	DLO-RIKILT	R	PU	M40
D3.4	Report on trials with the DSS platform in	WP3	CRANFIELD	R	PU	M40

[55] If your action taking part in the Pilot on Open Research Data, you must include a data management plan as a distinct deliverable within the first 6 months of the project. This deliverable will evolve during the lifetime of the project in order to present the status of the project's reflections on data management. A template for such a plan is available on the Participant Portal (Guide on Data Management).

[56] Type: R: Report; DEM: Demonstration, prototype, pilot; DEC: Websites, patents, press & media action; Other: Software, technical diagram
Dissemination level: PU: Public, fully open; CO: Confidential, restricted; CI: Classified; information as referred to in Commission Decision 2001/844/EC

Delive-rable Nr	Deliverable name	Work package number	Short name lead participant	Type[56]	Dissemi nation Level	Delivery date
	different silos systems in different regions (including China) by post-harvest grain/nut managers					
D3.5	Tested and validated web-based MyToolBox platform for integrated mycotoxin management and decision support toolkit for all actors in the chains of small grain cereals, maize and figs available	WP3	HORTA	Other (soft-ware)	CO/PU	M48
D4.1	Report providing end-user feedback to WP1, WP2 and WP3 partners on preliminary assessment of proposed interventions and initial MyToolBox e-platform design	WP4	FLI	R	PU	M18
D4.2	Report on preliminary end user evaluation of MyToolBox for application to cereal industries and to the feed supply chain and alternative uses of contaminated commodities	WP4	FLI	R	PU	M34
D4.3	Report on standard-setting implications of animal feeding trials with detoxifiers conducted in China	WP4	FLI	R	PU	M36
D4.4	Report setting out basis for establishing processing factors for mycotoxins based on transfer rates and overview of end-users assessment of final version of MyToolBox	WP4	FLI	R	PU	M45
D4.5	Report covering socio-economic impact of MyToolBox including recommendations for incentives	WP4	WU	R	PU	M45
D4.6	Report with recommendations to regulators resulting from mycotoxin reduction strategies developed in MyToolBox	WP4	BFR	R	PU	M47
D5.1	Project website	WP5	ICC	DEC	PU	M4
D5.2	Data management plan	WP5	BOKU	R	PU	M6
D5.3	Initial Communication Plan	WP5	ICC	R	PU	M6
D5.4	Interim Plan for the dissemination and exploitation of the project's results	WP5	ICC	R	PU	M24
D5.5	Knowledge Management Report	WP5	IRIS	R	CO	M36
D5.6	Final Plan for the dissemination and exploitation of the project's results (with Communication report annexed)	WP5	ICC	R	PU	M48
D5.7	Report on Training activities (with materials annexed)	WP5	ICC	R, DEC	PU	M48
D5.8	Satellite workshop at final conference	WP5	ICC	OTHE R	PU	M48
D6.1	Minutes from KO, MB and PSC meetings	WP6	BOKU	OTHE R	CO	M1-M48
D6.2	MyToolBox Intranet management tool	WP6	ICC	DEC	CO	M2

3.2 Management structure and procedures

Management structure

Project Management Organisation structure and decision making procedures

The coordination tasks defined in WP6 concern the overall coordination of the technological and scientific developments within the different WPs to ensure alignment of the developed products and scientific findings. The **Coordinator**, Prof. Rudolf Krska (BOKU), will be responsible for the robust coordination of the MyToolBox consortium and ensuring that the project is within scope, is delivered on time, within budget and to a high quality to ensure impact. Prof. Krska is a well-renowned and well-respected academic from the mycotoxin field globally. **Prof. Dr. Krska is currently the world-wide most cited mycotoxin expert of the last decade, has been involved in many international projects and has been successful co-founder of two spin-off companies of his university.** He is an ideal figure to lead such a flagship project and he also has the organisation and leadership skill-sets to effectively perform this role, which is demonstrated through is role as the Head of the Department of Agrobiotechnology at BOKU with a team of 180 researchers under his leadership. Prof. Krska will also be responsible for communicating all results, scientific and financial, and related issues with the EU representative.

The Coordinator will be assisted by a **Project Manager** and a **Scientific Manager** (Deputy Coordinator) from RIKILT. The Project Manager will assist the Coordinator in administrative issues related to reporting and finance. The highly experienced project manager Dr Kurt Brunner (BOKU) will be responsible for the timely delivery of draft scientific and financial reports to the Coordinator. The Scientific Manager Monique de Nijs (DLO-RIKILT) will monitor the progress of the **milestones** (see table 3.2a) and deliverables (see table 3.1c) and will (pre-)review all project reports. The Researcher Manager will also consult the Project Coordinator especially in case immediate feedback is required. The **Dissemination Manager** will be responsible for leading the consortium in further planning and implementation of the project communication and dissemination activities, and ensuring the project reaches out to different audiences in the most effective way. The **IPR Manager** (Dr Mirta Pinilla, IRIS) will be responsible for knowledge management within the consortium. Will lead the consortium in protecting the project results where appropriate, and in maintaining confidentiality of the results until the IP is protected to safeguard the innovation impacts. Mirta has vast experience in relation to knowledge management in collaborative projects. She has experience of leading patent applications, open access publications, etc. in past projects.

Each WP is led by a **WP leader** who take routine decisions communicate those with the Coordinator, monitor progress of output and budget according to the work plan, and prepare all documents related to their WP.

Table 3.2a: List of Milestones

No.	Milestone name	Related WP(s)	Estim. date	Means of verification
M1.1	Selection of a) biopesticides for *Fusarium* head blight experiments, b) crop debris treatments, c) brassica species for biofumigation studies, d) maize cultivars for aflatoxin resistance experiments and e) of atoxigenic *Aspergillus* strains completed	WP1	M3	List (names) available of the selected biopesticide products, crop debris treatments, brassica species, maize cultivars and of appropriate atoxigenic *Aspergillus* strains
M1.2	Field experiments in years 2 and 3 to optimise the selected a) biopesticides, b) crop debris treatments & c) biofumigation and to d) validate cultivar resistance environmental stability and e) quantify biocontrol efficacy, host genotype and environment stability completed	WP1	M30	Results of year 2 & 3 years field experiments statistical analysed and reported including translation to crop management advises provided to MyToolBox
M2.1	Infra-red regions for CO_2 sensing in different grain types selected and boundary temperature x moisture content models for growth and mycotoxin production in wheat/barley/oats and peanuts optimised	WP2	M12	IR-wavenumber-region available and start of prototype testing
M2.2	Performance of mycotoxin-degrading organisms and enzymes during the fermentation of contaminated wheat and maize at laboratory scale evaluated	WP2	M15	Organisms and enzymes and fermentation parameters to achieve best performance at biofuel production at lab-scale and guidelines provided to MyToolBox

No.	Milestone name	Related WP(s)	Estim. date	Means of verification
M2.3	Testing pre-milling and innovative milling strategies in pilot plants completed	WP2	M22	Data Collection from pilot plant trials for achieved mycotoxin reduction in wheat batches available. including translation to advise on safe use options provided to MyToolBox
M2.4	Prototype of an automated sorting system for dried figs and other commodities such as apricots and pistachios ready to be assessed by end-users	WP2	M30	Feedback from end users (see also WP3)
M3.1	Overview of available GAP, GMP and HACCP protocols for small grain cereals, maize and dried figs and on forecasting models including plans for their further extension	WP3	M10	Presentation, discussion and further decisions at Steering Committee Meeting
M3.2	Integrated platform completed for testing in end user silos for post-harvest managers	WP3	M36	Platform implemented in several silos for testing
M4.1	Completion of MyToolBox end-user assessment of α-version of MyToolBox	WP4	M24	Written feedback provided to WP3
M4.2	Feed additives studies with cows and pigs undertaken in China completed	WP4	M24	The efficacy of feed additives to reduce the absorption of dietary aflatoxin B1 (AFB1) in dairy cows and to reduce the absorption of dietary fumonisin B1 (FB1) in pigs under local conditions in China is known and reported
M4.3	Completion of MyToolBox end-user assessment of prototype version of MyToolBox and completion of cost-effectiveness assessment of proposed interventions from MyToolBox and feedback of proposed incentives to relevant partners	WP4	M36	Feedback provided to WP3 and written recommendations to WP3
M5.1	The project IP is being protected as appropriate and robust business, exploitation and dissemination strategies to indicate post-project impacts	WP5	M36	Means of verification: IRP register

Two **project management bodies** are envisaged: The Project Management Board, which is an executive management body of the project, and the Project Steering Committee, which is the strategic steering and decision making body.

Day-to-day decision making will be the responsibility of the **Project Management Board (PMB),** chaired by the Coordinator**,** whose membership will comprise the **Coordinator** (chair), **Scientific Manager** (Deputy Coordinator), **Project manager** and **WP leaders**. The Project Management Board will meet as frequently as required with a minimum of every 6 months. In general, occasions will be sought to combine activities to minimise travel costs.

The **decision-making body** on **major strategic decisions** will be the **Project Steering Committee (PSC),** chaired by the Coordinator, whereby each partner organisation will appoint one representative and will include the **Coordinator** (chair), the **Scientific Manager**, the **Project Manager**, the **IPR Manager** (Dr Mirta Pinilla, IRIS) and the **Dissemination Manager** (Dr Michaela Pichler, ICC) as well as the **Deputy WP-leaders** (WP1: Prof Ferenc Bagi, UNS; WP2: Prof. Naresh Magan, CRANFIELD; WP3: Dr Vittorio Rossi, HORTA; WP4: Dr Carsten Fauhl-Hassek, BfR; WP5: Dr Ine van der Fels-Klerx, DLO-RIKLIT; WP6: Dr Monique de Nijs, DLO-RIKILT). Each partner is assigned one vote to ensure a democratically run project (2/3 majority voting). Major strategic decisions will be made by the Project Steering Committee, such as change of scope, IPR related decisions, conflicts, budget related decisions, etc. All matters of a strategic nature will be floated to the Steering Committee and will aim to be accompanied by a recommendation from the Project Management Board. The PSC suggests refocusing of research directions, proposes the publication plan and decides on conflicting issues such as prioritizing samples in case of a temporary breakdown or other contingency situation and subsequent mitigating actions (see table 3.2b). The Project Steering Committee will meet once per year.

In addition, a voluntary Advisory Board has been established that will provide guidance and advice to the consortium through its international experts in themes of high relevance for the project. To this end, the **Advisory Board (AB)** of MyToolBox involves **policy makers (**table 7), **scientific advisors** (table 8), and **stakeholders** (table 9). The Advisory Board will provide an important mechanism not only for providing guidance on the Project Steering Committee, but also for dissemination to their membership and most importantly on the needs of the end users. The Advisory Board will contribute particularly to activities in WP4 (Practical Implementation) and WP5 (Dissemination), which are both outward-facing project activities. It is noteworthy that the Advisory Board is a voluntary body. The Advisory Board will sign a confidentiality agreement with the Coordinator in order to safeguard the confidential aspects of the project that they might be exposed to. The members of the Advisory Board will have no decision making capacity in the project. The AB will meet three times with the PSC.

In addition to policy makers, the AB includes renowned scientists in the area of mycotoxin reduction from two other continents, i.e. the United States Department of Agriculture (USDA) and the Nelson Mandela African Institution of Science and Technology. It also consists of **external stakeholders represented by** the European Breakfast Cereal Association (CEEREAL), the European Farmers & European Agri-Cooperatives (COPA-COGECA), the European Feed Technology Center (EUFETEC), the European Feed Manufacturer's Federation (FEFAC), the EU Association of Specialty Feed Ingredients and their Mixtures (FEFANA), the DLG group, one of Europe's biggest exporters of seed, feed and milling grain, the International Nut and Dried Fruit Council (INC) and the Global Food Safety Partnership (GFSP), a unique public-private initiative dedicated to improving the safety of food in middle-income and developing countries. All these organisations have already expressed their willingness to join the Advisory Board (**LOIs available, see Annex**) should the MyToolBox project be selected for funding. These stakeholders will provide additional 'reality-checks' from the outset of the project and will also function as a forum for ensuring that the actions have 'end-user' support.

Table 7: Policy members of the Advisory Board (AB)

	Short name	Full name
1	DG Sante	EU Health and Food Safety
2	FAO	Food and Agriculture Organization of the United Nations
3	EFSA	European Food Safety Authority
4	MAO	Chinese Ministry of Agriculture, Division of Feed (Hongyang Wei)

Table 8: Scientific advisors in the Advisory Board

	Representatives name	Full name
1	Peter Cotty, USA	USDA
2	Martin E. Kymania, Tanzania	The Nelson Mandela African Institution of Science and Technology

Table 9: Stakeholder members of the Advisory Board

	Short name	Full name	Mission
1	CEEREAL	European Breakfast Cereal Association	CEEREAL represents the breakfast cereal and oat milling industries towards the European Union and its institutions, industry and consumer associations as well as consumers.
2	Copa Cogeca	European Farmers & European Agri-Cooperatives	The united voice of farmers and their co-operatives in the European Union
3	EUFETEC	European Feed Technology Center	The EUFETEC mission is to support feed sector-bound and sector-steered technological innovation
4	FEFAC	European Feed Manufacturer's Federation	Represent, defend and promote the interests of the European compound feed industry to the European Institutions;
5	FEFANA	EU Association of Specialty Feed Ingredients and their Mixtures	FEFANA (EU Association of Specialty Feed Ingredients and their Mixtures) is the organization representing the interests of the specialty ingredients industry that enter the food chain via feed, whether produced in Europe or imported from third countries. Manufacturers and traders of feed additives, functional feed ingredients, premixes and other mixtures of specialty ingredients are amongst our member companies.
6	DLG	DLG group, Denmark	The DLG Group is one of the largest agricultural companies in

			Europe and is currently owned by 30,000 Danish farmers.
7	INC	International Nut and Dried Fruit Council	The INC is the International source for information on nuts and dried fruits for Health, Nutrition, Food Safety, Statistics, Government Standards and Government regulations regarding trade barriers and trade quality standards.
8	GFSP	Global Food Safety Partnership	The World Bank´s Global Food Safety Partnership is a unique public-private initiative dedicated to improving the safety of food in middle-income and developing countries. It brings together farmers, business and industry, governments, regulatory bodies, international development organizations, and civil society to drive a globally-coordinated and locally-driven food safety approach.

The overall organisational structure for management of the project and the relationship between the EC, Project Coordinator, Project Management Board, Project Steering Committee and the Advisory Board is shown in Figure 5.

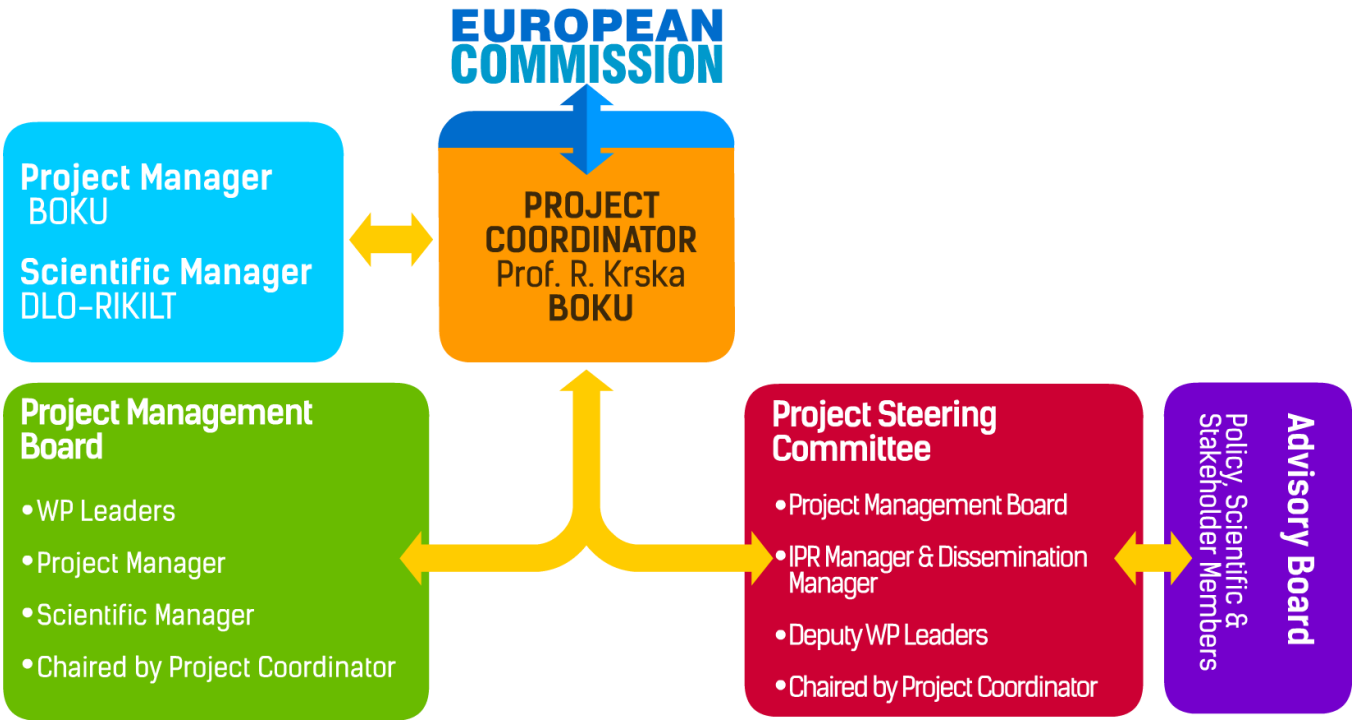

Figure 5 - Management structure for MyToolBox

Meetings

There will be nine face-to-face meeting points in the project during which one or several groups meet (Meetings are indicated in the GANTT Chart).

The first of nine **Project Management Board** (**PMB) meetings** will take place during the Kick-off meeting (M1**)**. BOKU will present the decision-making and financial procedures and will put the corresponding documentation on the internal website. The other eight PMB meetings will take place in months M6, M12, M18, M24, M30, M36, M42. Comprehensive progress reports and plans for the next 4-6 months will be presented by the WP leaders. These meetings serve to solve practical and scientific issues of non-strategic nature and to discuss immediate actions to be taken case milestones or deliverables cannot be delivered on time.

The five **Project Steering Committee** (**PSC) meetings** will take place during the kick-off meeting in M1 and in months M12, M24, M36 and M48. At the PSC meetings, major strategic decisions are made, the progress of the entire project is discussed in detail in cooperation with the Advisory Board (see below) and plans for the next 12 months are reported. The Project Steering Committee will verify that the project meets both deliverable and milestone targets. If there is a major decision point in a WP which could affect the scope of the work for example, the issue will be brought

first to the Project Management Board and then to the Project Steering Committee for decision making. The Project Steering Committee will have an agenda which covers at least the following points: i) Update of progress against implementation plan; ii) Update on next stage of implementation; iii) Financial issues; iv) Contractual issues; v) Intellectual property issues; vi) Update on meetings with DG Research and other stakeholders; vii) Gender implementation plan; viii) Dissemination of project results

The **Advisory Board** (**AB) meetings** will take place in months M1, M24 and M48. In addition, annual teleconferences after year 1 (M12) and 3 (M36) will be organised between the Project Steering Committee and the Advisory Board.

Each meeting will be held at the location of one of the WP leaders. The host will provide meeting rooms, lunches and one formal dinner for all attendants. The meeting in Month 12 will be held in China. The meetings in months M36 and M48 where the stakeholders will be present will be organised by ICC.

Teleconferencing will be used to address specific issues that might arise during the project or where specific decisions are required. WP leaders will also need to hold separate *ad-hoc* meetings of those partners involved in activities, and again these will be combined wherever possible with other meetings.

The Coordinator will chair each meeting and will produce written minutes of each meeting, which shall be the formal record of all decisions taken. Minutes of each meeting will be distributed by e-mail to all project partners within a week.

All Project Management Board Members and Project Steering Committee members can suggest topics for the meeting agenda.

Additionally, it is expected that WP leaders could be invited to specific stakeholder association meetings to describe specific commodity related aspects of the project. The Agenda will be a mixture of progress reports from WPs where there are non-confidential scientific and innovation achievements to report and items generated by the Project Steering Committee that require end-user input.

Reporting Procedures

In addition to formal deliverables identified in table 3.1c, WP leaders will provide the Coordinator with 6 monthly and annual internal progress reports as follows:

- 6 monthly reports summarising progress on milestones and deliverables, reporting any unexpected results and future strategies
- 12 monthly report reporting on progress, highlighting any planning issues and also providing a summary of financial resources for the WP

The internal reports will feed into the Contractual reports required by the Grant Agreement.

In addition, all partners will contribute to the formal project progress reports and will submit formal financial statements as appropriate in line with the requirements of the Grant Agreement. Communication amongst the consortium will be enabled through a secure section of the MyToolBox website (by password access), which will also serve as a structured repository for project documents and will be the main project platform for internal exchange of information.

Appropriateness of the organisational structure and decision-making mechanisms

This large consortium includes pan-European and cross-continental members that has a remit to address a pressing societal challenge in a 48 month work plan through the delivery of a series of ambitious scientific and technological advances and ensuring high impact. To this end, we are confident that a core project management unit made up of the Coordinator, Scientific Manager and Project Manager, each with key roles and responsibilities is a realistic and robust approach. As this core unit will be supported by the Project Management Board, is sufficient decentralised leadership to enable this ambitious project highly manageable. In the interest of a democratic and inclusive project for all partners, the existence of a Project Steering Committee that can focus of the strategic direction of the project will ensure that all partners have a voice in how the project is run and all are afforded an opportunity to steer the project ensuring it delivers impacts for all partners and for the wider stakeholder communities (through external engagements and interaction with a highly relevant and influential Advisory Board).

Innovation Management

As MyToolBox is focussed at delivering innovations that will lead to real changes in agricultural and food processing practices, it is critical that the MyToolBox partners from the academia and industry are strongly linked to 'end-users'.

As such, innovation Management is firmly embedded in all aspects of our project. Importantly, the development of MyToolBox will be done in close interaction with the end-user stakeholders. On the one hand this is achieved through the inclusion of five end-user partners in the MyToolBox project. While these five **SME and large industries** will have direct practical involvement in the project by providing access to facilities and providing materials, ultimate success will

require uptake by much larger sections of the agricultural community and the food industry. To this end, in addition to collaboration with our end-user partners, engagement with a wider community of end-users from beyond the consortium will be mobilised, whereby they will be afforded the opportunity to provide their detailed user aims and requirements that will feed into and have an influence on the design and development of the MyToolBox e-platform. They will also be afforded the opportunity to be involved in testing its various iterations so that continuous feedback loops from the users to the development teams are established. The first alpha version of MyToolBox will available for feedback on use by stakeholders (M15) in WP4. Subsequent iterations of the beta platform will also be tested by them so that they can provide their feedback so that a fully operational and debugged version of the platform can be deployed in the latter stages of the project (WP3). This will ensure there is an effective innovation cycle to our project, where the end-user and stakeholders are central to the development to ensure a market and user-centric approach. Cross-cutting activities in our project such as Knowledge/IPR management, business and exploitation planning are all central to Innovation Management.

Risk management

During the project we will systematically analyse and manage risks that could jeopardise the implementation of the project, its scientific and technological outcomes, as well risks in relation to IPR, or the ability to deliver the projects innovation and long term impact. The tasks and the organisational structure were analysed for their specific risks per WP. The severity of each risk was analysed and a plan for a solution was suggested. In case a risk does happen, the partner will immediately inform the work package leader who will inform the Coordinator. The Coordinator will directly inform the EU representative. Solutions will be discussed between the coordinator, the WP leader and the partner and proposed to the EU representative and only be implemented after discussion with and written approval of the EU representative. Consequences for distribution of budget will be identified and agreed upon with EU representative before the implementation of the proposed solution.

At proposal stage, a number of critical risks have been identified and discussed in table 3.2b below including details on risk mitigation action to be taken.

Table 3.2b: Critical risks for implementation

Risk nr.	Description of risk	WP (s) involved	Proposed risk-mitigation measures
R1.1	Field experiment crop failure (e.g. poor crop establishment due to adverse environmental conditions)	WP1	Multiple experiments conducted in multiple years and environments reduce impact of individual crop failure in single year/environment.
R1.2	Lack of efficacy of biopesticides against *Fusarium or inconsistency of biopesticides between environments*	WP1	Screen large number and diverse range of biopesticides including some with preliminary data showing efficacy. Screen within various environments in UK and Norway. Output maybe environment specific and identify most suitable biopesticides for Northern and Central Europe
R1.3	Lack of efficacy of crop residue management against *Fusarium* or Inconsistency in efficacy of crop residue management between environments	WP1	Multiple strategies to be investigated including some with preliminary data showing efficacy Due to environmental condition this can be expected and using different strategies (competition for residues and biofumigation) will overcome this
R1.4	Lack of aflatoxin resistance within European maize genotypes tested	WP1	Large number of genotypes will be screened and known differences within two traits (drought tolerance and maturity class), key to aflatoxin resistance, are known
R1.5	Lack of efficacy of atoxigenic biocontrol against aflatoxin	WP1	Atoxigenic biocontrol to reduce aflatoxin has been successfully developed in America, Africa and Asia. This is application of proven technology will be implemented with the aid of International Experts from each of these continents
R1.6	Complexity of matrices beyond those validated for quantitative multi-target analysis may cause significant signal suppression in LC-MS/MS analysis	WP1, WP2	Apply ^{13}C-labelled internal standards and clean-up strategies for major mycotoxins to correct for matrix effects; severity sensitivity and accuracy of the method by regular participation in international ring-trials.
R2.1	Lack of conformity between integration of abiotic/biotic data and process-	WP2	This technical risk will be mitigated re-analysing all output information and exploiting new variables.

Risk nr.	Description of risk	WP (s) involved	Proposed risk-mitigation measures
	engineering solutions within integrated management system.		
R2.2	Technical failure to apply the screening model to diversified raw materials characterized by different properties	WP2	Evaluation of alternative screening parameters/properties not strictly related to hyperspectral imaging.
R2.3	Failure to obtain the desired increase in yields according to pre-milling/milling mycotoxin mitigation protocols	WP2	Special measures (applications and instrumental conditions/parameters) will be newly set.
R2.4	Identified and tested mitigation strategies (exploiting food processing effects) do not provide expected results in terms of percentage of mycotoxin reduction	WP2	Special measures (applications and instrumental conditions/parameters) will be newly set. Further pilot/industrial trials will be designed to strengthen the predictive models.
R2.5	Failure of the established degrading organisms related to biogas and bioethanol production.	WP2	Partner BIOMIN is EU leader in the field. Test on different species/strains according to time/budget re-allocation.
R2.6	Variable results obtained from animal feeding trials in China upon specific mycotoxin detoxifying feed additives due to inter-individual differences.	WP2	A repetition of the experimental design could be set in order to reduce number of involved animals and inter-individual variability.
R2.7	Degradation / conjugation products of mycotoxins formed during thermal processing, cannot be chromatographically retained under applied LC-MS/MS conditions due to i) polarity, ii) covalent binding or iii) lack of standards	WP2	Analytical method will be extended to i) hydrophilic interaction chromatography, ii) solvolysis will be applied with automated labelling and iii) synthesis of standards at BOKU from cell cultures
R3.1	ZEN in wheat data available from only a few EU countries, or not linked to farm management	WP3	ZEN will be included in current DON models only for those countries for which ZEN data is available, or using weather data only (not linked to farm management)
R3.2	Available DON-wheat forecasting models from EU show bad fit for China	WP3	Models will be re-parameterized for China
R3.3	Pilot scale testing of post-harvest DSS shows no large differences (< 10-15 %) between using and not using the DSS	WP3	Further tests will be planned to confirm this and consultations will be made with testers in order to identify the value of using the DSS and to establish additional user requirements that should be met.
R3.4	Few data on the same batch sampled / analysed in different ways is available	WP3	Data from different batches will be used, and Monte Carlo simulations run to take account of variability and uncertainty
R3.5	The MyToolBox e-platform fails to fulfil on usability and user experience	WP3	The iterative development involving close engagement with target users (including farmers) will ensure any failures are quickly addressed in upcoming development sprints.
R4.1	Poor response to consultations via questionnaires	WP4	Increase level of telephone and face-to-face interviews to be conducted
R4.2	End-users indicate severe motivational barriers to implementation of interventions despite possible incentives or provide adverse feedback from end-users from stakeholder focus groups on practical viability of implementation of measures	WP4	Seek compromises with WP leaders (WP1, WP2 and WP3) to find means to accommodate end-user concerns
R4.3	Interventions in WP2 do not achieve	WP4	The wider potential impact of improved understanding the

Risk nr.	Description of risk	WP (s) involved	Proposed risk-mitigation measures
	sufficient reductions in mycotoxin levels to justify changes in regulatory limits		effects of processing factors on limits for processed foods will explored with regulators.
R4.4	Results from *in-vivo* experiments in China with cows and pigs with feed additives are inconsistent with EU data	WP4	Facilitate meeting between EU and CAAS-FRI researchers to identify and reconcile differences in data interpretation– if necessary repeat experiments
R4.5	Failure to reach informal agreement on standards to be applied in China for testing feed additives	WP4	Consider taking results from experiments in China to Codex to seek some form of arbitration.
R5.1	Contrasting visions among different partners/stakeholders with regard to strategies, methodologies or intellectual property issues.	WP5	Any conflicts will be managed by Management Board, assisted by the IPR Manager, in keeping with the information contained in Section 2.2b of this bid and in line with a previously signed Consortium Agreement in order to keep confidentiality requirements yet maintain effectiveness of the developed strategies and protocols.
R5.2	The business model for the MyToolBox platform is unconvincing and will not lead to sustainable post-project exploitation	WP5	This will become apparent after an analysis of the business plans at Project Interim and the Project Management Board will propose a contingency Business model and consultations with external experts will be made if required.
R6.1	Misunderstanding among partners (i.e. China/Europe).	General (managed in WP6)	Partner experiences in cross-cultured context.
R6.2	Estimated resources (i.e. P/M) might not be sufficient.	General (managed in WP6)	Special measures with time (i.e. milestones/ deliverables) and budget re-allocation, if necessary.
R6.3	Key personnel, scientists or laboratory personnel involved can no longer participate due to unforeseen changes in work due to food safety crises, illness, change of job or position or any other reasons	General (managed in WP6)	Key personnel will be quickly replaced with another similar profile, scientist or laboratory personnel with relevant expertise. The CV will be discussed with the coordinator for approval.
R6.4	Unforeseen delays in the workplan due to the close connection among different sub/tasks	General (managed in WP6)	Re-scheduling of the affected parts will be immediately planned with the Management Board.

3.3 Consortium as a whole

Description of the consortium

The MyToolBox consortium consists of **leading experts from 23 partner organisations**, **43%** of which are **from industry including farmers**, who together have jointly developed the MyToolBox proposal. The consortium is founded on high quality scientific work and long experience in the proposed fields of innovation from the partners.

The **multi-actor partnership** in MyToolBox is exemplified by the involvement of **five prominent end users** representing the food industry (Barilla, Italy), food and crop energy (Südzucker, Germany), grain industry (AGRO LV, Ukraine with operations in Poland), fig industry (TARIS, Turkey) and the feed industry (D.O.O. Agrocentrum Becej in Serbia, which also involves the farm of its owner). Hence, through our consortium end users from the farming community, agronomists, actors along the food & feed supply chain, professionals working in agriculture and food manufacturing, along with policy partners, will collaborate closely with a group of 5 technology SMEs and with the partners from academia. This is a complete value chain of actors. MyToolBox has EU partners from Austria (Coordinator), the Netherlands, Germany, Great Britain, Spain and Italy, from associated countries of Norway, Serbia, Turkey and Ukraine (in cooperation with Poland) and **three partners from China** (ASAG, CAAS-FRI and CAAS-IAPPST) who are tightly connected with several end users representing the Chinese grain and peanut industry.

The MyToolBox project will build on the positive achievements of the substantial EU-funded research previously conducted and mentioned before which dealt with mycotoxin contamination in the food and feed chain. Through the

Project Advisory Board representing policy makers and seven Stakeholder associations from the European food and feed industry (see 3.2) valuable additional input will be provided especially in relation to the needs of end users all across Europe and beyond.

Participation from outside EU

Besides the strong involvement of our partners from the 'Other Country' **China** in line with recommendations from the call text (see below), the MyToolBox consortium has strong linkages in research and innovation with many other national, European and international projects around the world. Some 40 countries are represented through the participation of ICC, mainly through governmental organisations, federal ministries, or national research institutions. The continents are also connected via linkage to the International Society for Mycotoxicology (ISM), of which the MyToolBox Coordinator is General Secretary.

Through the Project Advisory Board, MyToolBox will receive **valuable advice from outside the EU** including the Food and Agriculture Organization of the United Nations (FAO) and the Chinese Ministry of Agriculture, Division of Feed (MOA) who have also confirmed their participation in the MyToolBox Advisory Board. Moreover, the Advisory Board will obtain scientific support through the involvement of renowned scientists in the area of mycotoxin reduction from two other continents, i.e. the United States Department of Agriculture (USDA) and the Nelson Mandela African Institution of Science and Technology. By connecting with USDA and its projects funded by the Bill & Melinda Gates Foundation, advice on the formulation and application of biological control agents (see WP1) will be provided.

The involvement of the Chinese Academy of Agricultural Sciences (CAAS-FRI and CAAS-IAPPST) and the Academy of State Administration of Grain (ASAG), MyToolBox allows to **link to a number of ongoing research projects in China** that deal with mycotoxin formation and its prevention and control in grains and cereals, as carried out in WP1 - 3. By doing to, numerous Chinese universities, academies and end users who have either agreed to contribute to MyToolBox or will be willing to become part of the EU-China-Mycotoxin-Forum to be established in MyToolBox:

- **Academy of State Administration of Grain (ASAG)***
- **Chinese Academy of Agricultural Sciences (CAAS)** *
  - Feed Research Institute (CAAS-FRI)*
  - Institute of Agro-Products Processing Science and Technology (CAAS-IAPPST)*
- Institute of Food Quality and Safety, Jiangsu Academy of Agricultural Sciences
- Department of Public Health, Chengdu Medical College
- College of Light Industry and Food Sciences, South China University of Technology
- Xihua University
- Institute of Medicinal Plant Development, Chinese Academy of Medical Sciences
- Shandong Academy of Agricultural Science
- China National Center For Food Safety Risk Assessment (CFSA)
- Chinese Ministry of Agriculture, Division of Feed
- Hebei Baoding Qinghuaxufeng Trade of Grains and Oil Co. Ltd.
- China Grain Reserves Corporation
- Hualong (Rushan) Foodstuffs Enterprise Co., Ltd.

* MyToolBox partner

In addition, the following Chinese institutions will be connected to MyToolBox and is EU-China-Mycotoxin-Forum via Prof Liu Yang´s (CAAS-IAPPST), who is coordinator of numerous mycotoxin related projects in China:

- Agricultural Resources and Regional Planning of the Chinese Academy of Agricultural Sciences
- Beijing Entry-Exit Inspection and Quarantine Bureau
- China National Center for Food Safety Risk Assessment
- Fujian Agriculture and Forestry University
- Henan University of Technology
- Huazhong Agricultural University
- Institute of Plant Protection, Safety and Quality Institute of Agricultural Products of Anhui Academy of Agricultural Sciences
- Jiangsu Academy of Agricultural Sciences
- Jiangxi Academy of Agricultural Sciences
- Jinan University
- Liaoning Technical University
- Nanchang University

- Northwest Agriculture & Forestry University
- Oil Crops Research Institute of the Chinese Academy of Agricultural Sciences
- Jiangnan University
- Nanjing Institute of Agricultural Mechanization of National Ministry of Agriculture
- Nanjing Agricultural University
- Peanut Research Institute of Shandong Academy of Agricultural Sciences
- Qiqihar University
- Sericulture & Agri-Food Research Institute of Guangdong Academy of Agricultural Sciences
- Shandong Agricultural University
- Shanghai Academy of Agricultural Sciences
- Southwest University
- Zhejiang University

New relations and collaborations will be forged with other relevant projects over the course of the project with the view to maximising synergies and establishing an enduring scientific relationship with China. The EU-China partnerships established through MyToolBox will contribute to creating transparent supply chains for food and feed imported in the EU and exported to China, which will have far reaching implications for food safety and consumer confidence.

Roles of each member, management and leadership

The roles of each participant are described in the partner profiles. MyToolBox will be **coordinated by BOKU** (lead) The Coordinator Prof Dr Krska is the world-wide most cited mycotoxin expert of the last decade, has been involved in many international projects and is successful co-founder of two spin-off companies of his university. His expertise will be applied in leading the project and supervising the analytical work and metabolomics studies carried out in WP1-2. Furthermore, **Prof Dr Krska** will play a key role in the relations with the Chinese partners thanks to his **distinguished professorship at CAAS**; **WP1** will be led by Partner **HAU**. Within WP1, HAU will contribute to the tasks on alternative chemical control and novel cultural control of *Fusarium* Head Blight in small grain cereals through the use of field experiments. **BIOFORSK (SME)** will participate in WP1 to identify treatments (chemical/biological) that can reduce the inoculum potential of *Fusarium* spp. on oat straw residues by aiding residue decomposition and/or directly targeting the pathogen. **AXEB (SME)** will provide commercial bio-pesticides as active ingredients for the screening of products in field experiments in wheat (UK) and oat (Norway) in WP1 and will advise on the selection of alternative control products for *Fusarium* control in small grain cereals. **UNS** will investigate the possibility of reduction of aflatoxins in maize through genetics (drought resistance, early maturity), and secondly finding the way of biocontrol for aflatoxins in maize in WP1. **Agrocentrum (SME)** will be responsible for providing field experiments at one locality in Serbia, near Bečej. It will assist UNS in WP1 during inoculation, infection assessments, harvest and storage of maize from field experiments to reduce aflatoxins in maize through genetics and through biocontrol. **AGRO LV (IND/End User)** will provide advice on end-user suitability of components of the integrated mycotoxin management system developed in WP1 and will conduct demonstration trials of successful control strategies.

Barilla (IND/End User) will coordinate **WP2** including the research into post-harvest measures including innovative milling strategies and the investigation of the effects of baking processing on mycotoxin levels. **Cranfield** will be involved in WP2 in relation to the development of real time monitoring of biological activity in stored cereals and peanuts and in WP3 to provide a dynamic DSS tool for post-harvest grain storage managers. **IRIS (SME)** will work with Cranfield in the development of real time monitoring system for stored grains/peanuts (WP2), and in planning the post-project exploitation pathways for the solution. In order to facilitate safe use options for contaminated cereal grains, mycotoxin degrading enzymes/microorganisms provided by **BIOMIN (IND)** will be evaluated in cooperation with BOKU for their efficacy during the biogas and bioethanol production. For the latter, **Südzucker (IND/End User)**, a sugar producing company, will be involved in WP2 as end user, to test the efficacy of these employed enzymes and microorganism in their industrial facilities, to minimize the mycotoxin content in the by-product dried distillers' grain soluble (DDGS).

SDU is another experienced partner in WP2 and will contribute to developing non-invasive screening and sorting systems for aflatoxin contaminated figs by using NIR and VIS spectra. Finally, in WP2, **TARIS (IND/End User)** will contribute to the development of real-time fig sorting systems by providing consultancy to partner SDU who is the responsible partner for system development. TARIS will also be the responsible partner to announce the developed system to the other fig processing companies which are a member of TARIS fig Union.

DLO-RIKILT will lead **WP3** to develop, test and validate the web-based ICT MyToolBox platform. Moreover, DLO-RIKILT will lead WP3 task related to forecasting optimal farm management and control, and optimization of mycotoxin monitoring. **HORTA (SME)** will lead the design and testing of MyToolBox and will contribute to WP3 task related to

forecasting optimal farm management and control strategies. HORTA will also be involved in WP1 to perform field experiments to test different maize genotypes. In WP3, **IRIS (SME)** will build the infrastructure for the web-based MyToolBox web-based cloud platform, and its mobile platforms. They will also test and service the platform over the course of the project, and, together with HORTA they will plan a sustainable business model for the platform in WP5.

FLI (SME) will provide leadership within **WP4** and will interact with WP1, WP2 and particularly WP3 to ensure the involvement of end-users is maximised at all stages of development. As a standard setting authority, **BfR** will contribute to WP4 by carrying out a scientific compliance assessment of maximum levels in the feed/food chain for selected examples which will lead to an identification of points to be changed in legislation/standards in order to implement and apply the MyToolBox and analysis/discussion of benefits and drawbacks and making regulatory impact assessment. In WP4, **WU** will analyse how the risk behaviour of individual actors along a supply chain can affect the cost-effectiveness of proposed control and prevention measured and will study the potential socio-economic impact of MyToolBox including recommendations for incentives.

ICC (SME) will act as leader of **WP5** and will implement tasks related to communication and dissemination of information about the project and its results to a wide audience and by multi-actor engagement in connection with a MyToolBox identity, website and promotional material. The development of effective exploitation plans and of a sustainable business model for the platform in WP5, as mentioned above, will be carried out by IRIS.

With respect to the strong involvement of **Chinese partners**, the Institute of Agro-Products Processing Science and Technology (IAPPST) of **CAAS** will provide information and share experiences on the production, formulation, application, standardisation and long-term monitoring of biocontrol agents for peanuts and maize (**WP1**). Prof Liu Yang, chief scientist within IAPPST is coordinator of numerous national projects which are dealing with the prevention, formation and detoxification of mycotoxins. In cooperation with ICC, the International Society for Mycotoxicology and the Division of Mycotoxicology, Mycological Society of CAAS and the MyToolBox coordinator, Prof Yang will be coordinating a **network of knowledge** through the EU-China-Mycotoxin Forum between the institutions and stakeholders involved in the MyToolBox project including the Chinese institutions CAAS-IAPPST and CAAS-FRI and ASAG and their cooperation partners from the academia and industry (WP5). Based on the guidelines established in the EU, *in vivo* experiments in dairy cows and pigs will be conducted at the research facilities of the Feed Research Institute of CAAS (CAAS-FRI), in order to assess the efficacy of mycotoxin detoxifying feed additives under local Chinese conditions (**WP4**). In cooperation with several end users from China, **ASAG**, the second major Chinese partner within the MyToolBox consortium will provide access to its silos in four different provinces to install the real-time sensors for the post-harvest monitoring of grains and peanuts to be developed in **WP1**. ASAG will facilitate collaboration with two end users from the Chinese grain industry in post-harvest monitoring, i.e. Hebei Baoding Qinghuaxufeng Trade of Grains and Oil Co. Ltd. and the China Grain Reserves Corporation, which owns most of the grains and closely cooperates with ASAG. In the field of peanuts, ASAG will facilitate collaboration with to Hualong (Rushan) Foodstuffs Enterprise Co., Ltd., the leading company of the Chinese peanut industry both in production and export. ASAG will also collaborate in development of forecasting models to predict potential fungal contamination of small grain cereals at an early growth stage (**WP3**). In this area ASAG will involve two end users, i.e. the farmers Luo Jiakui and Lin Qiyin who are located near Mingguang City in the Anhui province, where mostly wheat is grown.

In short, this is a very strong consortium that brings together relevant academic, industry and policy actors from the field of mycotoxins, food and feed production, food processing, ICTs, photonics, process engineering, standardization and consultancy, who are all stakeholders in the outcomes and the medium to long term impacts of this project. The consortium mobilises the necessary **complementary scientific excellence, industry involvement**, commercial ambition and regulatory insights to be able to effectively and sustainably tackle a major societal challenge related to the biological contamination of crops and the food chain in the EU (including associated countries), China and beyond.

Leadership

One of the major strengths of the consortium is that it joins **outstanding expertise covering all aspects of the call text**. **The partners are complementary to each other** in several ways such as:

- Prof Dr Krska is currently a leading expert in **mycotoxin research world-wide** (greatest number of citations in the last decade and greatest number of papers in the area of mycotoxins since 1961) and involved in many international projects. His expertise will be applied in leading the project and exploring the field of masked mycotoxins. Furthermore, Prof Dr Krska will play a key role in the relations with the Chinese partners thanks to his distinguished professorship at CAAS;
- BOKU (Coordinator) is highly experienced in project coordination and has all necessary administrative and financial tools available to perform this task. DLO-RIKILT) is a well experienced coordinator and deputy coordinator of many EU projects e.g. COLLAB4SAFETY, CONffIDENCE; NANODEFINE.Prof Dr Yang (CCAS-IAPPST), Dr.

Wang Jin Quan (CAAS-FRI) and Dr Wang Songxue (ASAG), are highly respected scientist in food and feed science in China. They lead internationally known research groups and will carry out research on forecasting, real-time monitoring of grain silos and feeding trials in their institutions;

- Dr. Schatzmayr, Dr. Suman, Mr. Gortva, Mr. Casals and Dr Klingeberg are well known experts from the **private food and feed industry** whose expertise will be implemented in technical area's as well as for advise on policy making;
- AGRO LV, AGROCENTRUM and HAU are **farmers and represent farmers**, their immense expertise in crop management regarding mycotoxins will be implemented for the field trials but also for the translation of the scientific findings into applicable relevant and clear crop management advises;
- IRIS, BIOFORSK and HORTA have vast experience in i**ntegrating relevant prediction models in e-platforms**
- DLO-RIKILT and BfR are internationally well known for their knowledge in the food safety area, with Dr. de Nijs as an expert in mycotoxins and Dr. Fauhl-Hassek in the field of risk assessment. The expertise of Dr de Nijs will be implemented as Scientific Manager.
- FLI has considerable experience working with the food and feed industry as well as regulators. FLI has got the ability to network with a wide range of end-users and to be flexible in working with the primary agricultural producers as well as processors at different stages in the chain and with different size organisations ranging from multi-nationals to SMEs.
- **Dissemination activities** through press articles, articles in professional magazines, peer reviewed publications, exhitions and conferences, websites, social media and other dissemination channels, will ensure wide diffusion of the project results under the leadership of ICC who has got members in over 40 countries (and who will be heavily supported by IRIS).
- CRANFIELD and UNS and SDU are leading knowledge institutes with access to fungal collections, sensor technologies. They will design, coordinate and evaluate field trials and prototypes of sensors;
- Dr. van der Fels-Klerx and Dr. Mourits are leading experts in the area of bioeconomic modelling and their expertise will be used to maximise the acceptance of MyToolBox.
- IRIS brings vast expertise in vanguard and widely accessible **Information and Communication Technologies**, which are a fundamental practical vehicle for channelling the project results into practical decision support tools that can be used by farmers and professionals from the food and feed chains.